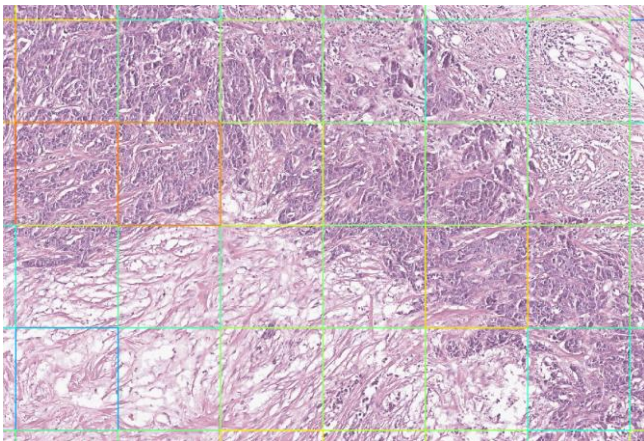


슬라이드 별 ROI 영역 선정 절차 및 참고 사항

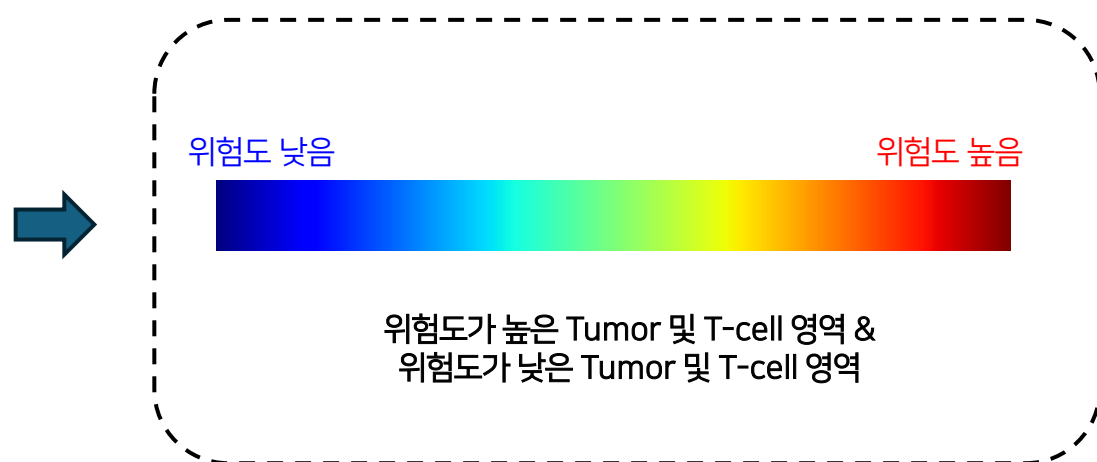
환자 : 1_152_19

- H&E 염색 슬라이드 이미지에서 작은 패치 단위로 분할하여 분석 수행
- 작은 패치의 크기는 개략적으로 약 $270\mu\text{m}$ (정확한 수치는 아니고 근사값, 패치 간에 오차가 있을 수 있음)
- .tmpprotocol 파일을 통해 annotation 파일 전달 ("[환자ID]_ROI_Updated.tmpprotocol")
- 한 튜브에 2개 이상의 패치가 들어감 (평균적으로 4개) -> 이전의 한 개 튜브에 한 패치가 아닌 여러 개의 패치가 한 튜브에 들어감
- 한 튜브에 들어갈 이웃 패치들은 위치적으로 근접하고 (경우에 따라서는 아닐 수도 있음) AI 예측 점수가 유사한 애들끼리 클러스터로 묶음
- 한 환자 당 48개의 튜브를 구성할 수 있는 ROI 영역 선정 완료
- ROI는 Tumor 영역과 T-cell 영역, AI 점수가 높은 곳과 낮은 곳이 구분되는 영역을 위주로 선별 (AI 예측 결과 활용, tiatoolbox)

사각형 사이즈 하나당 약 $270\mu\text{m}$ 크기를 가짐



각 사각형 색상의 의미 -> "AI 재발 예측 위험도"

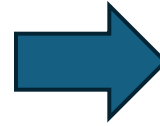
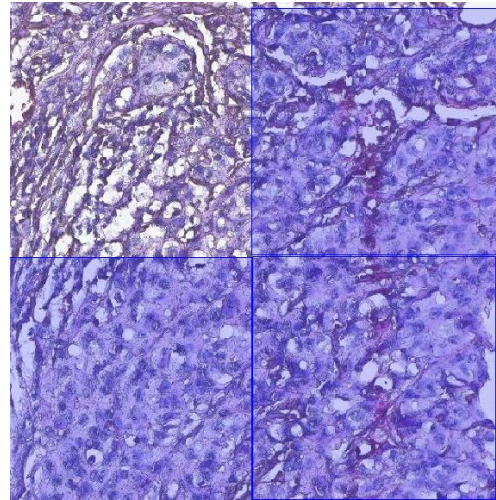
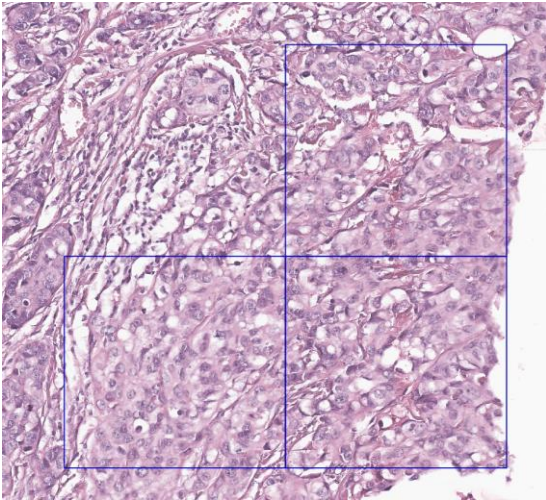


- 1_152_19 환자에서 점수 대에 따른 Region 별 분포

	0.00-0.10	0.10-0.20	0.20-0.30	0.30-0.40	0.40-0.50	0.50-0.60	0.60-0.70	0.70-0.80	0.80-0.90	0.90-1.00	
High score & Tumor (section e)	-	-	-	-	-	-	9	4	-	-	13
Low score & Tumor (section f)	-	5	10	-	-	-	-	-	-	-	15
Tumor Control (section v)	-	-	-	-	2	3	-	-	-	-	5
High score & T-cell (section g)	-	-	-	-	1	3	2	1	-	-	7
Low score & T-cell (section h)	-	4	2	2	-	-	-	-	-	-	8
T-cell Control (section w)	-	-	-	-	-	-	-	-	-	-	-
											48

총 48개의 well 에 들어갈 region 선정!

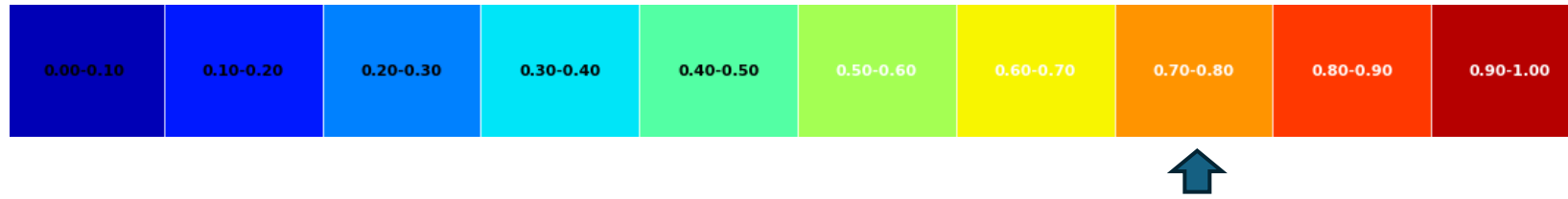
- 아래와 같이 영역이 선정된 경우, 이 경우 b1으로 naming이 되어 있으면, .tmpprotocol 파일에도 annotation 되어 있음



68	☒	b1	B12	☒
69	☒	b1	A12	☒
70	☒	b1	B12	☒

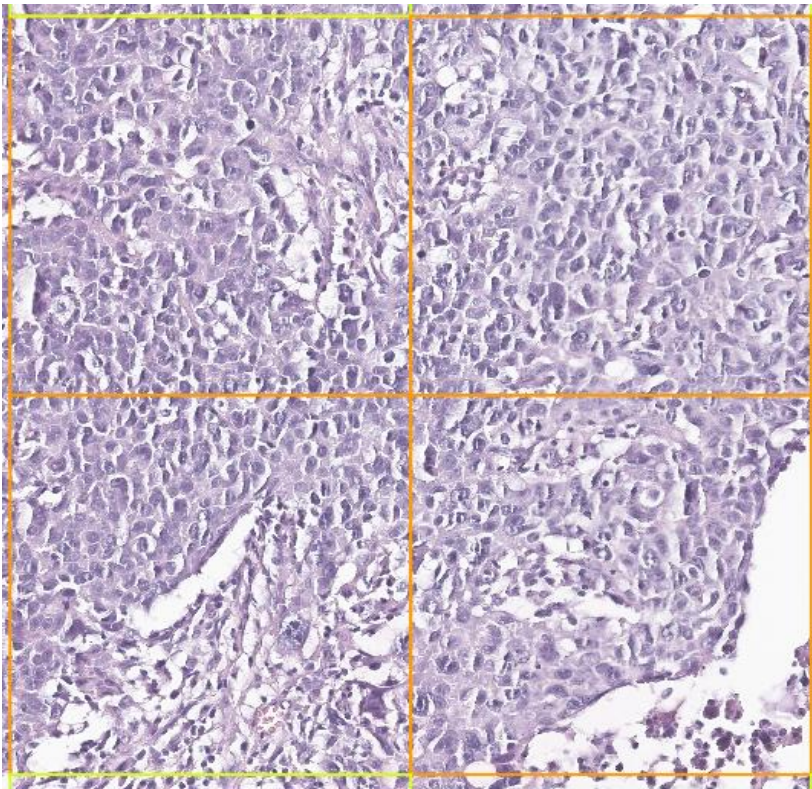
ROI Regions

OmixAI 1-152-19 환자 샘플 high-score tumor regions

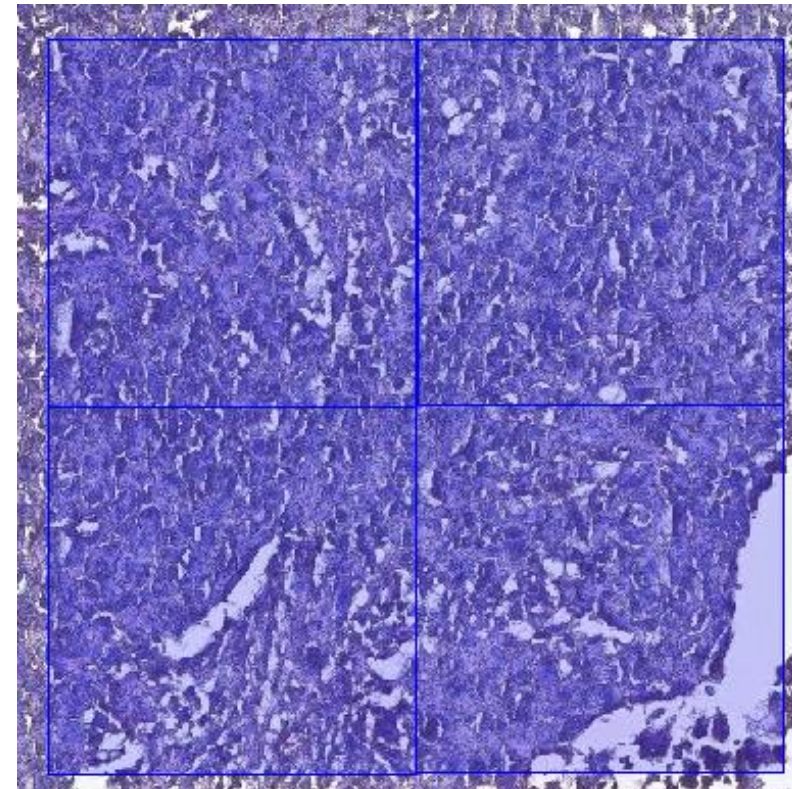


- Section **e1** => High Score & Tumor

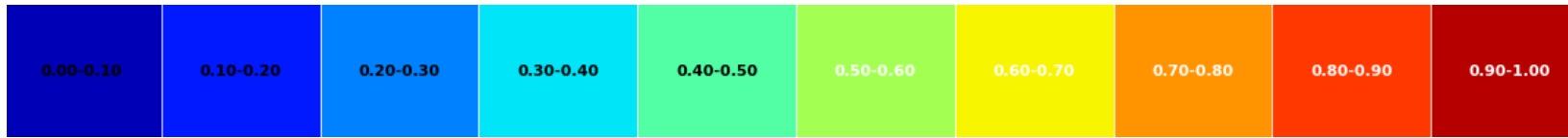
AI 스코어



Astromapper 매핑 결과

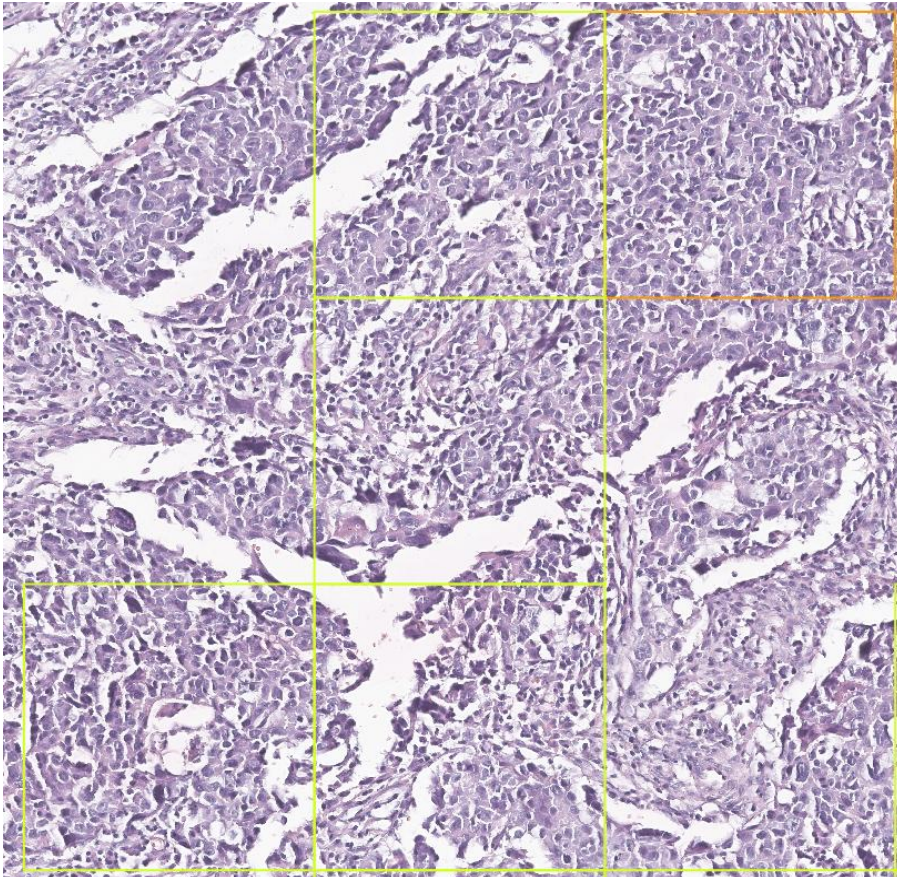


OmixAI 1-152-19 환자 샘플 high-score tumor regions

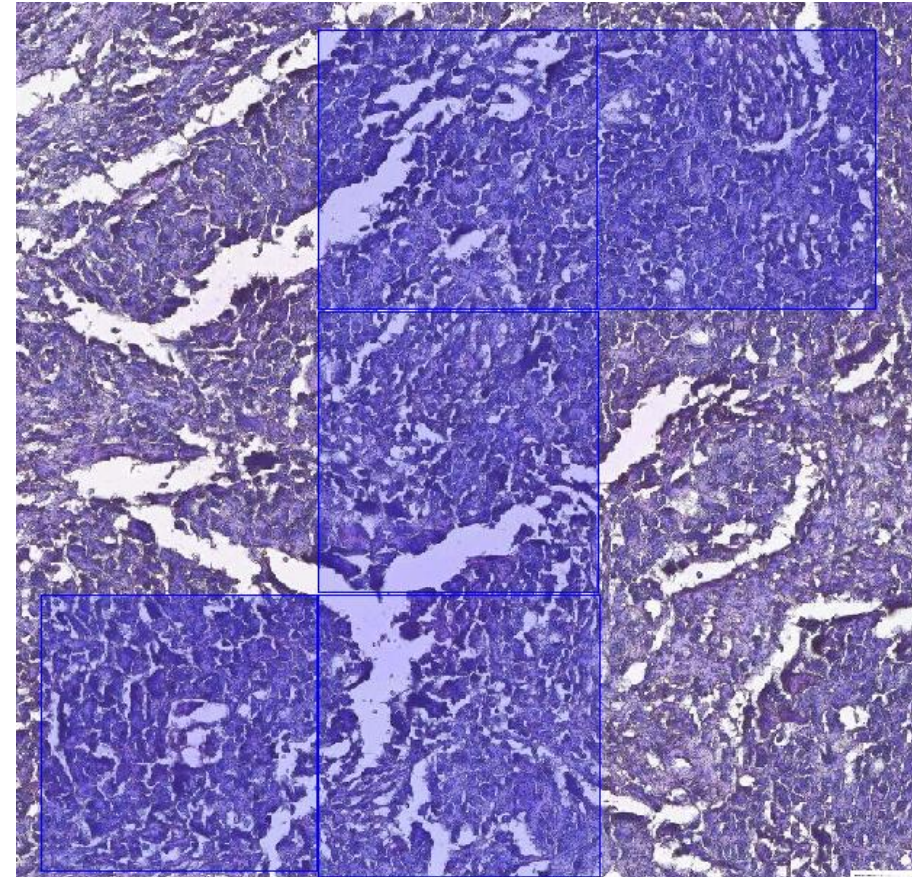


- Section **e2** => **High Score** & **Tumor**

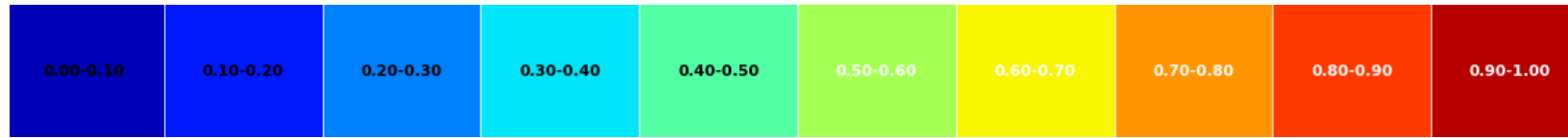
AI 스코어



Astromapper 매핑 결과

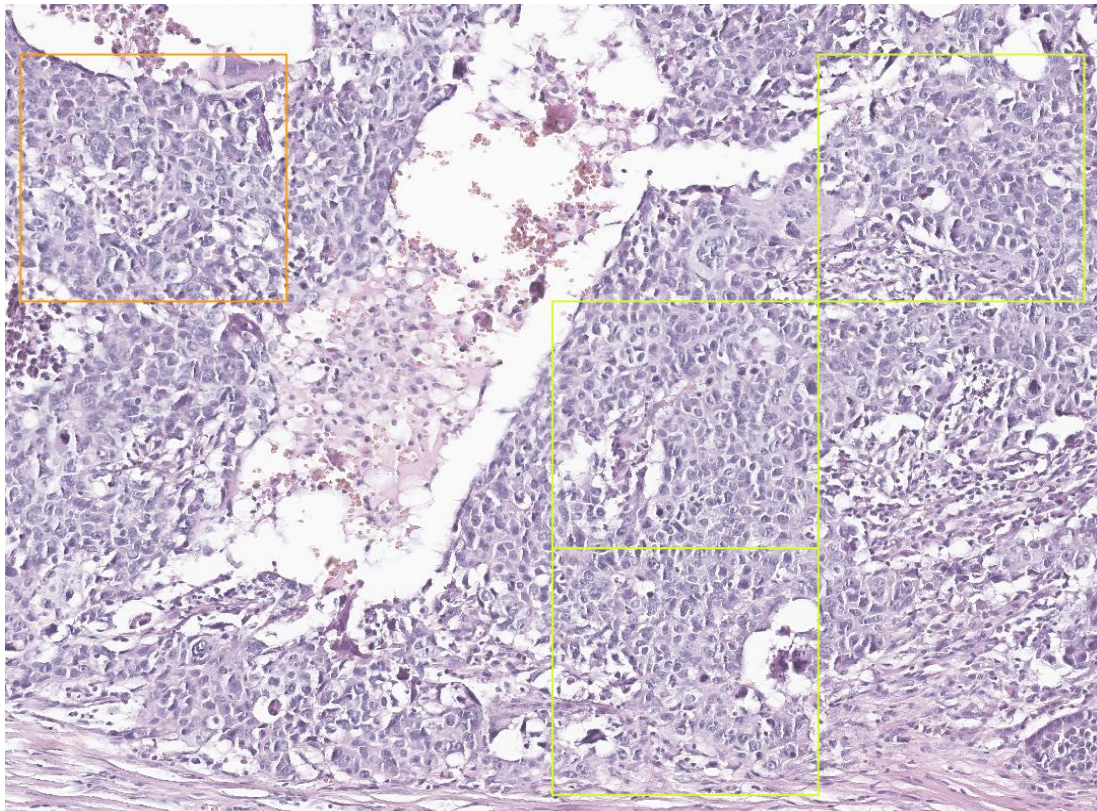


OmixAI 1-152-19 환자 샘플 high-score tumor regions

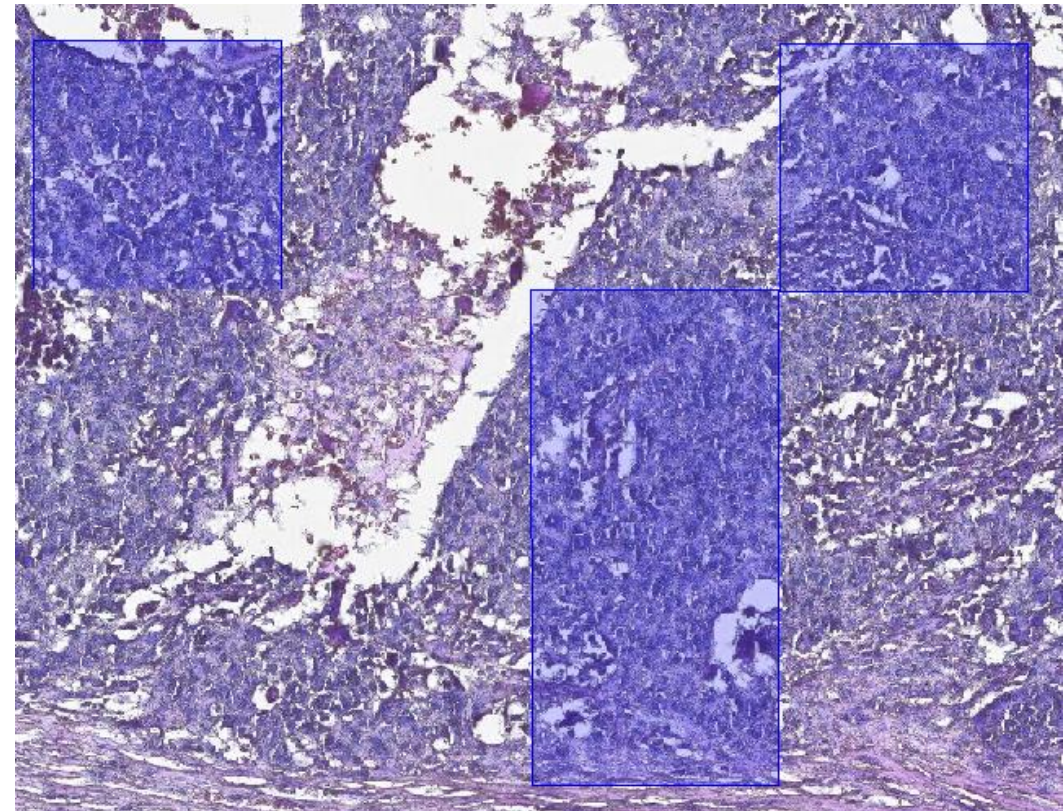


- Section **e3** => **High Score & Tumor**

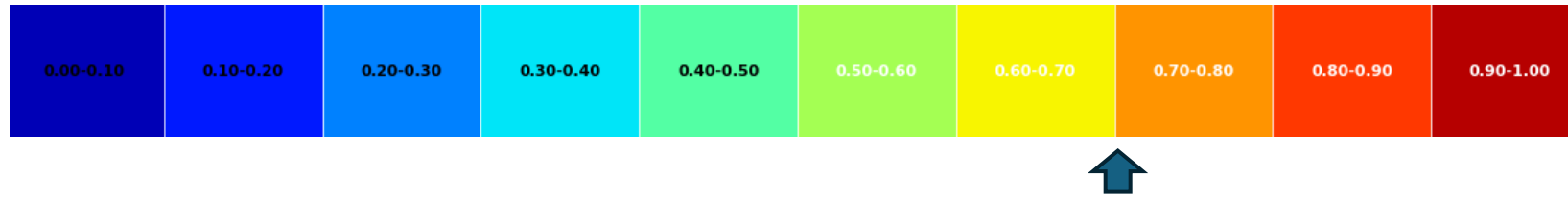
AI 스코어



Astromapper 매핑 결과

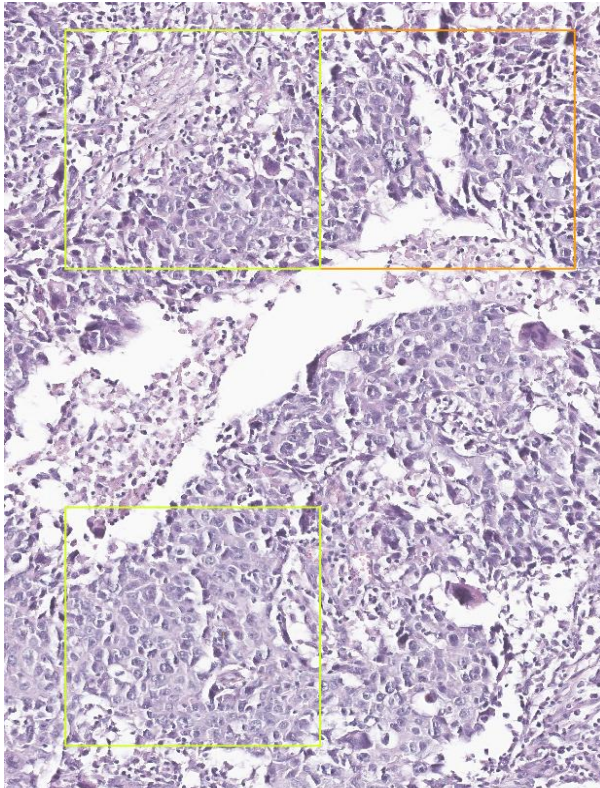


OmixAI 1-152-19 환자 샘플 high-score tumor regions

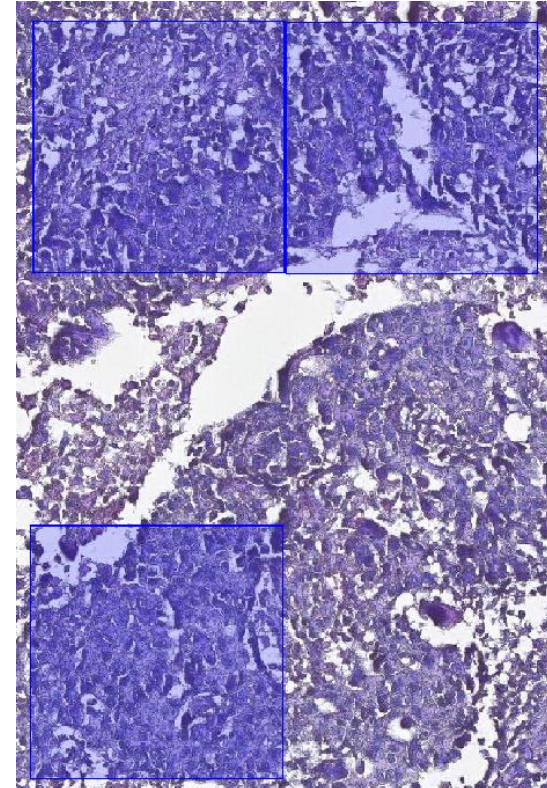


- Section **e4** => **High Score & Tumor**

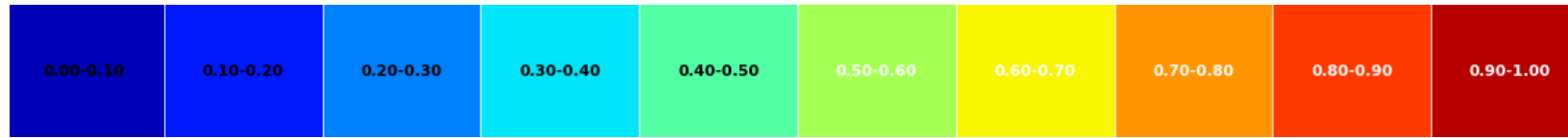
AI 스코어



Astromapper 매핑 결과

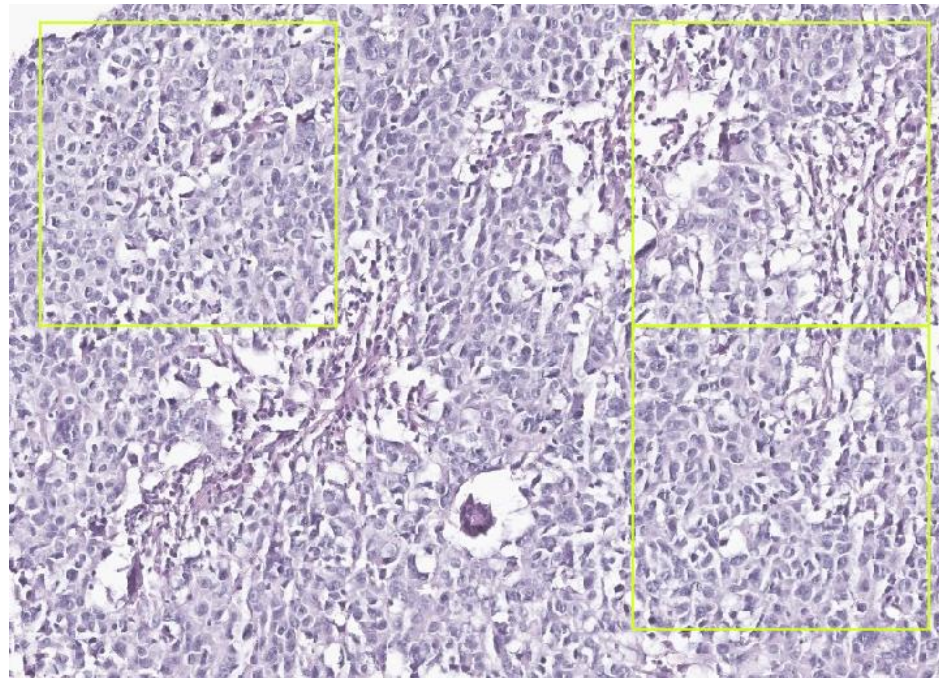


OmixAI 1-152-19 환자 샘플 high-score tumor regions

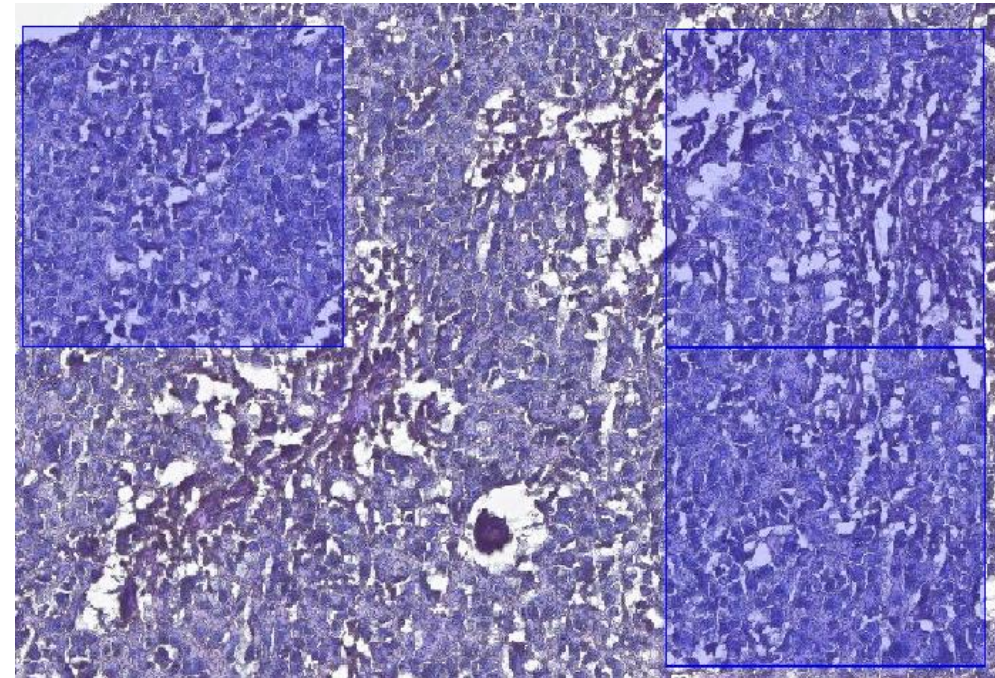


- Section **e5** => **High Score & Tumor**

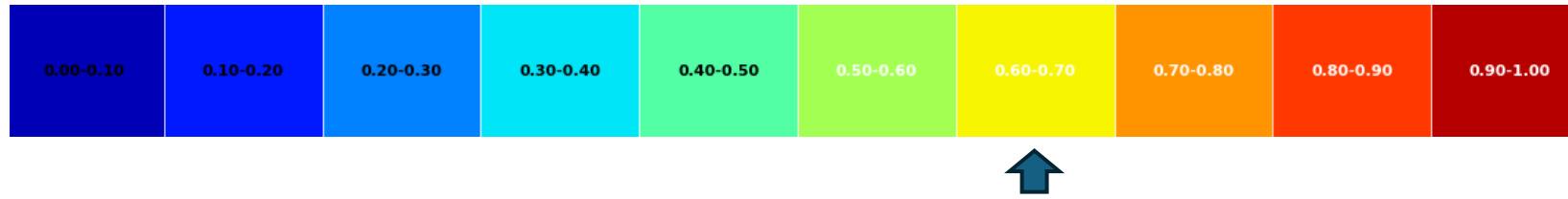
AI 스코어



Astromapper 매핑 결과

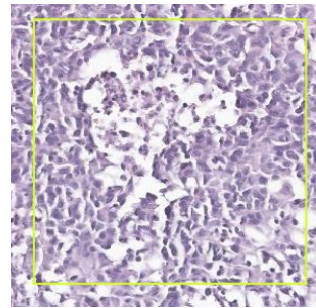
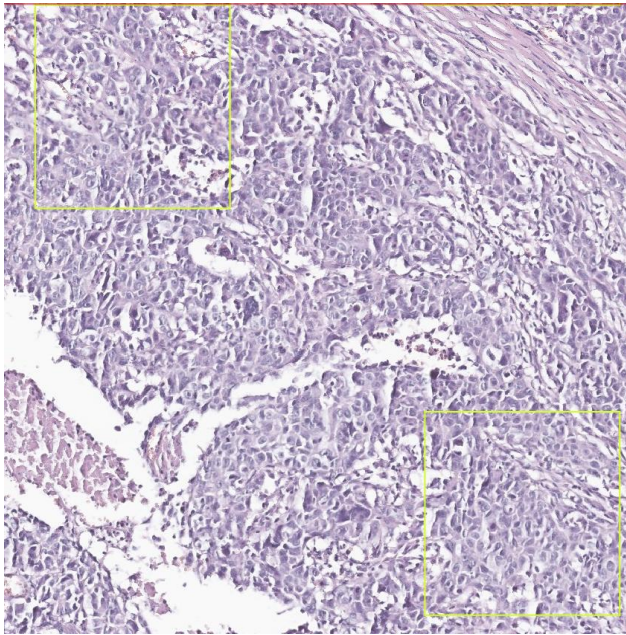


OmixAI 1-152-19 환자 샘플 high-score tumor regions

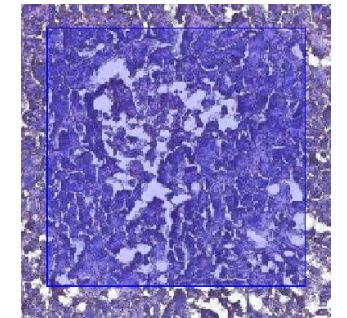
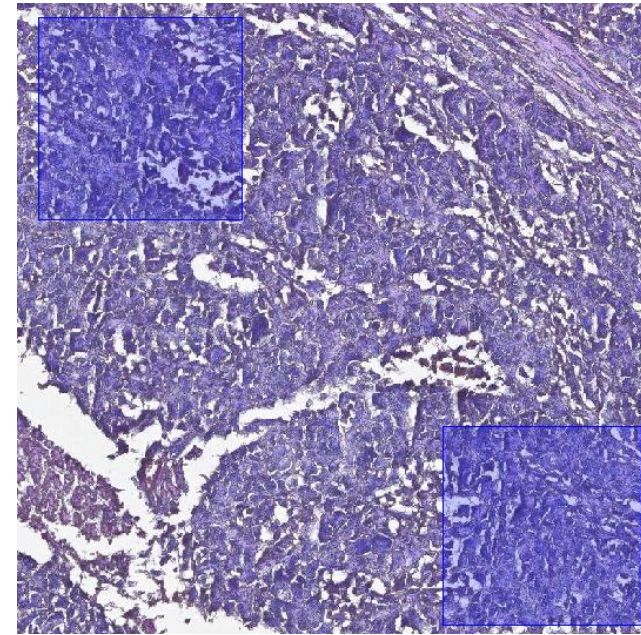


- Section **e6** => **High Score & Tumor**

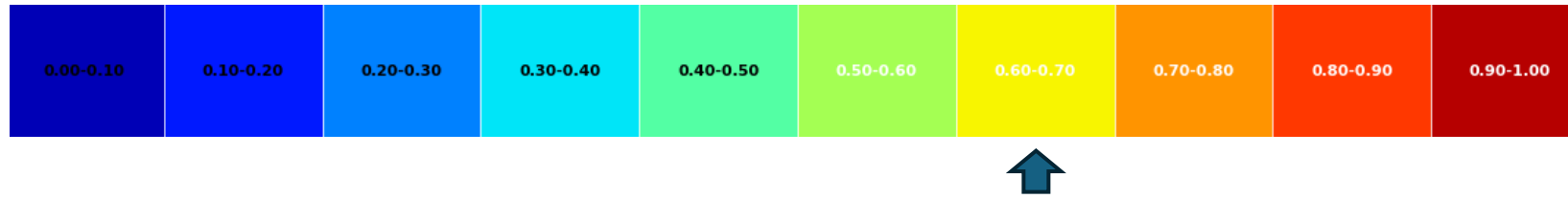
AI 스코어



Astromapper 매핑 결과

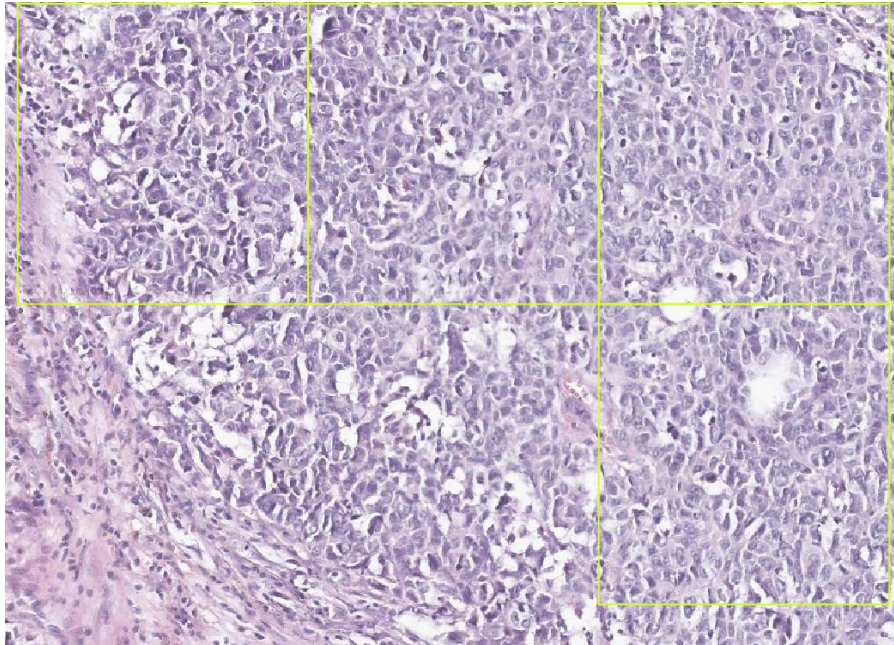


OmixAI 1-152-19 환자 샘플 high-score tumor regions

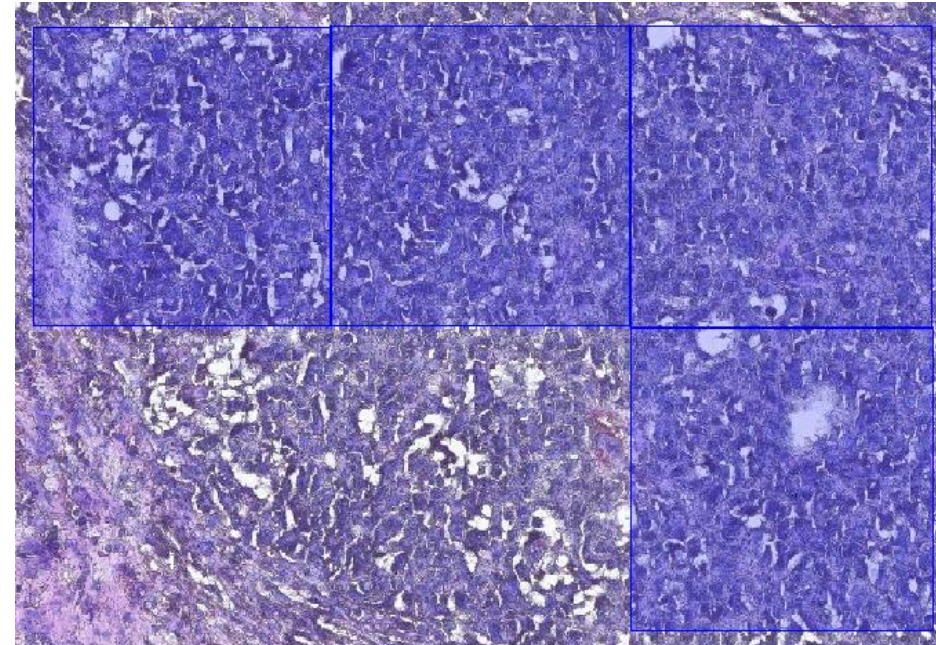


- Section **e7** => High Score & Tumor

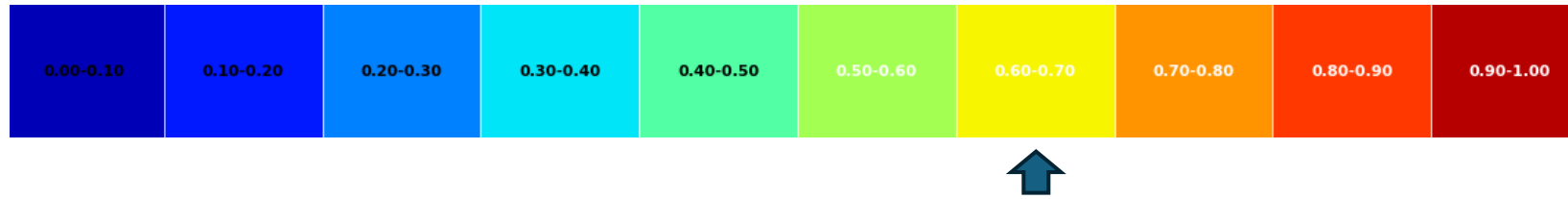
AI 스코어



Astromapper 매핑 결과

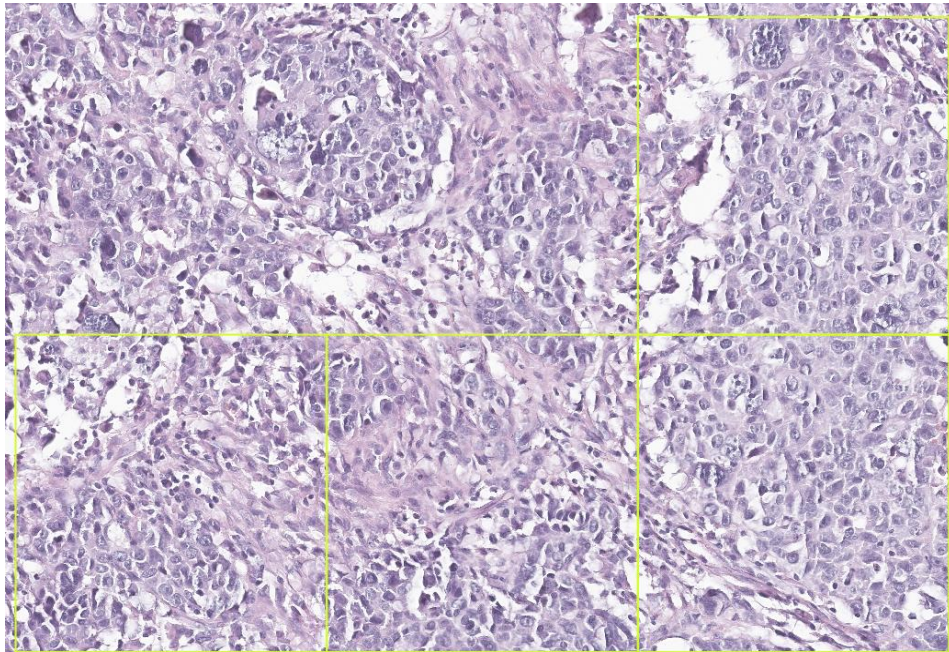


OmixAI 1-152-19 환자 샘플 high-score tumor regions

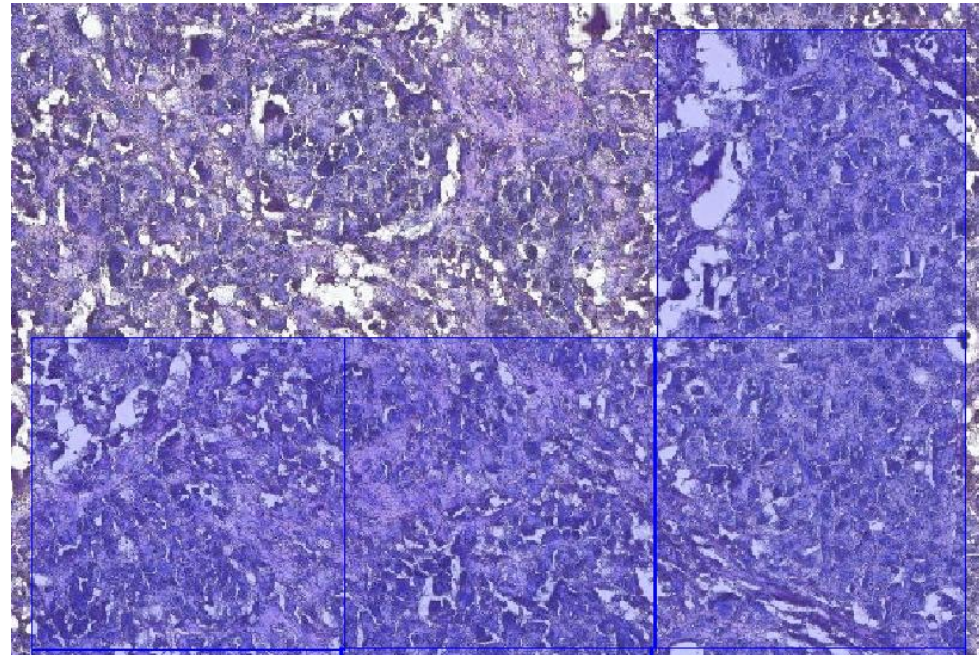


- Section **e8** => **High Score & Tumor**

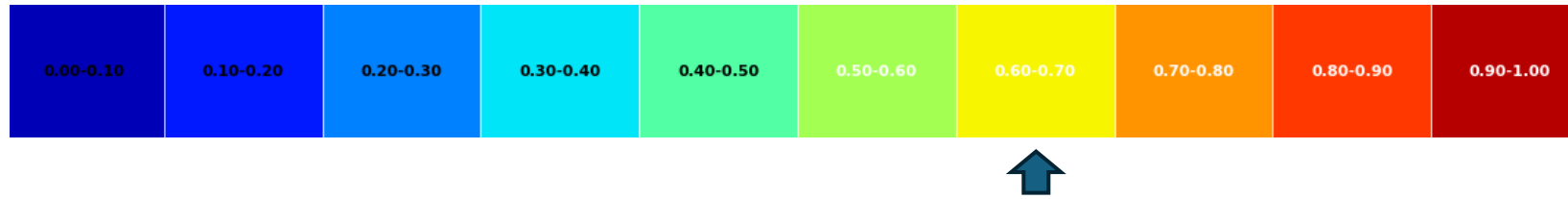
AI 스코어



Astromapper 매핑 결과

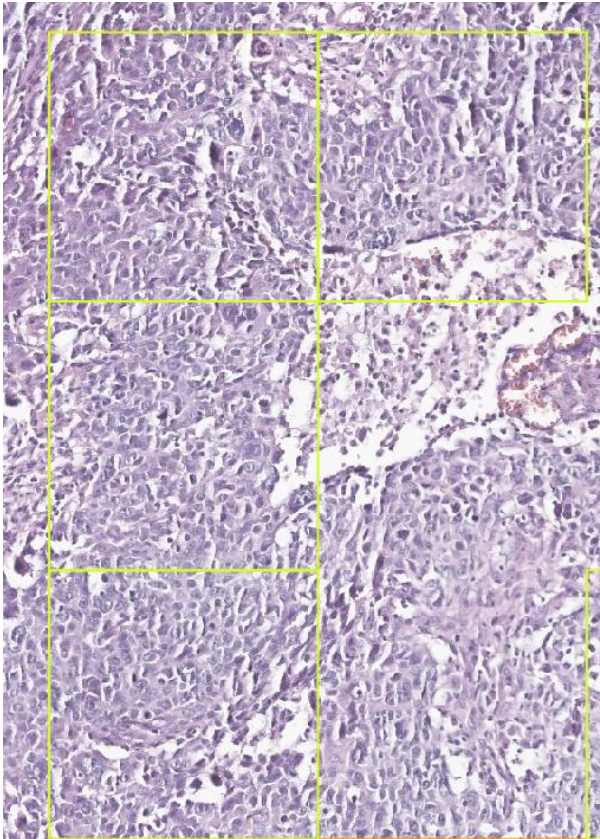


OmixAI 1-152-19 환자 샘플 high-score tumor regions

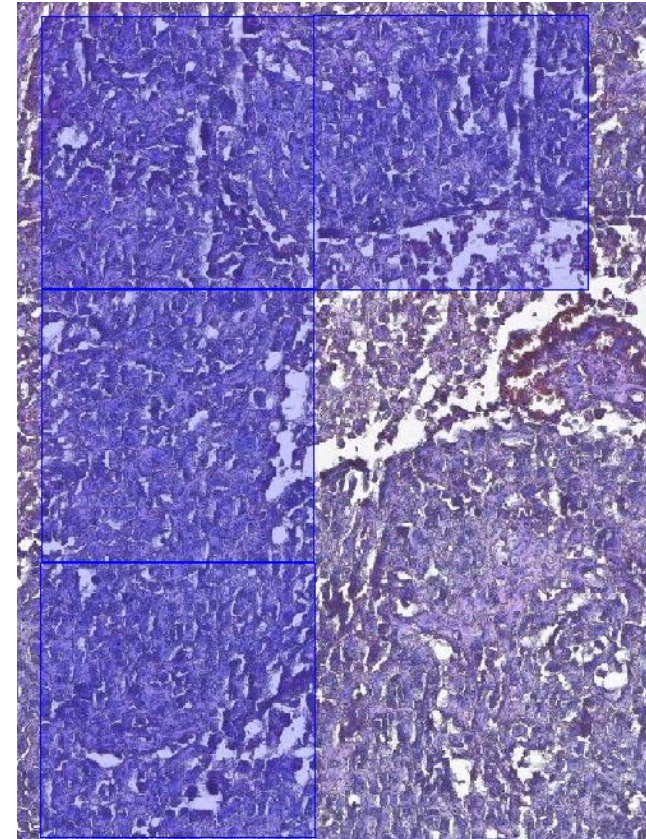


- Section **e9** => **High Score & Tumor**

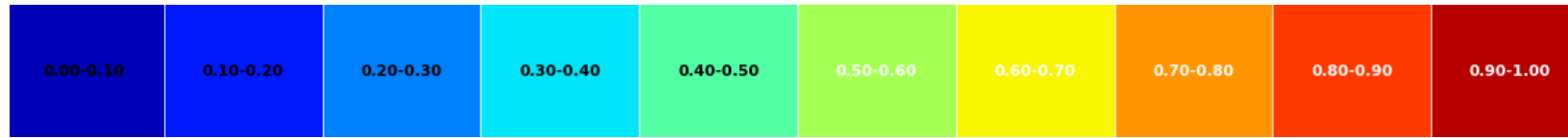
AI 스코어



Astromapper 매핑 결과

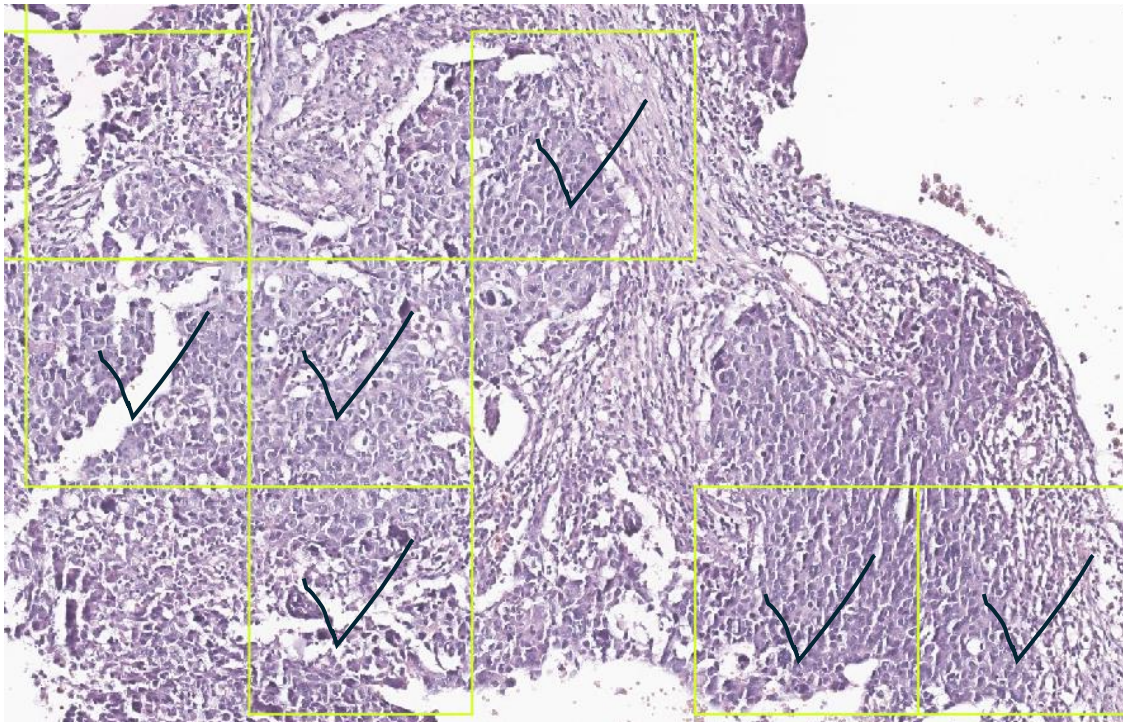


OmixAI 1-152-19 환자 샘플 high-score tumor regions

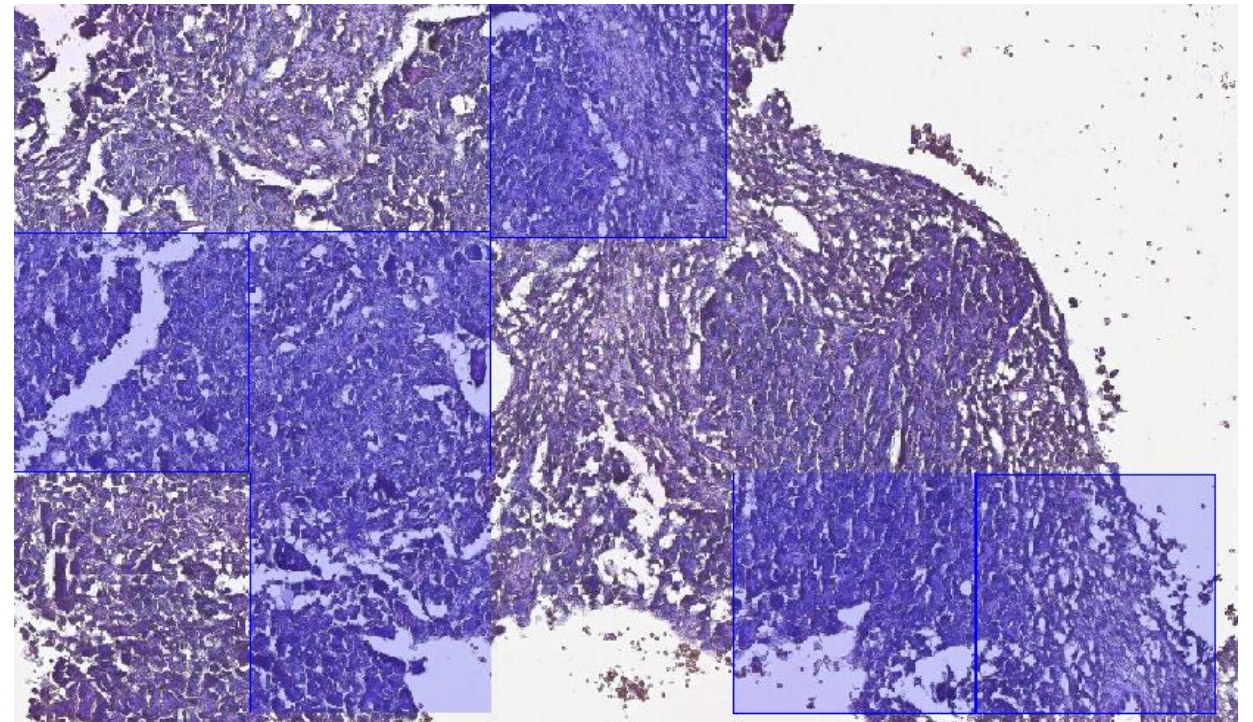


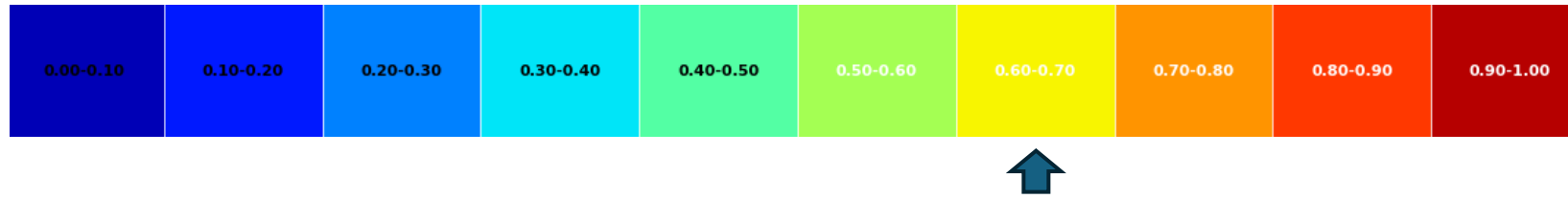
- Section **e10** => **High Score** & **Tumor**

AI 스코어



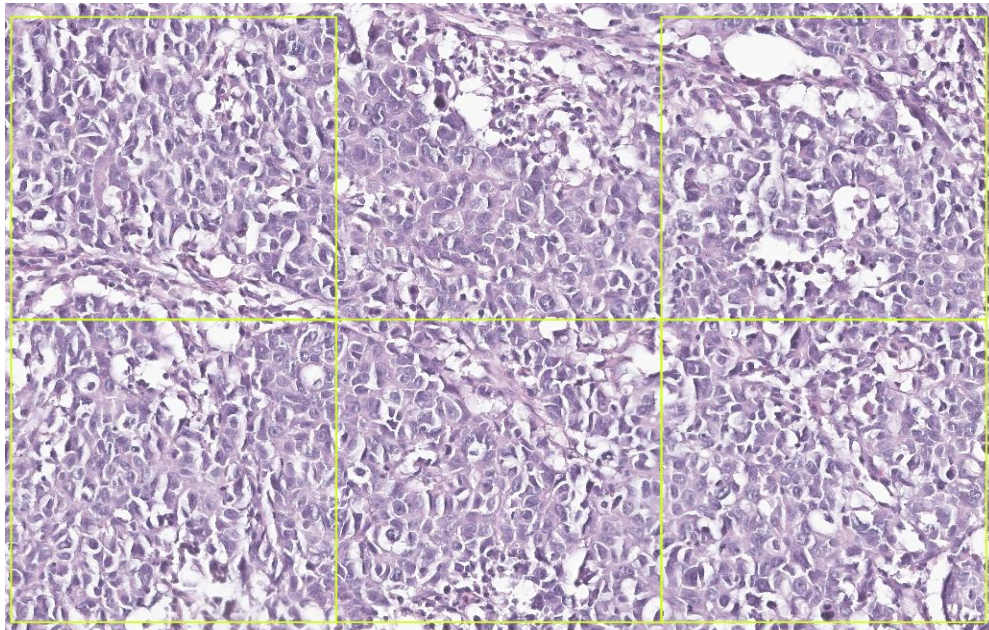
Astromapper 매핑 결과



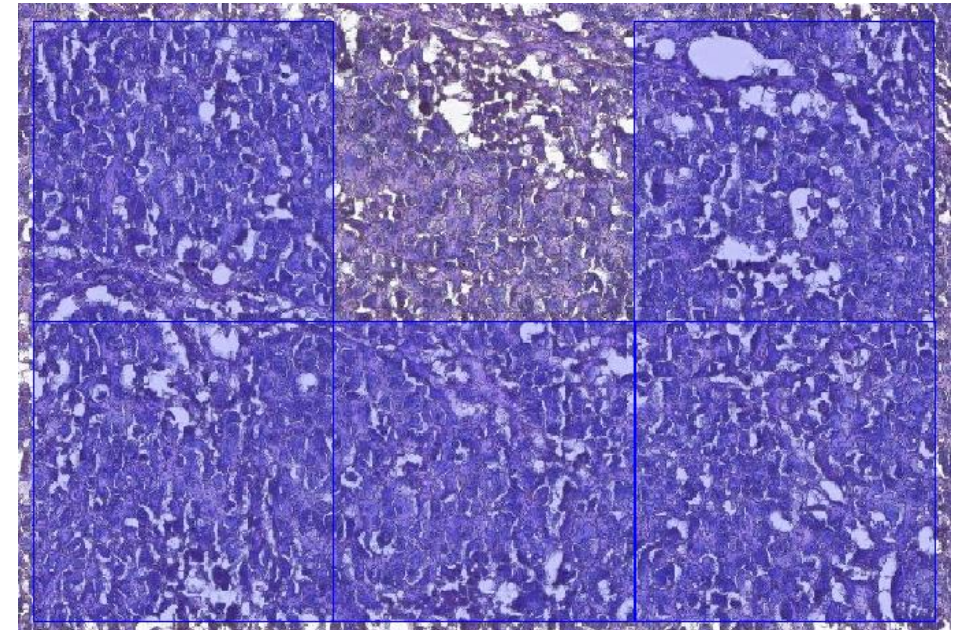


- Section **e11** => High Score & Tumor

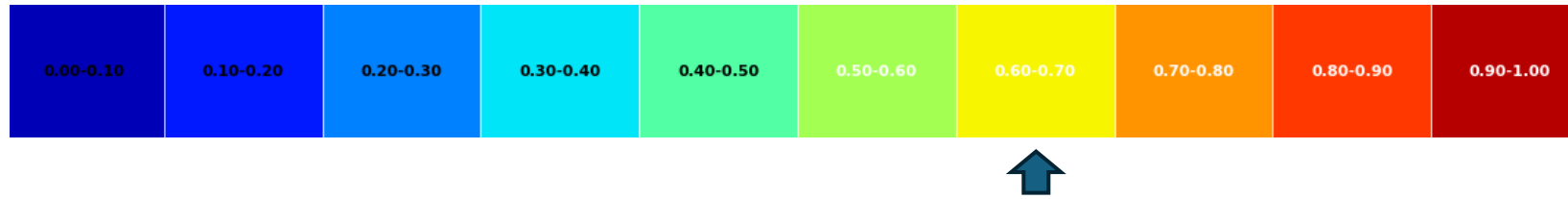
AI 스코어



Astromapper 매핑 결과

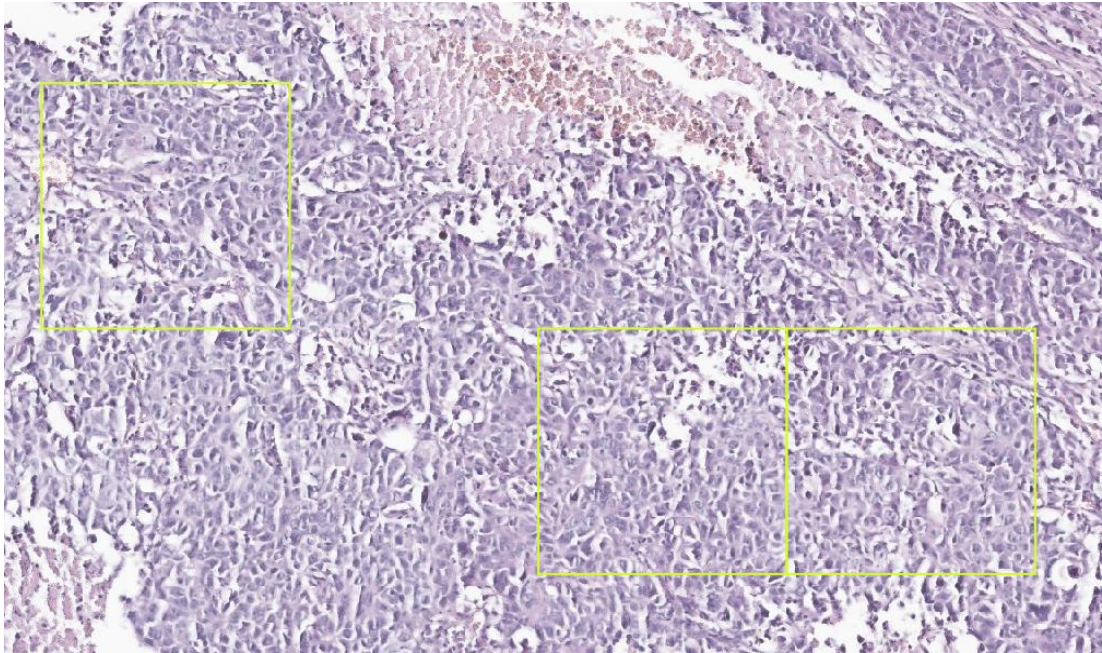


OmixAI 1-152-19 환자 샘플 high-score tumor regions

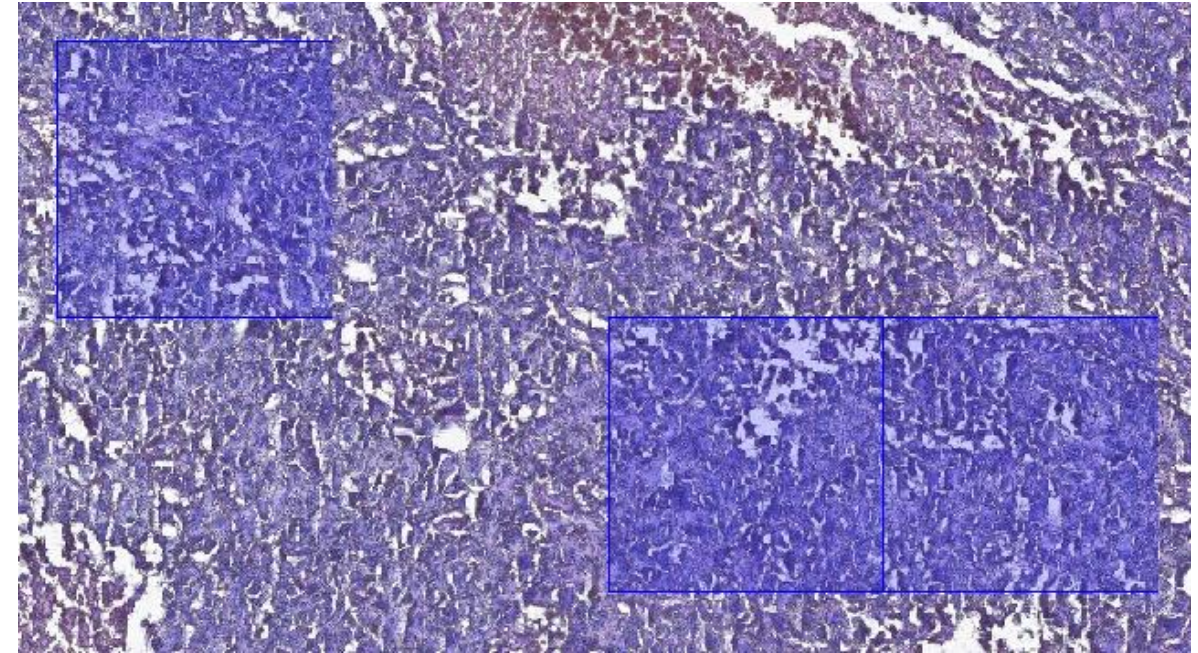


- Section **e12** => **High Score** & **Tumor**

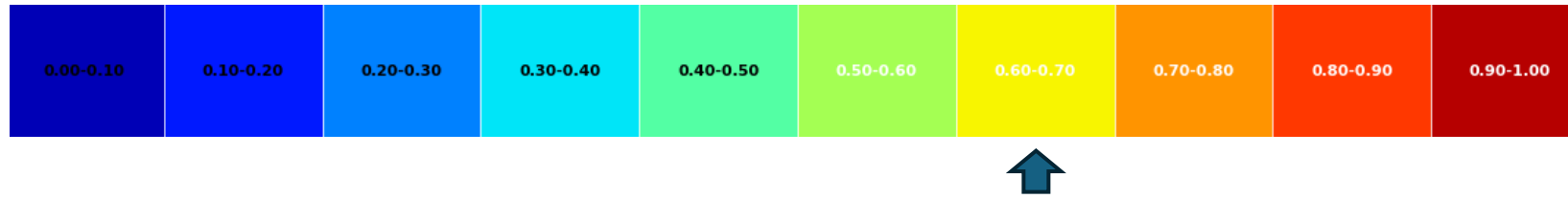
AI 스코어



Astromapper 매핑 결과

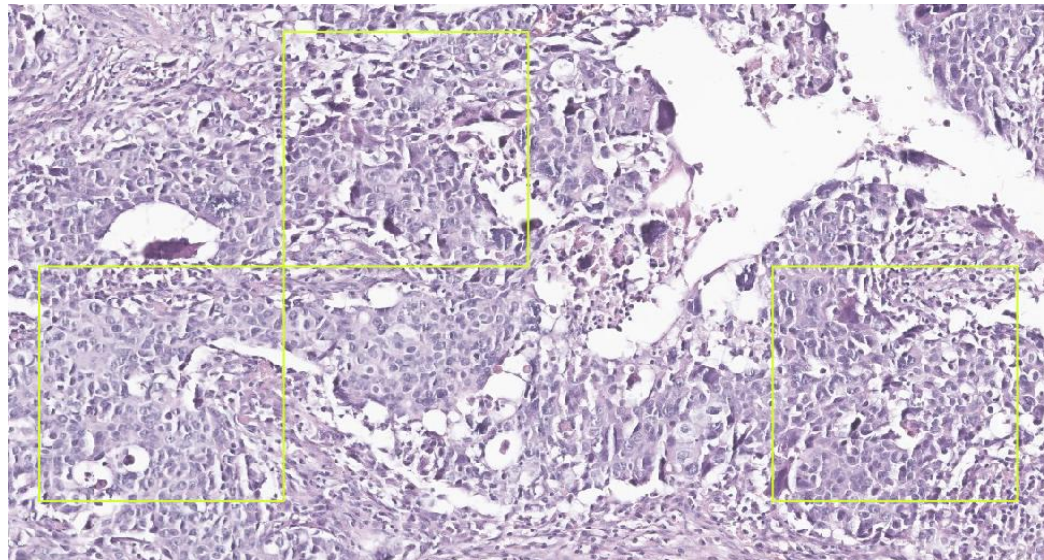


OmixAI 1-152-19 환자 샘플 high-score tumor regions

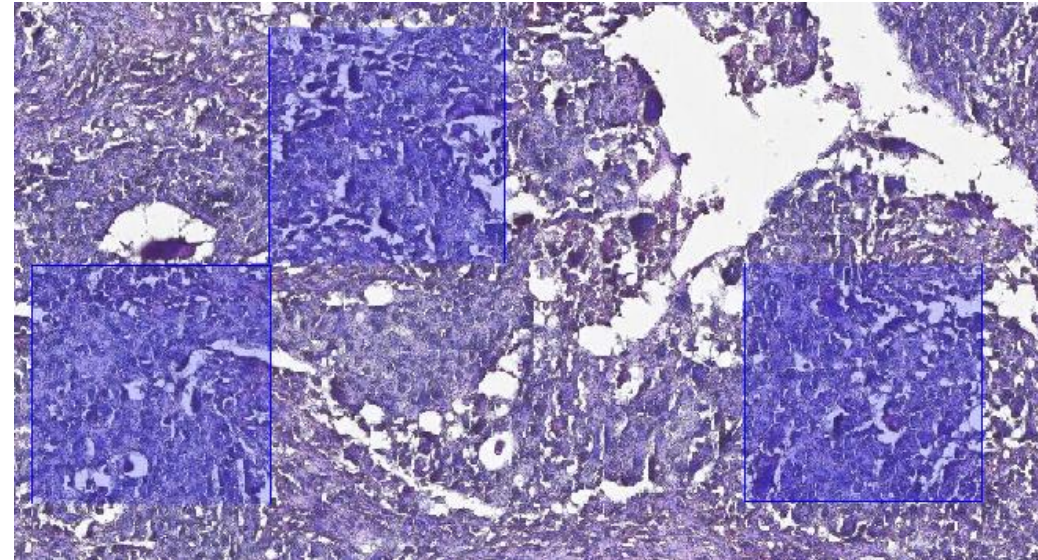


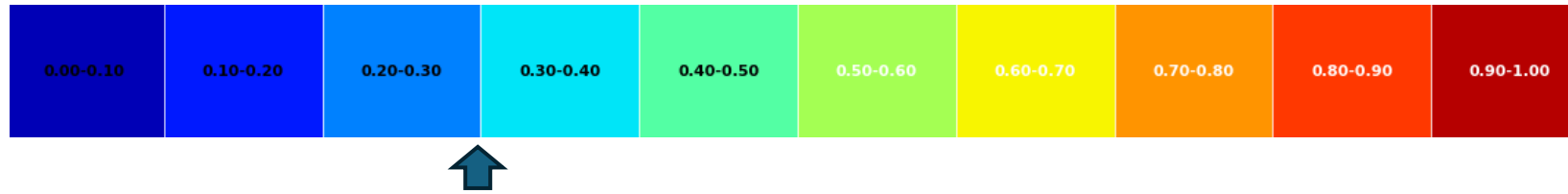
- Section **e13** => **High Score** & **Tumor**

AI 스코어



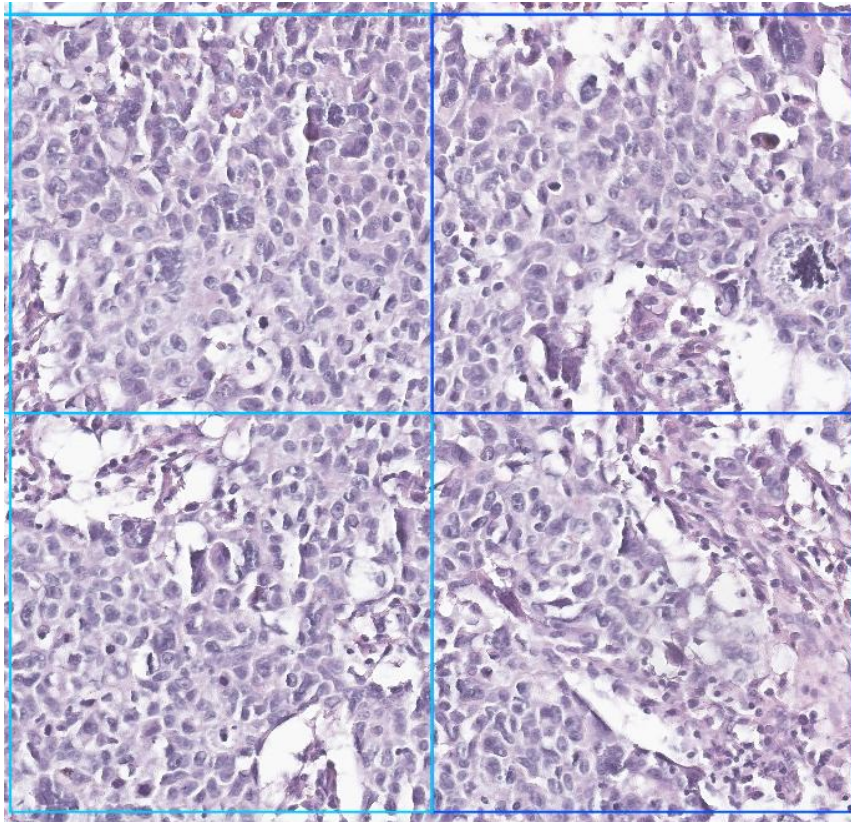
Astromapper 매핑 결과



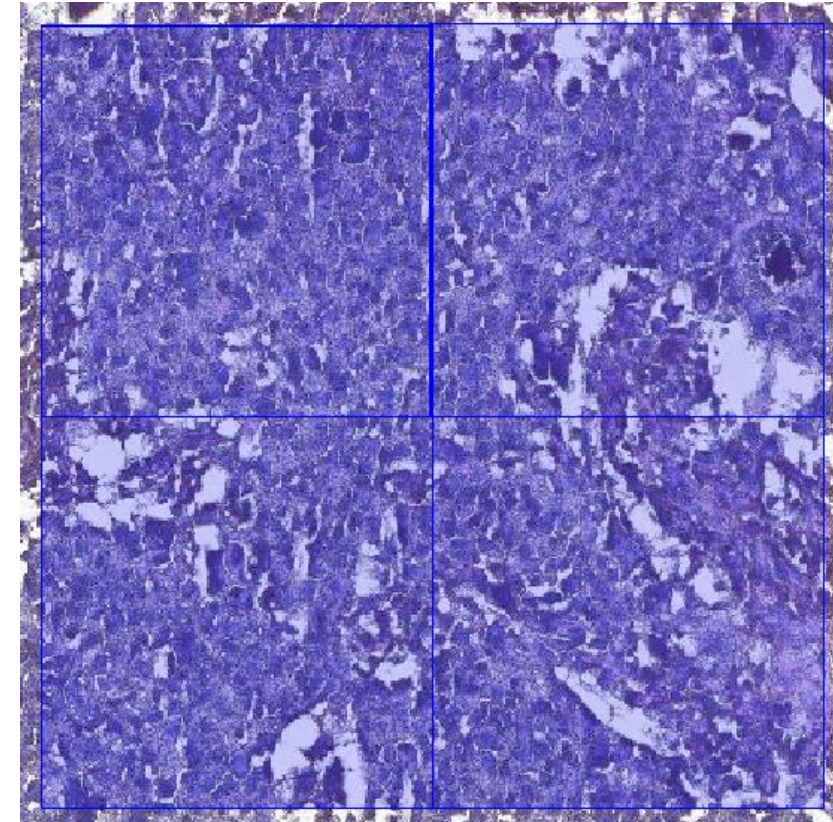


- Section **f1** => Low Score & Tumor

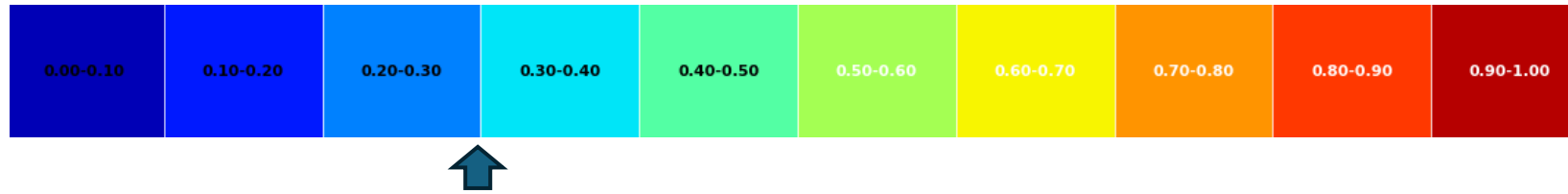
AI 스코어



Astromapper 매핑 결과

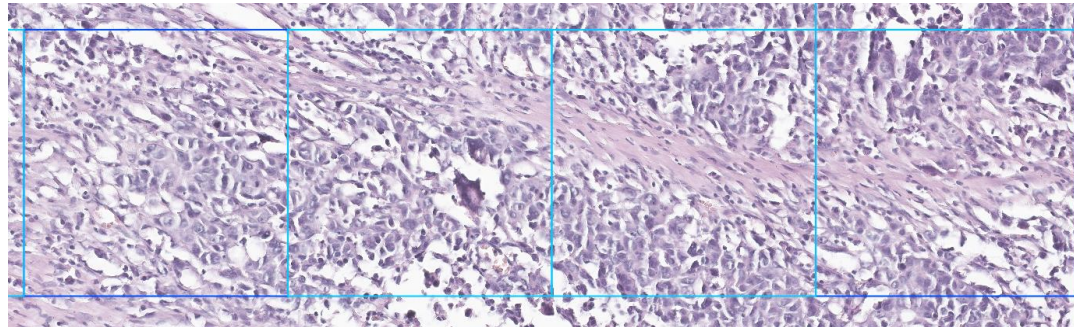


OmixAI 1-152-19 환자 샘플 low-score tumor regions

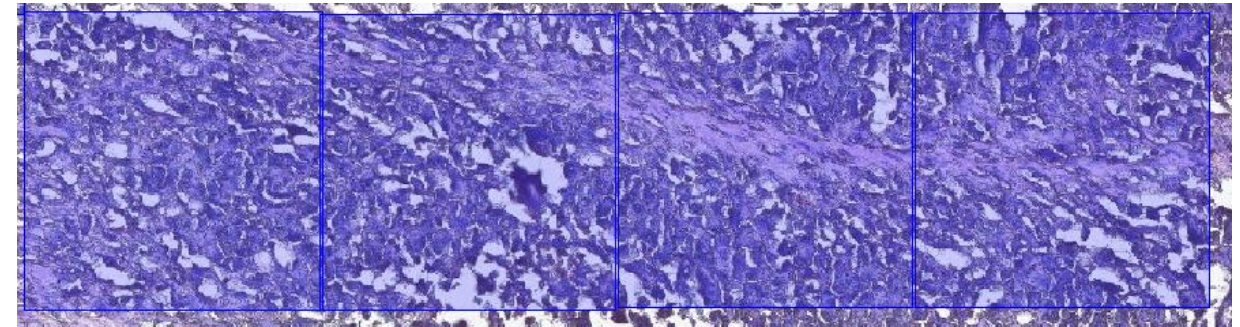


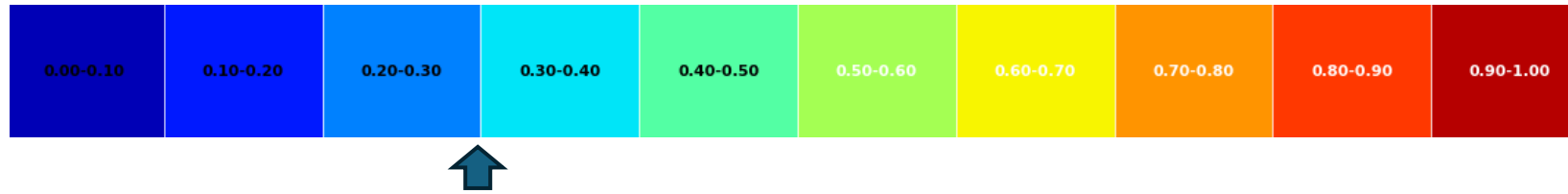
- Section **f2** => Low Score & **Tumor**

AI 스코어



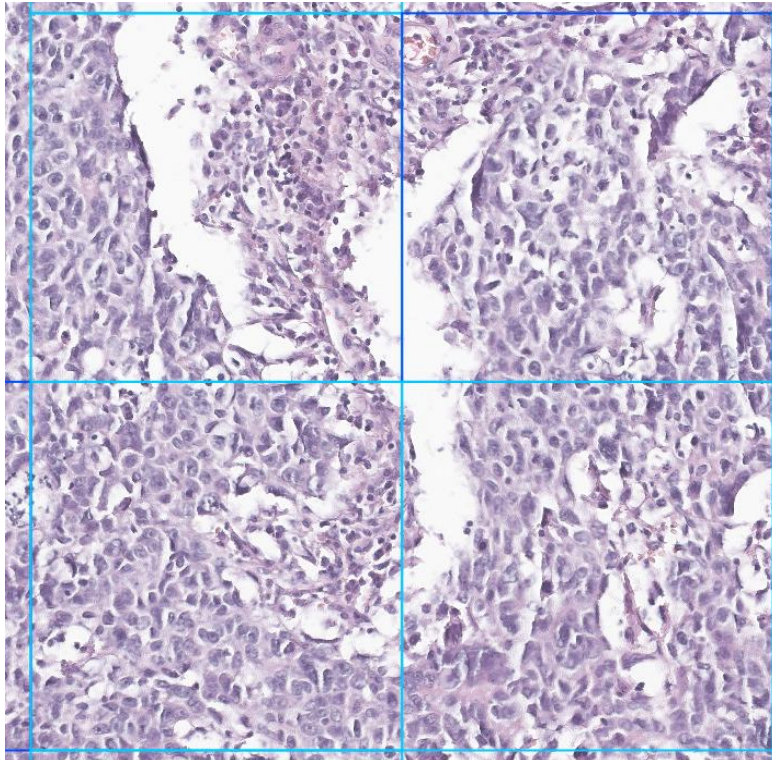
Astromapper 매핑 결과



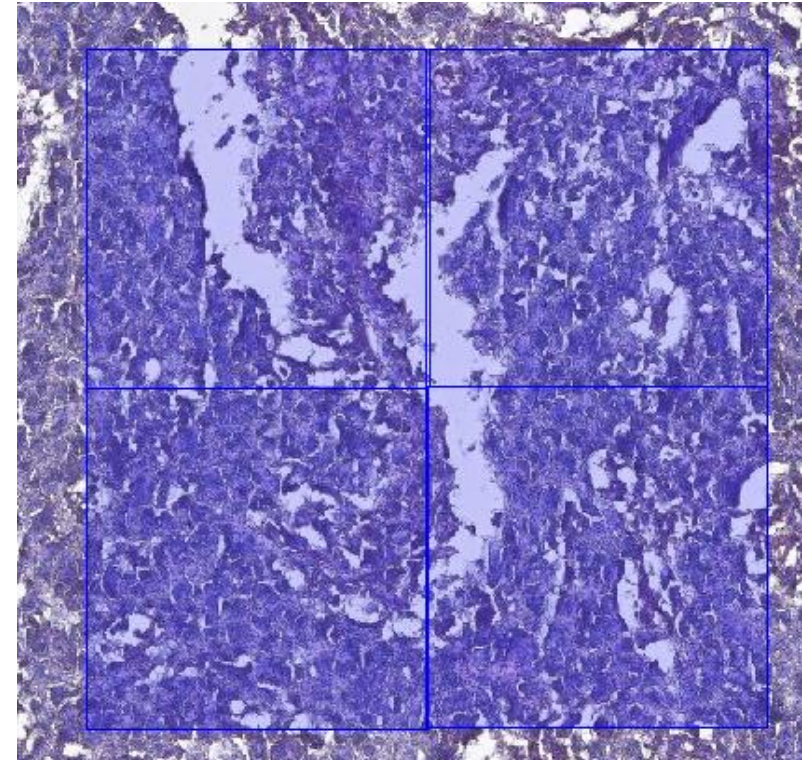


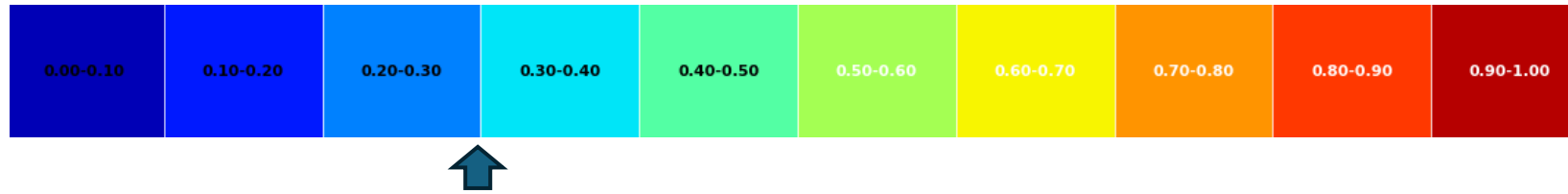
- Section **f3** => Low Score & Tumor

AI 스코어



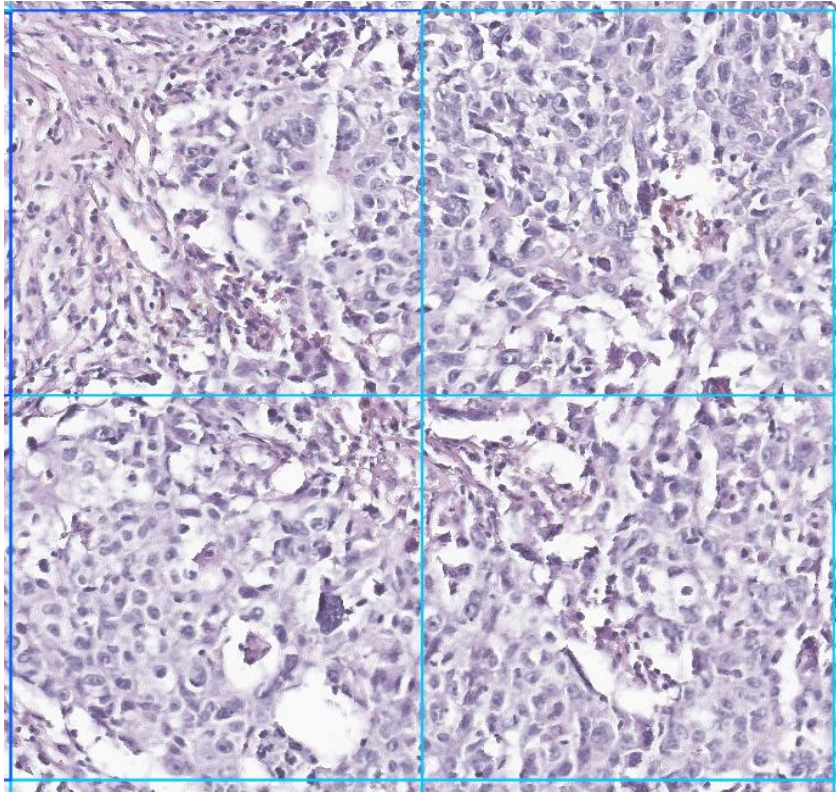
Astromapper 매핑 결과



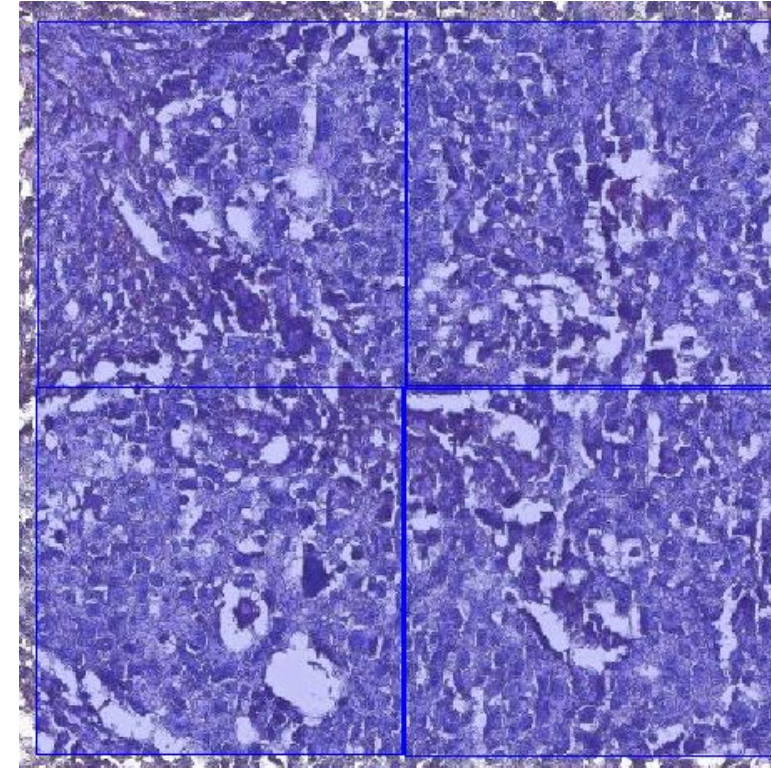


- Section **f4** => Low Score & Tumor

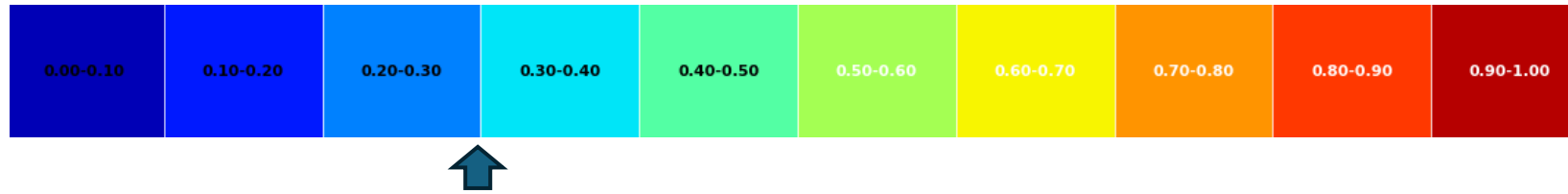
AI 스코어



Astromapper 매핑 결과

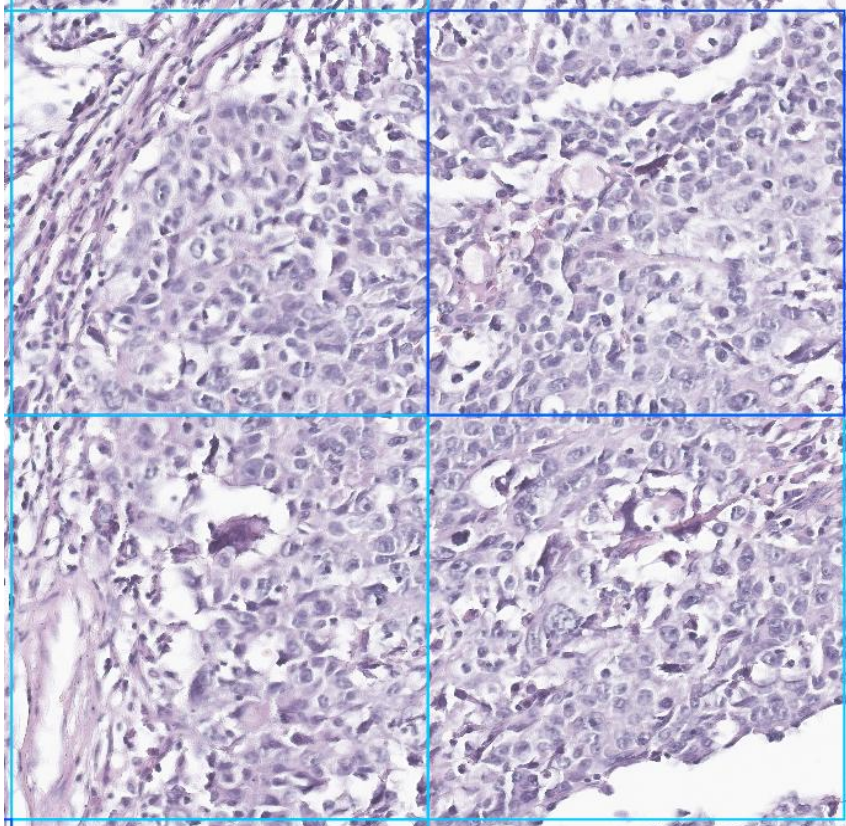


OmixAI 1-152-19 환자 샘플 low-score tumor regions

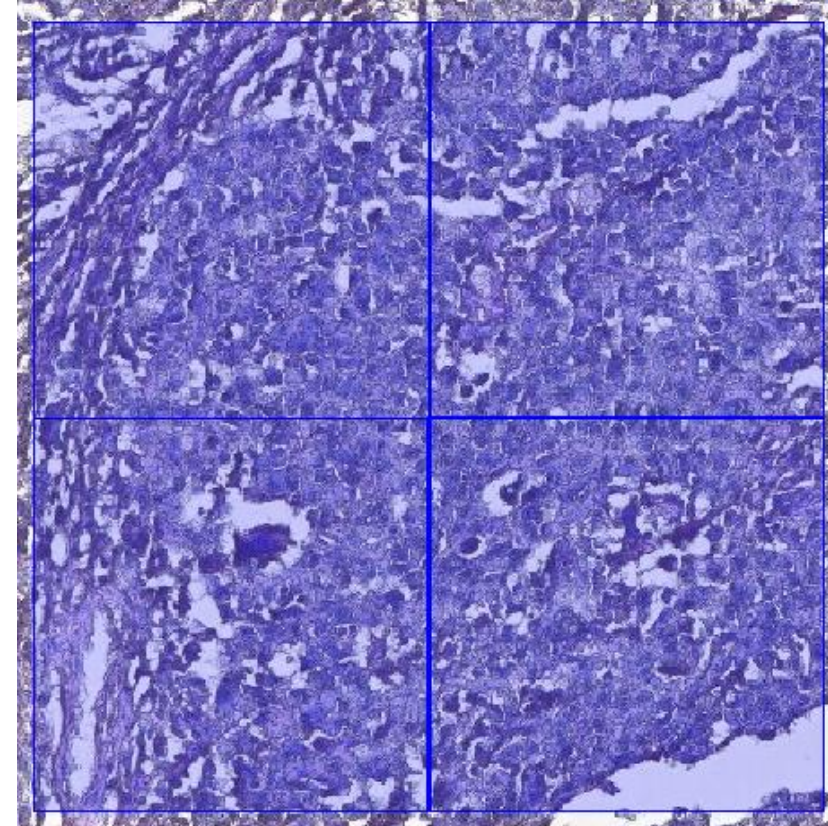


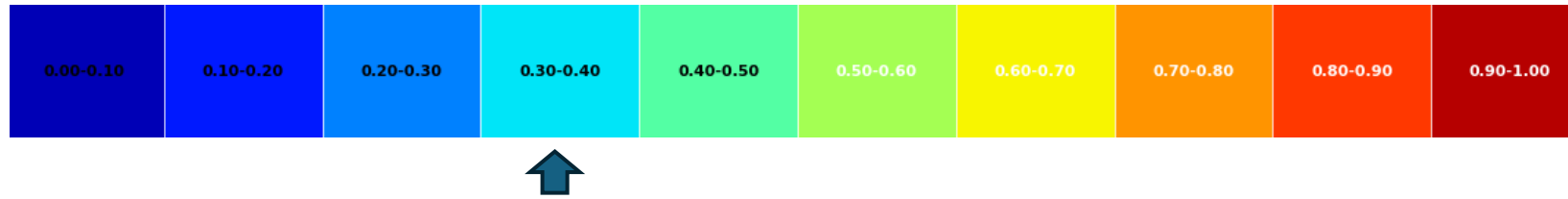
- Section **f5** => Low Score & **Tumor**

AI 스코어



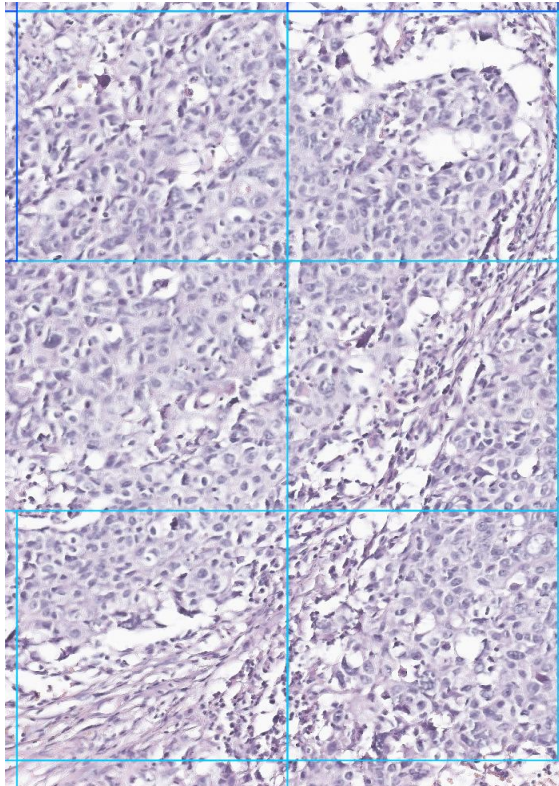
Astromapper 매핑 결과



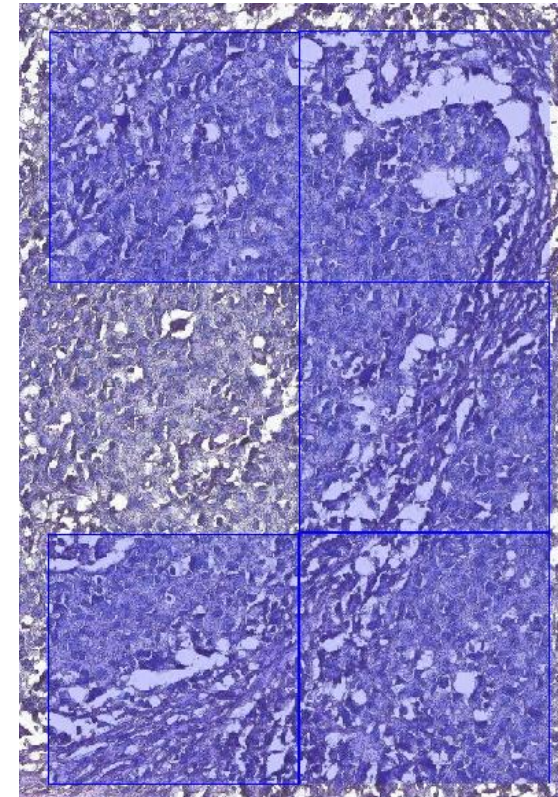


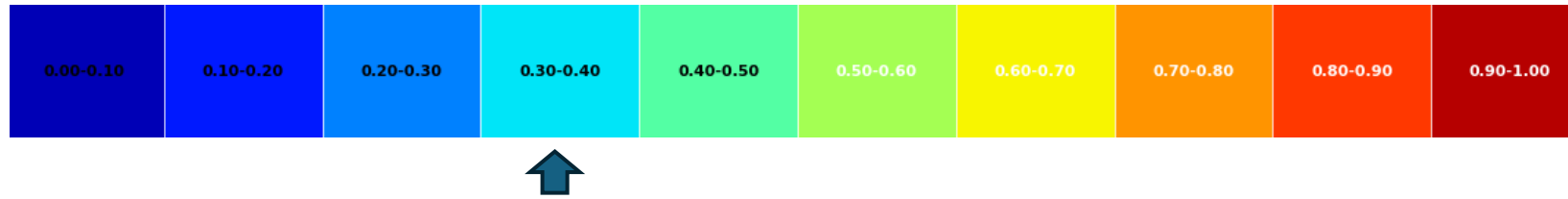
- Section **f6** => Low Score & Tumor

AI 스코어



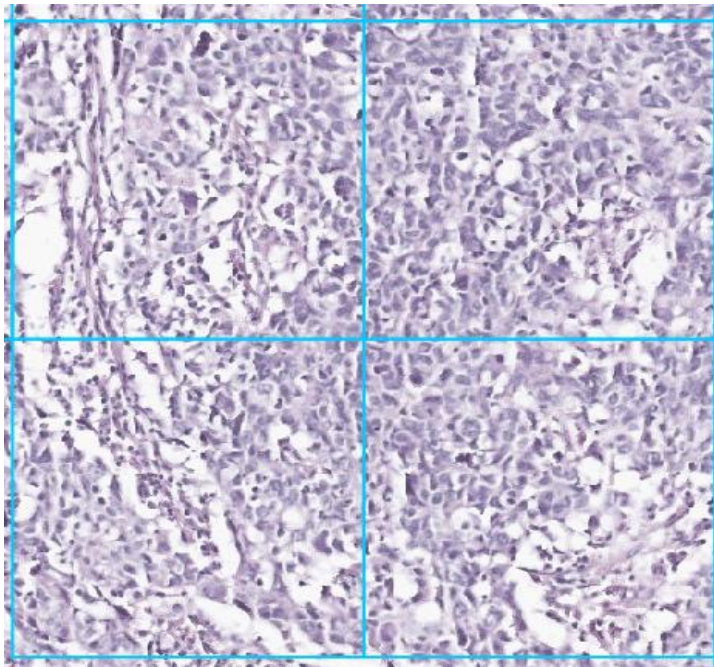
Astromapper 매핑 결과



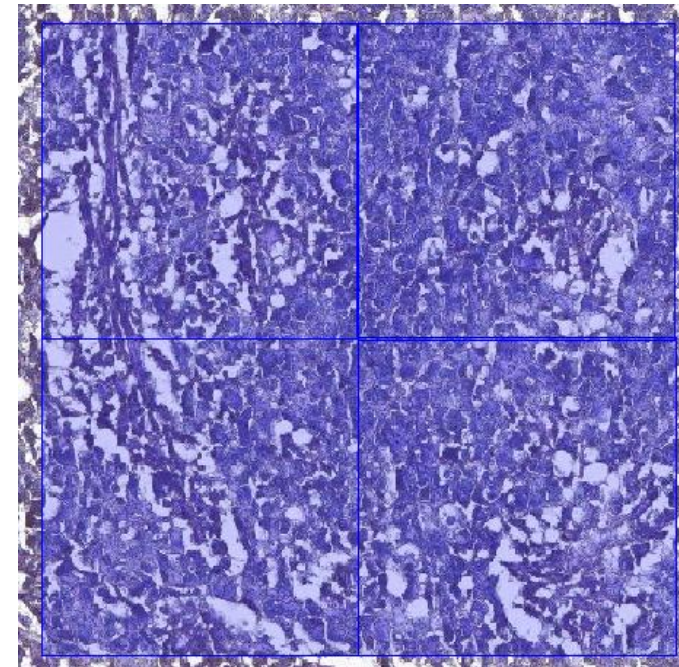


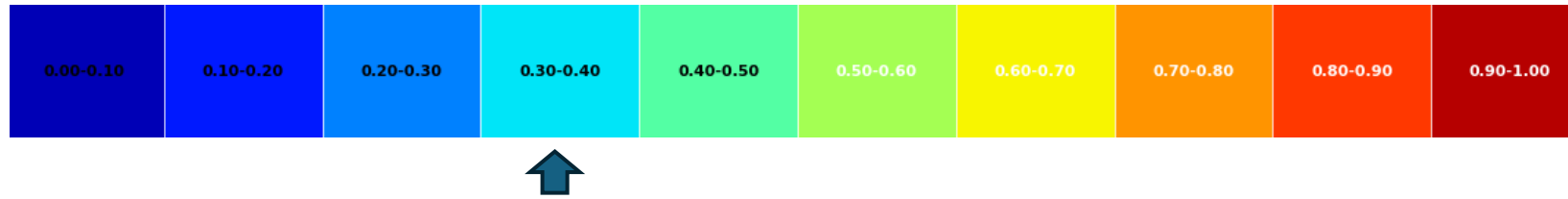
- Section **f7** => Low Score & Tumor

AI 스코어



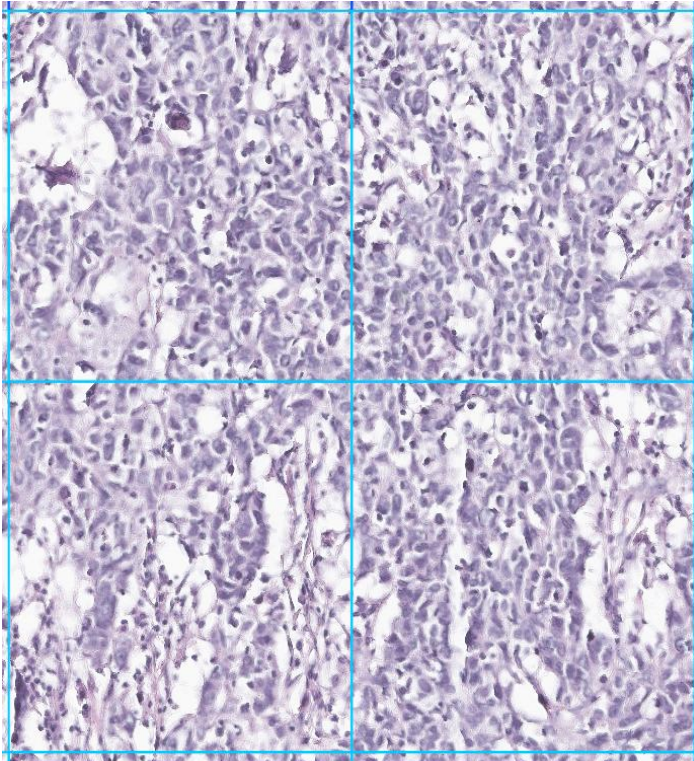
Astromapper 매핑 결과



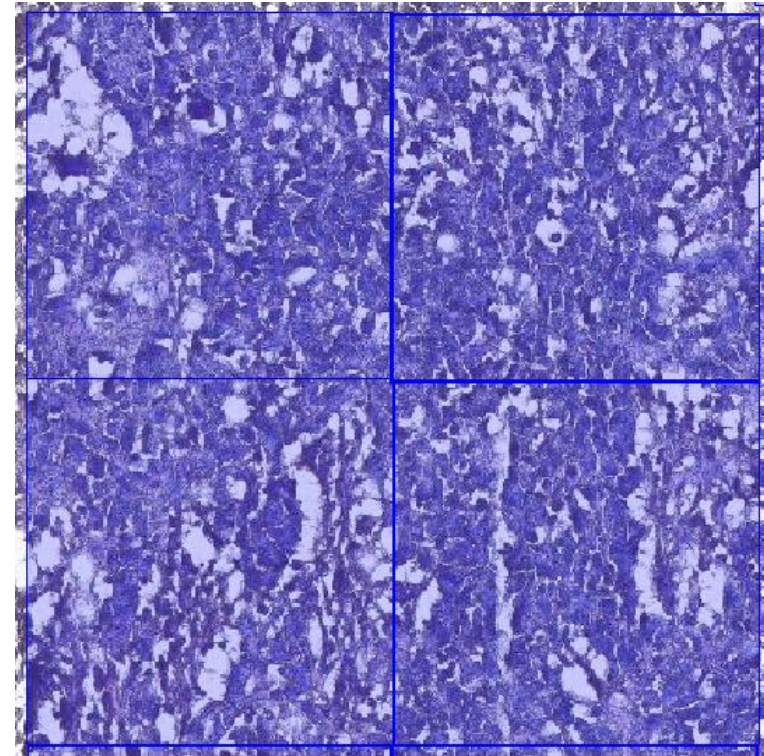


- Section **f8** => Low Score & Tumor

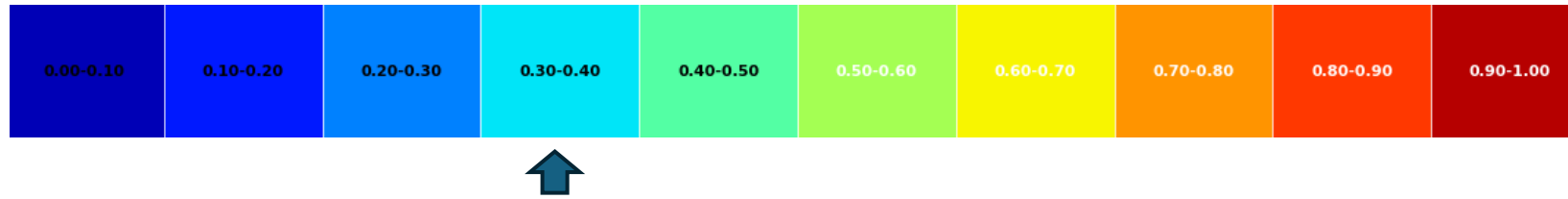
AI 스코어



Astromapper 매핑 결과

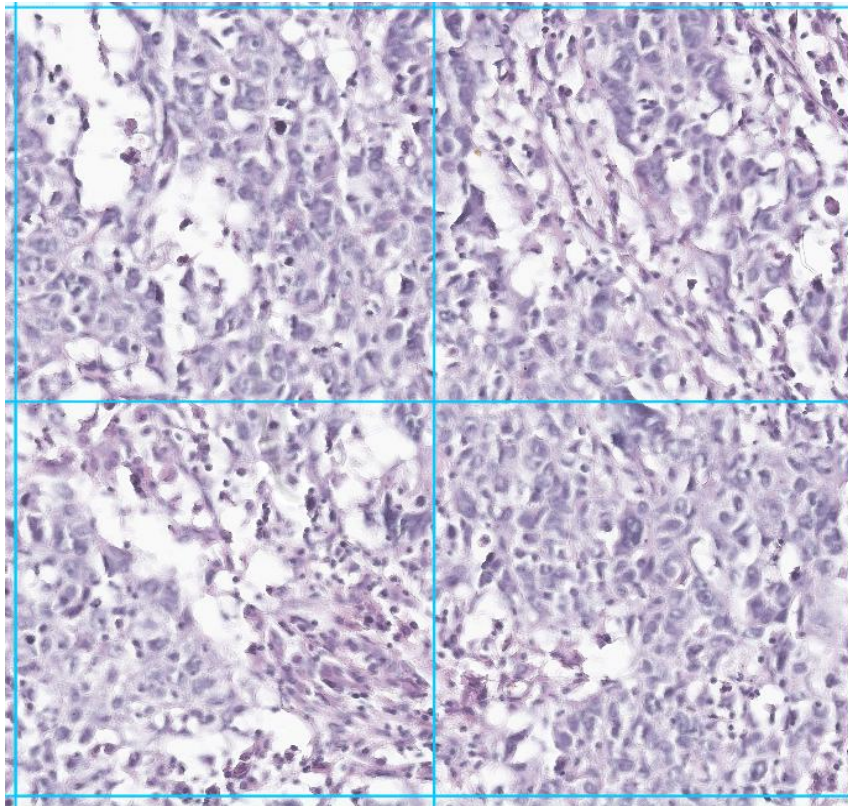


OmixAI 1-152-19 환자 샘플 low-score tumor regions

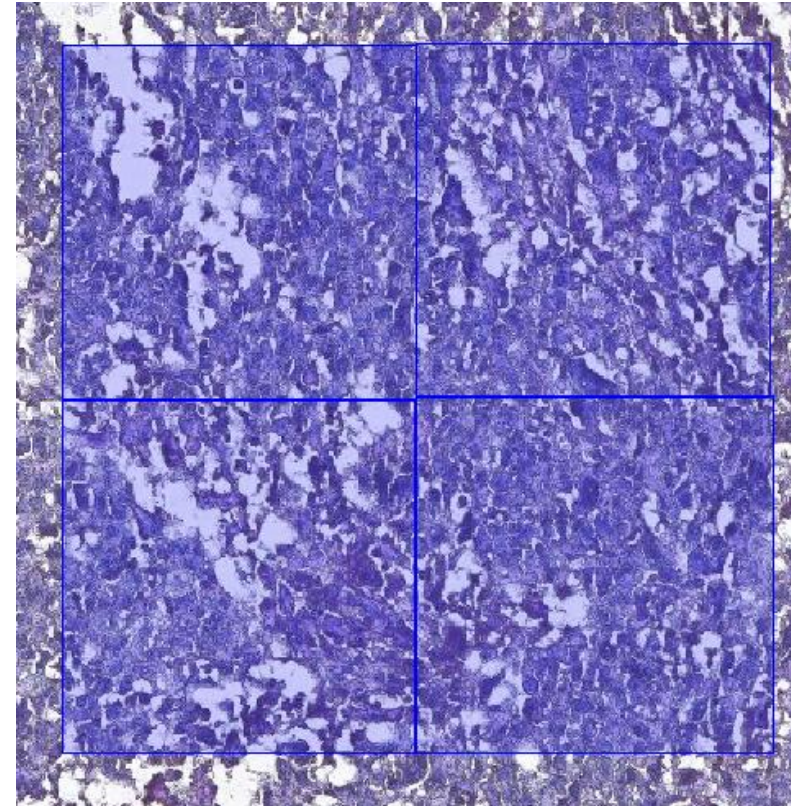


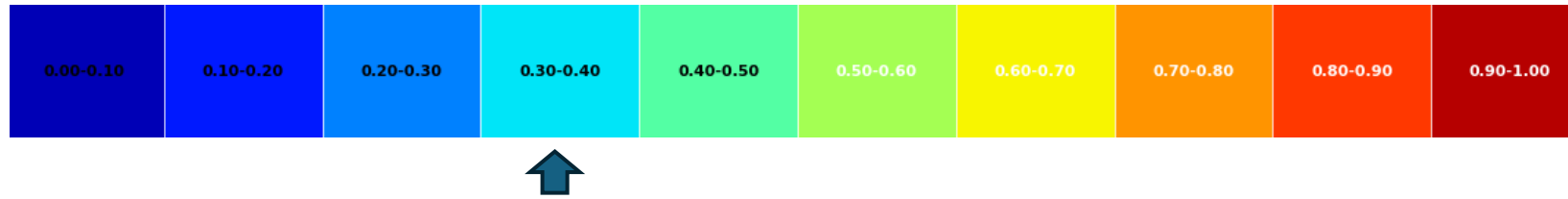
- Section **f9** => Low Score & Tumor

AI 스코어



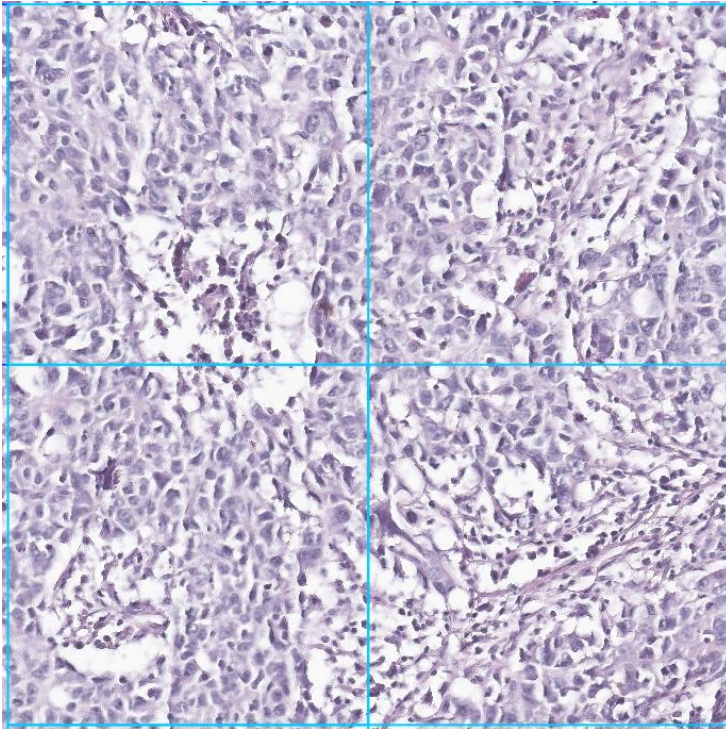
Astromapper 매핑 결과



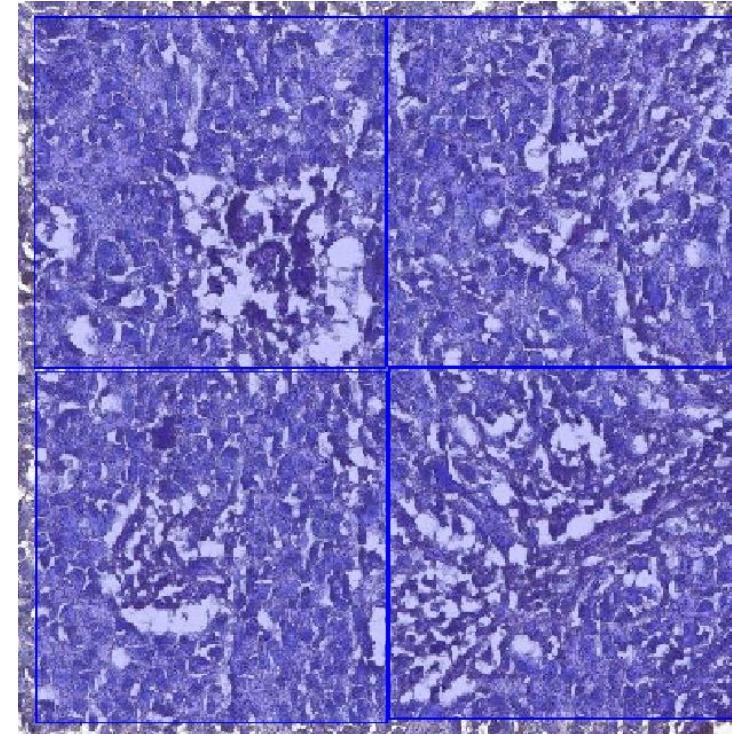


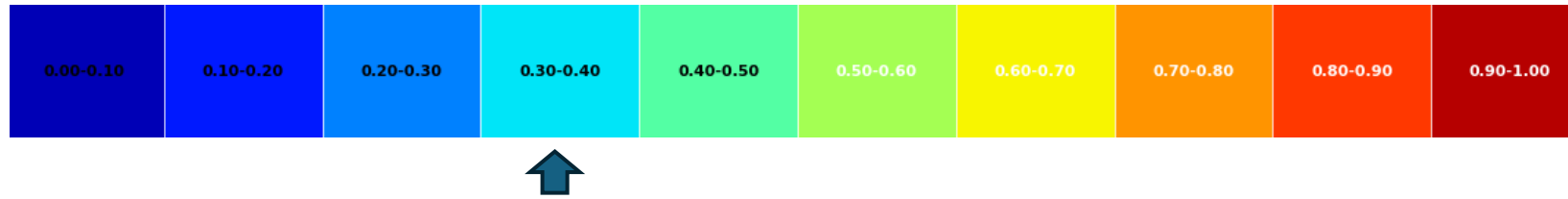
- Section **f10** => Low Score & Tumor

AI 스코어



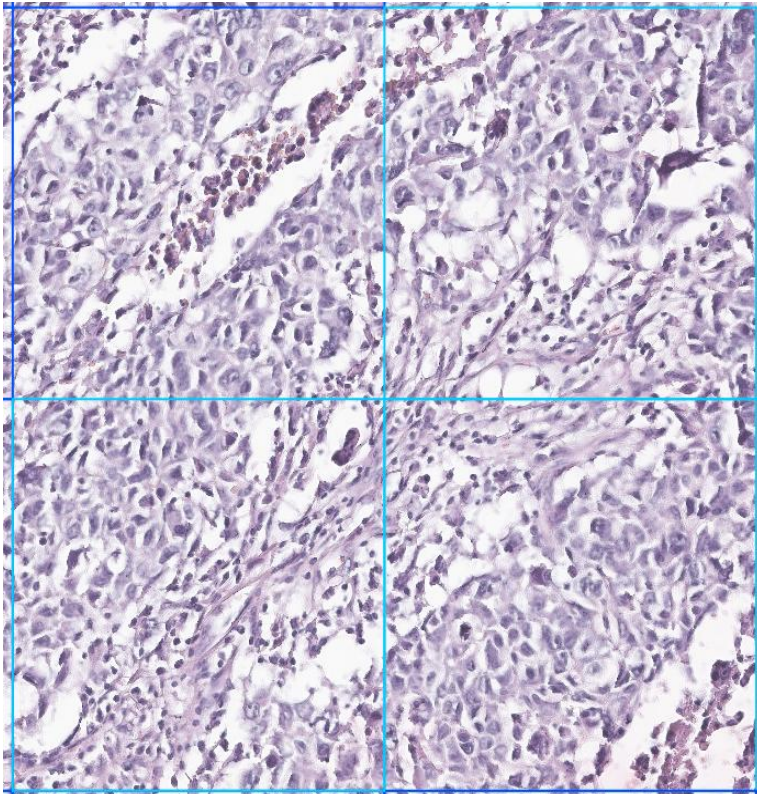
Astromapper 매핑 결과



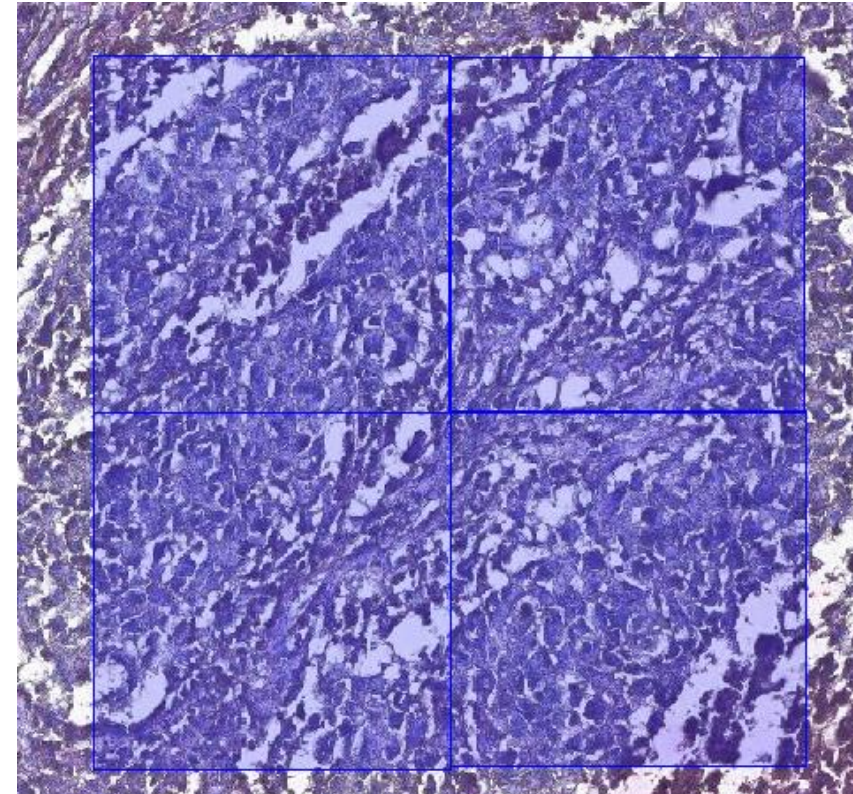


- Section **f11** => Low Score & Tumor

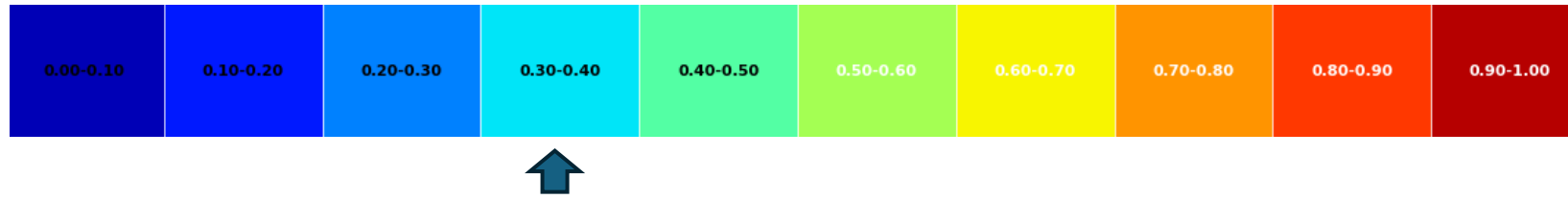
AI 스코어



Astromapper 매핑 결과

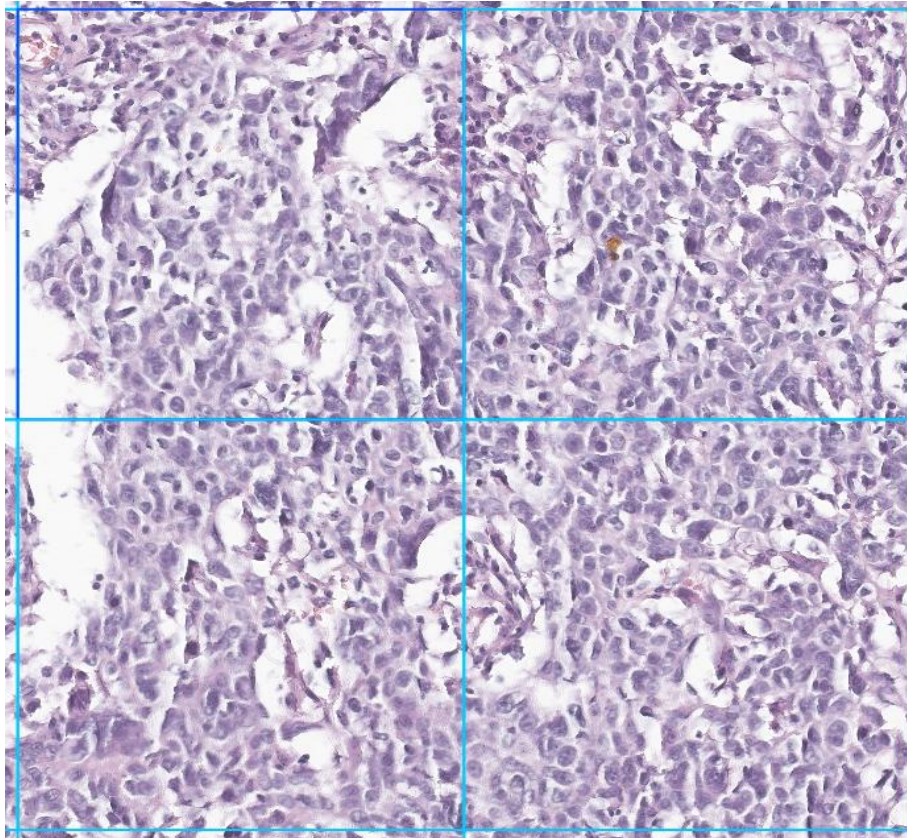


OmixAI 1-152-19 환자 샘플 low-score tumor regions

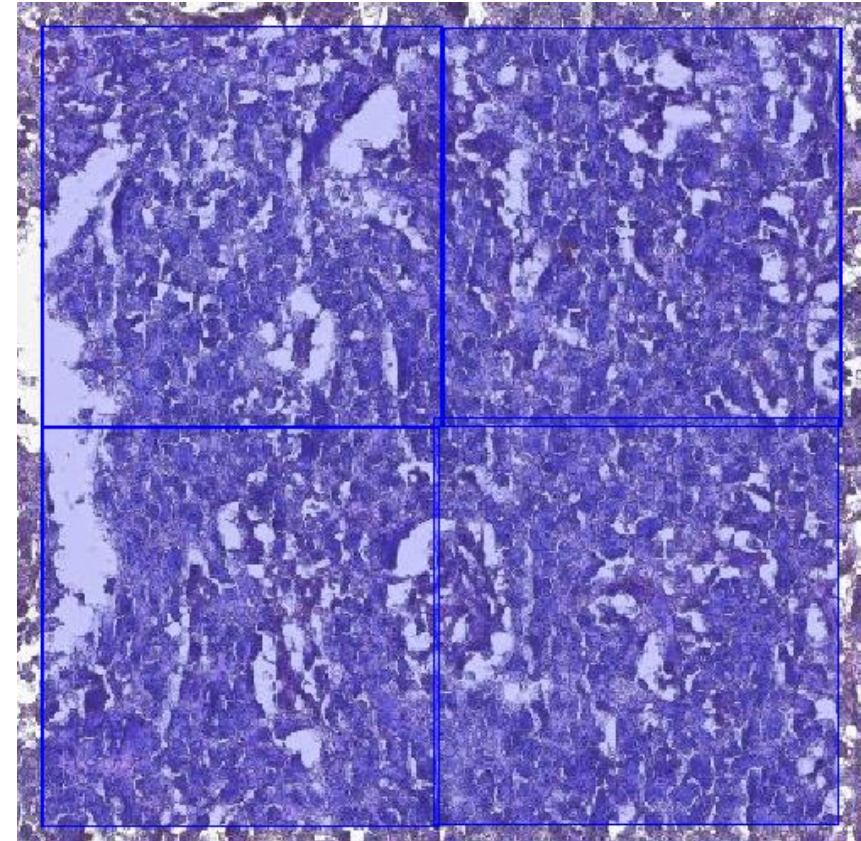


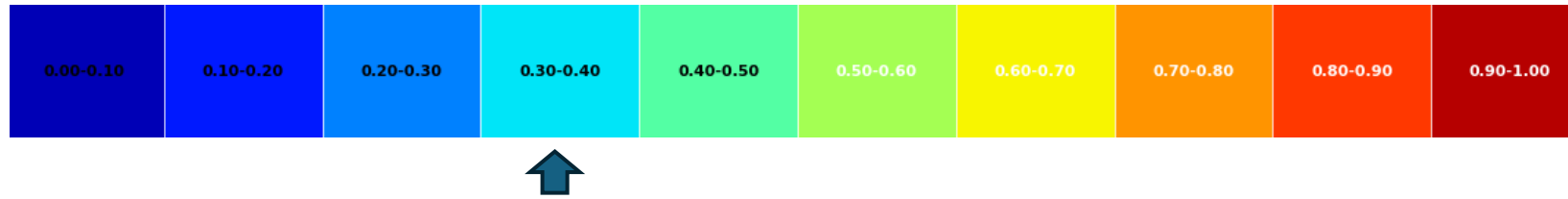
- Section **f12** => Low Score & Tumor

AI 스코어



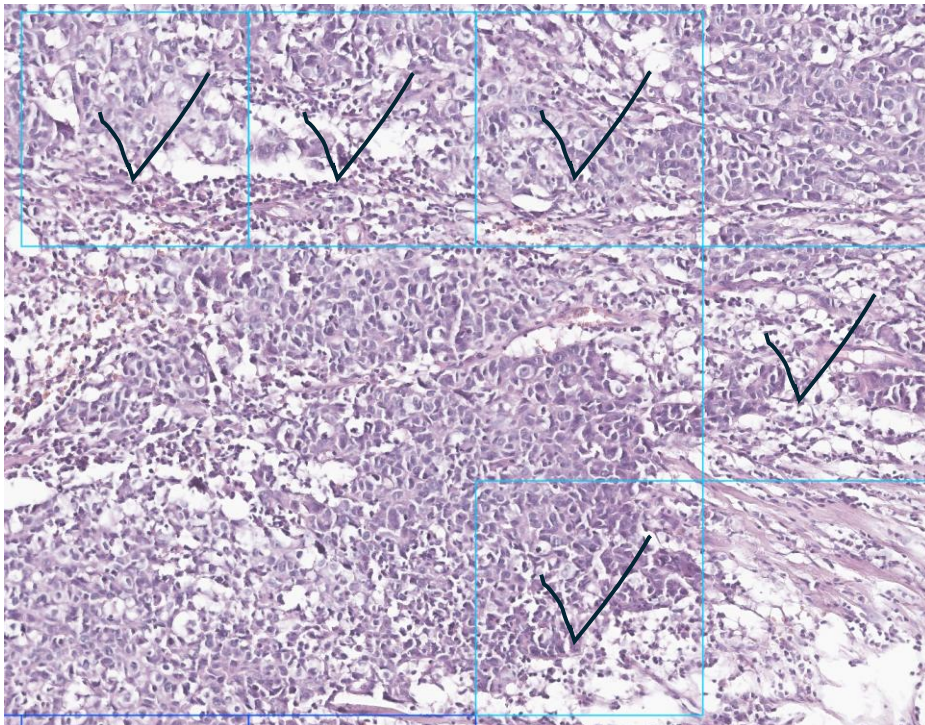
Astromapper 매핑 결과



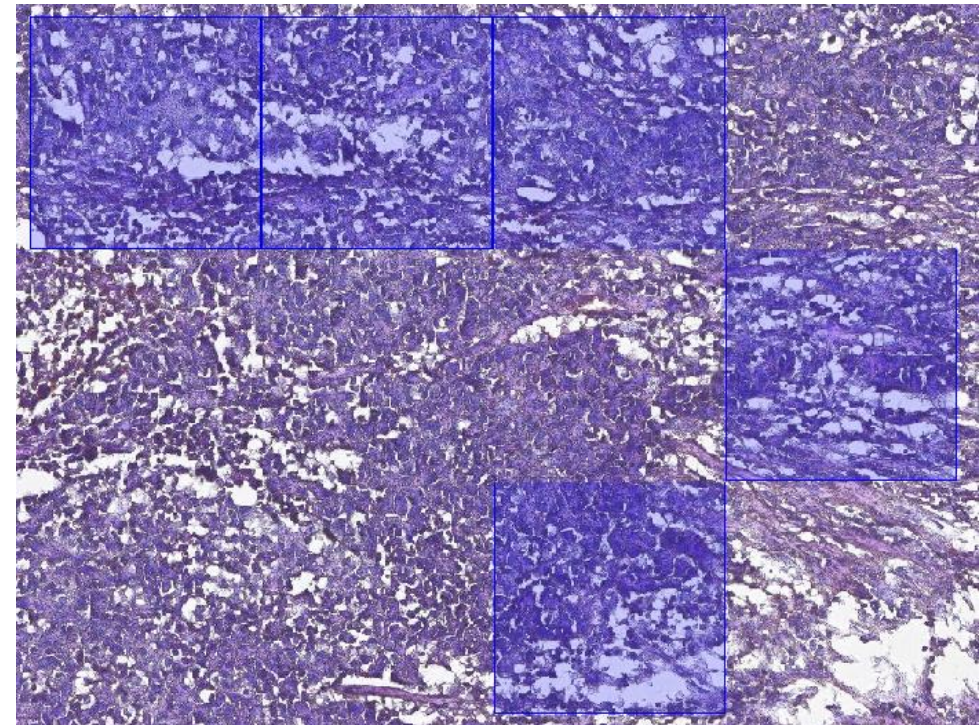


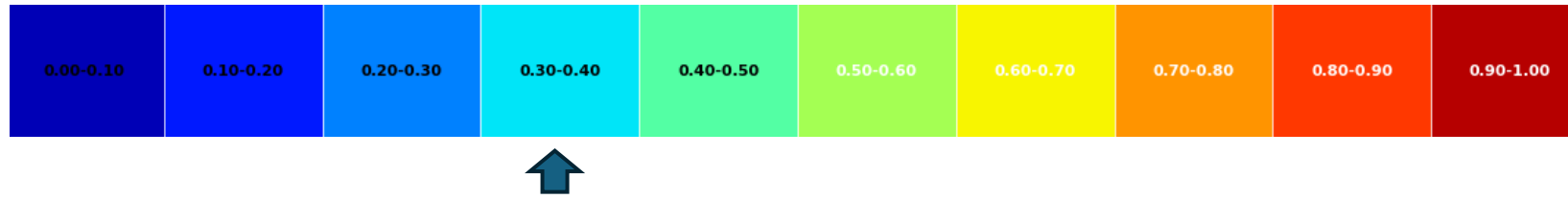
- Section **f13** => Low Score & Tumor

AI 스코어



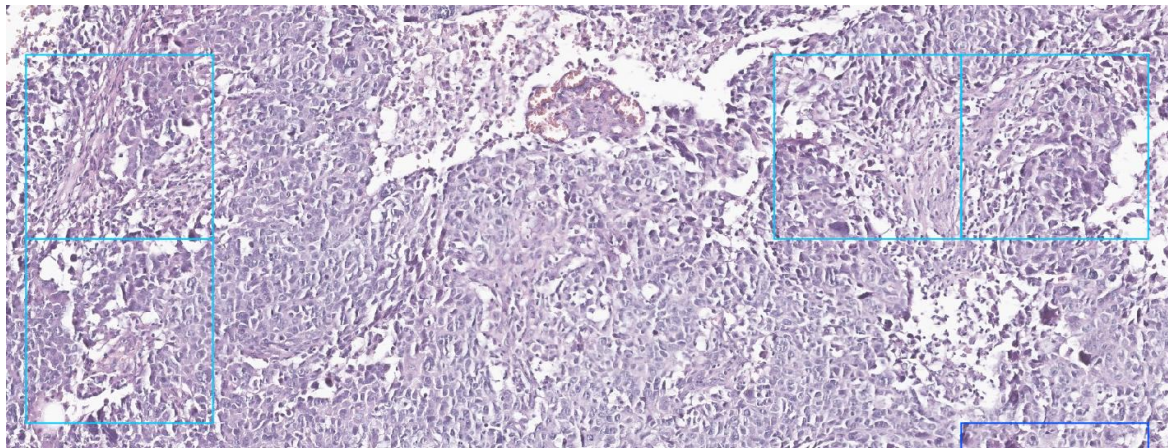
Astromapper 매핑 결과



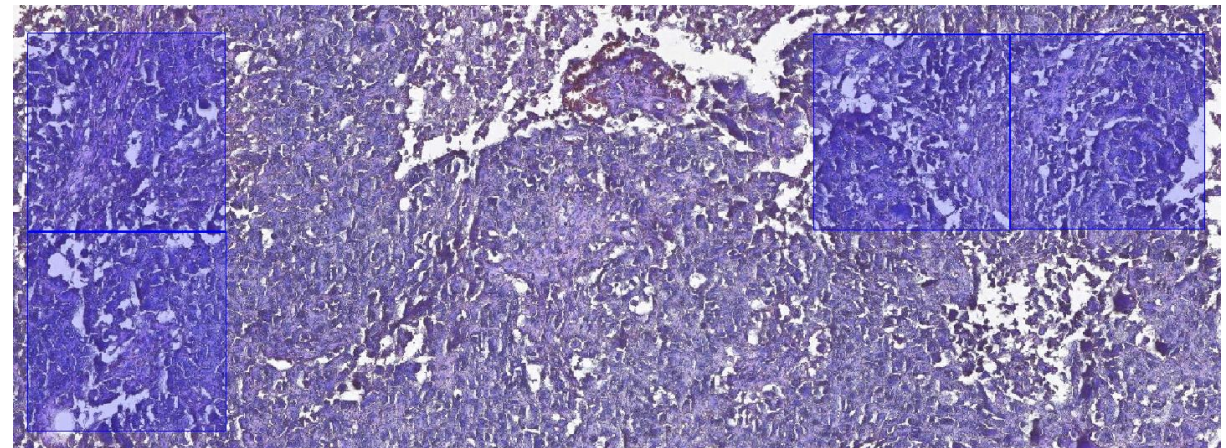


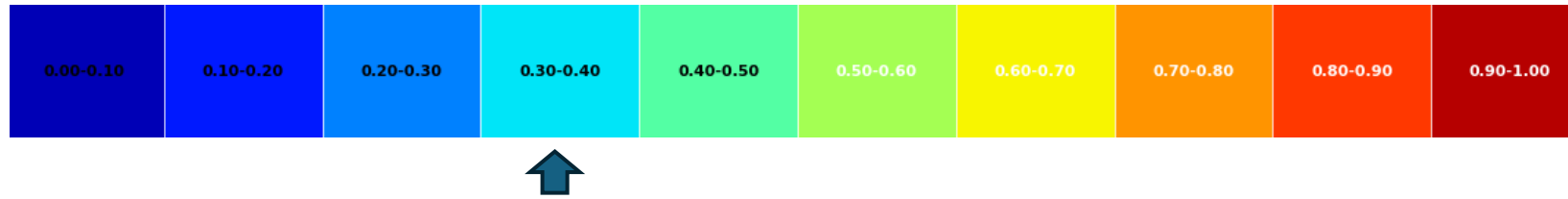
- Section **f14** => Low Score & Tumor

AI 스코어



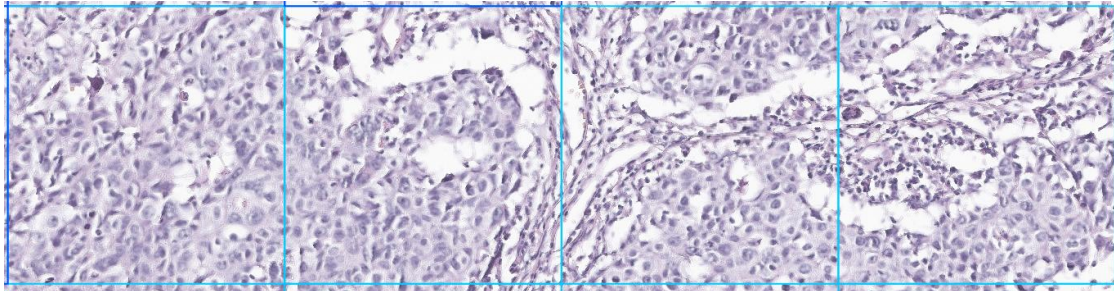
Astromapper 매핑 결과



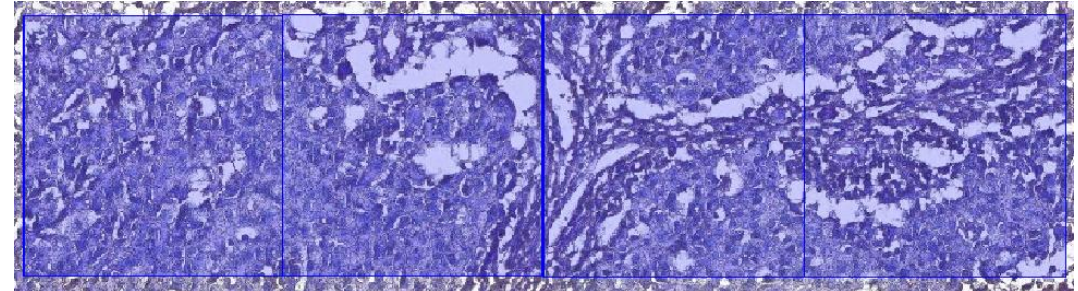


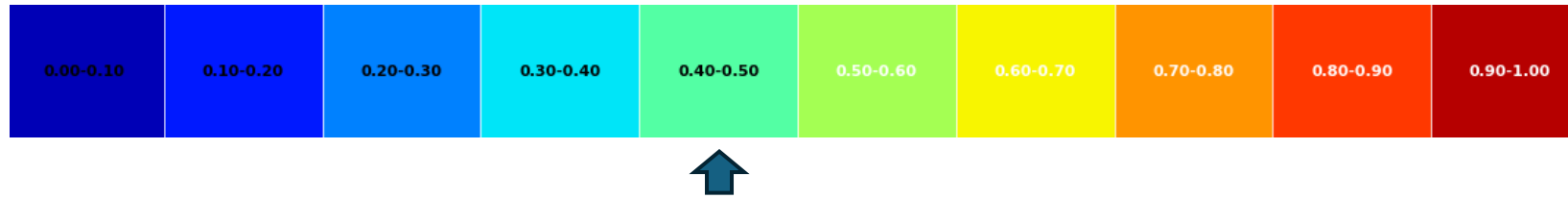
- Section **f15** => Low Score & Tumor

AI 스코어



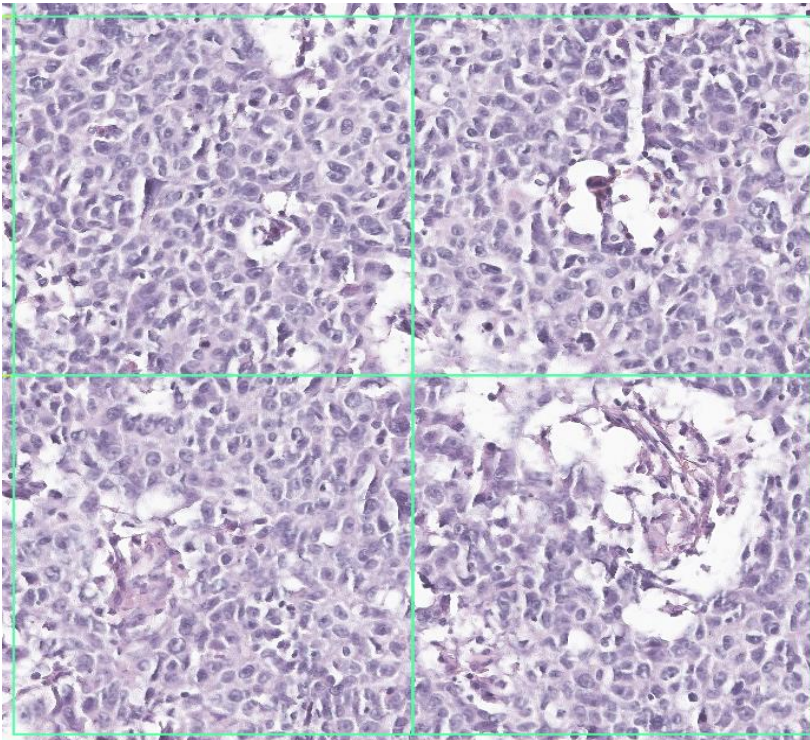
Astromapper 매핑 결과



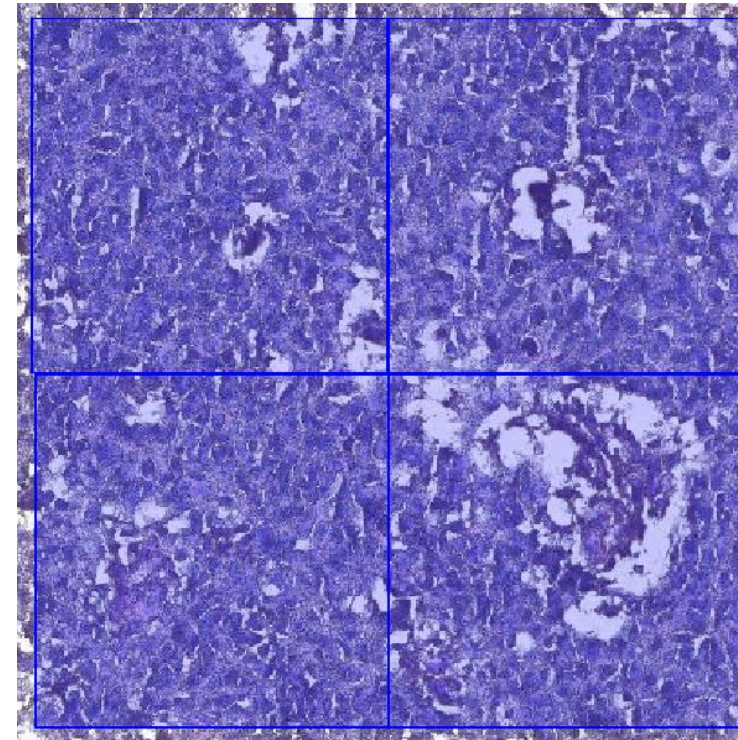


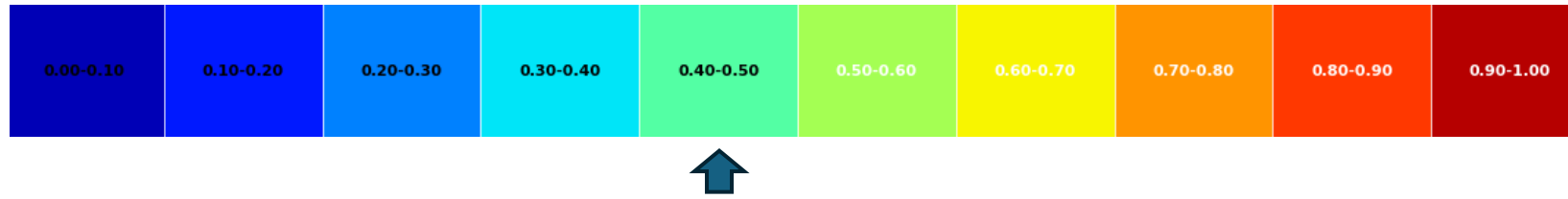
- Section **v1** => Low Score & Tumor

AI 스코어



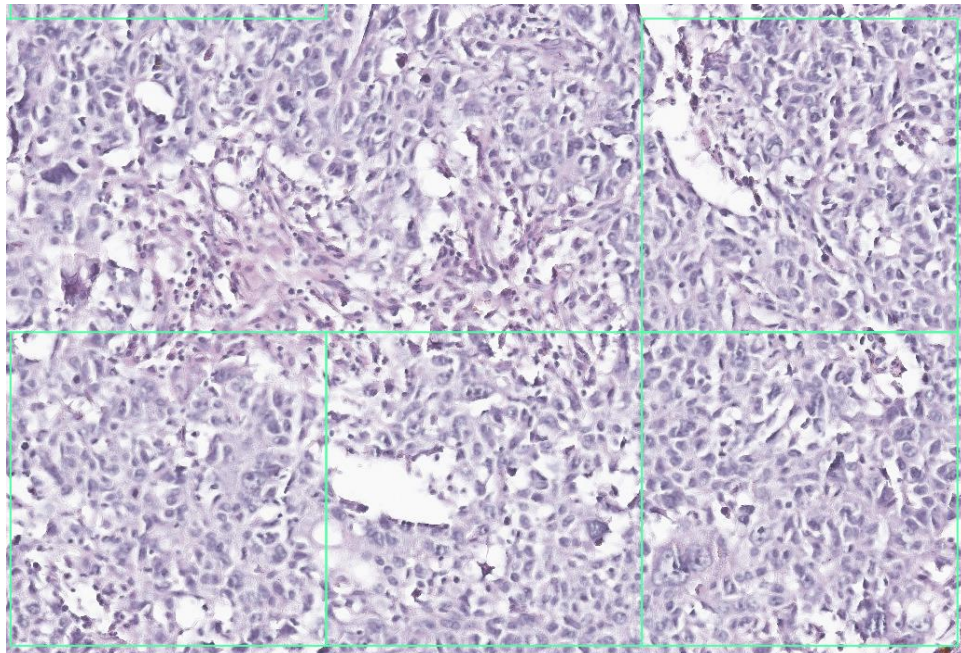
Astromapper 매핑 결과



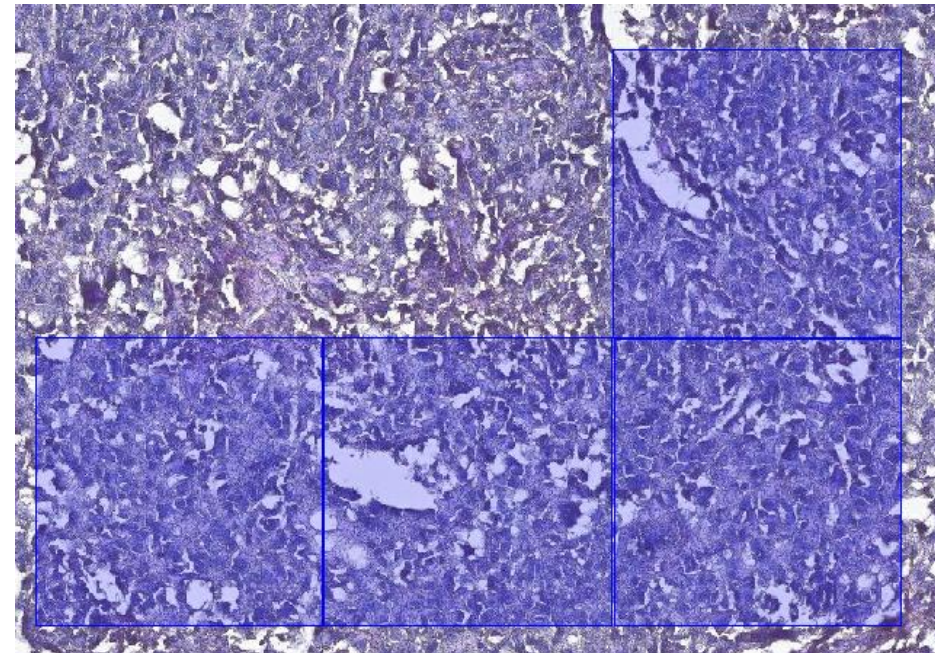


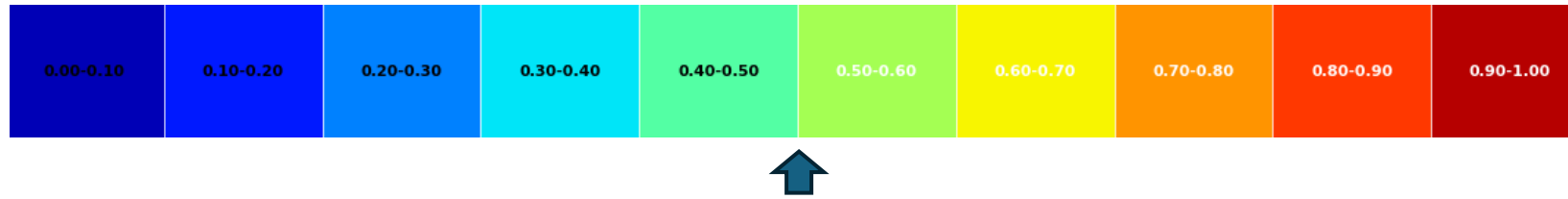
- Section **v2** => Low Score & **Tumor**

AI 스코어



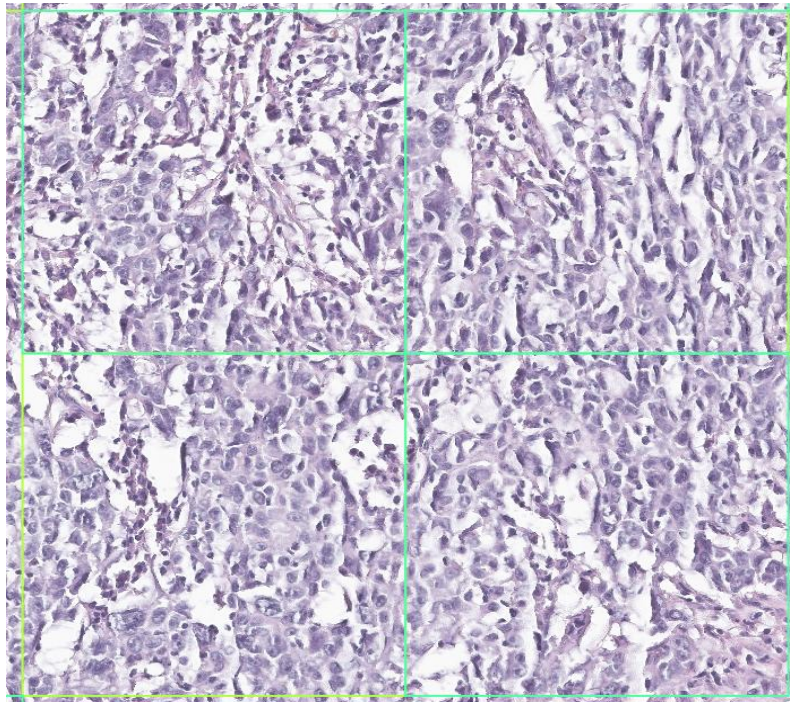
Astromapper 매핑 결과



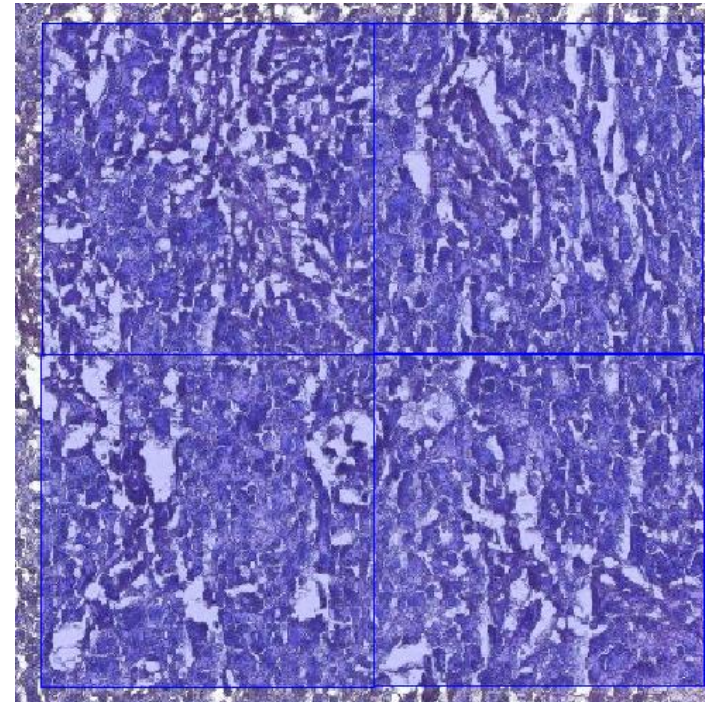


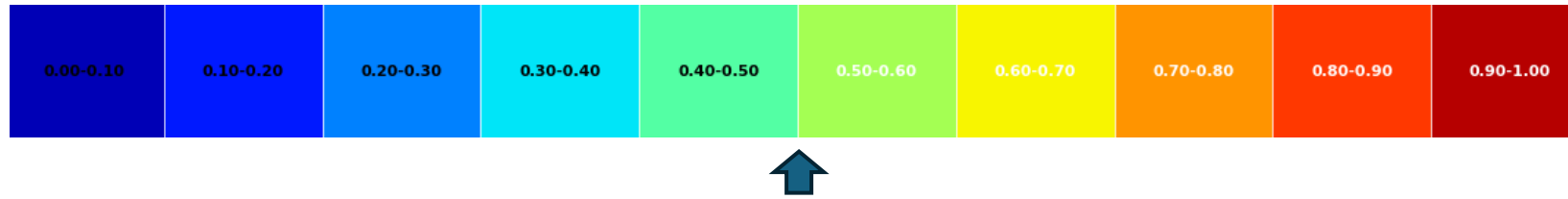
- Section v3 => Low Score & Tumor

AI 스코어



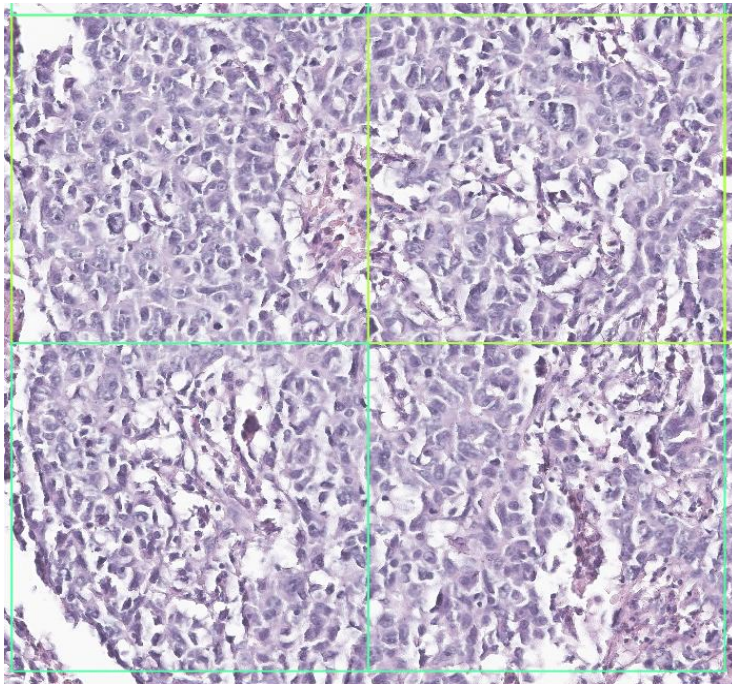
Astromapper 매핑 결과



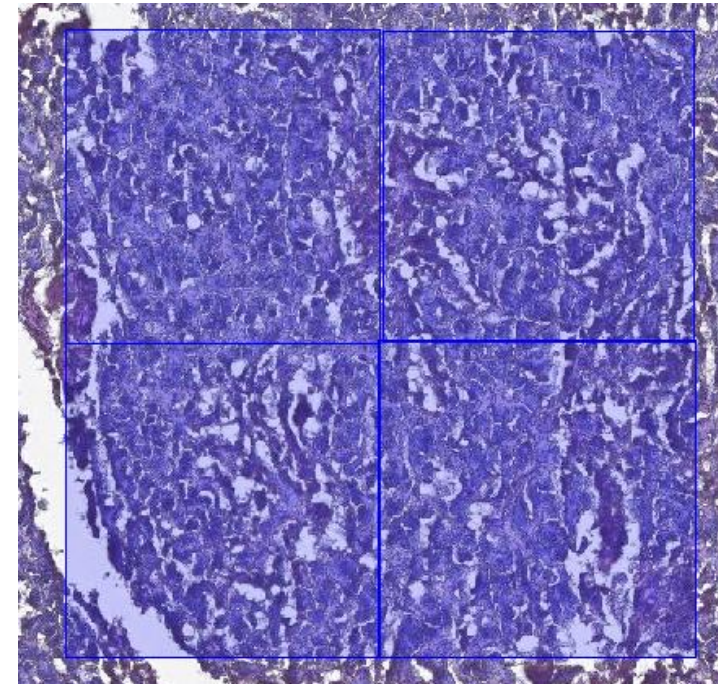


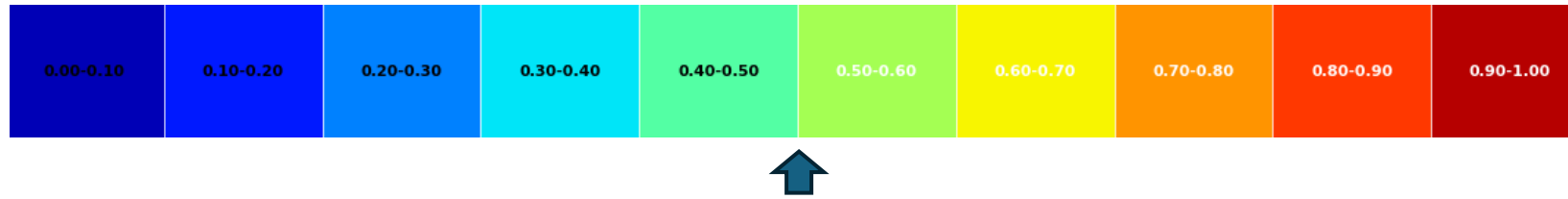
- Section **v4** => Low Score & Tumor

AI 스코어



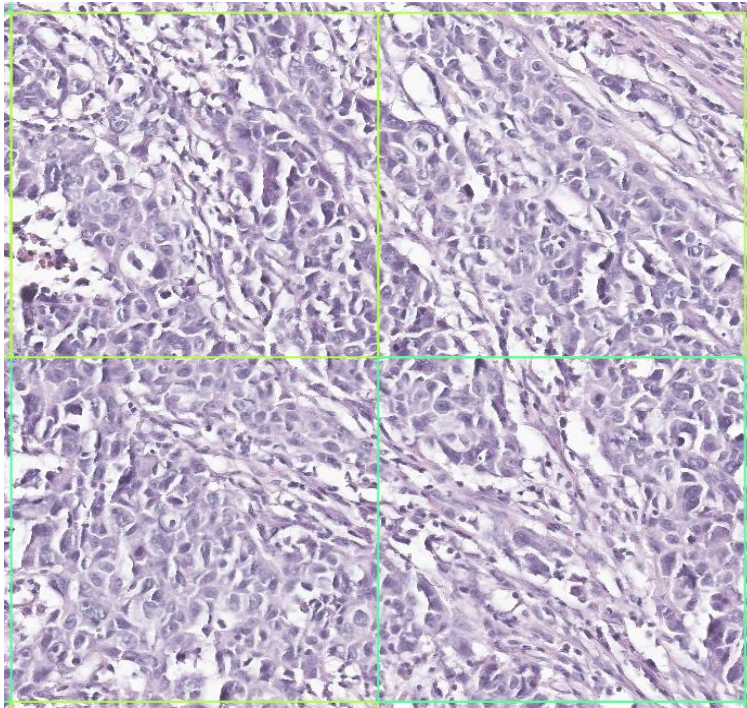
Astromapper 매핑 결과



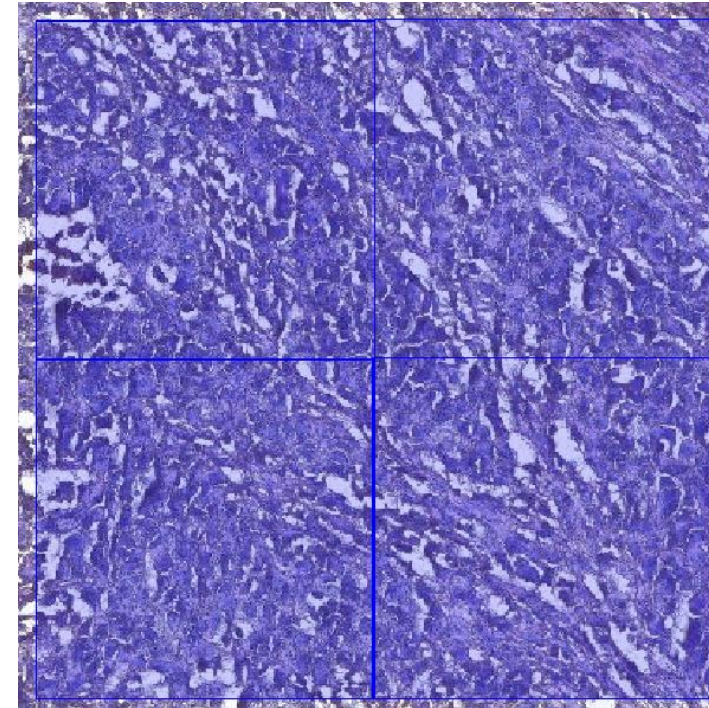


- Section **v5** => Low Score & **Tumor**

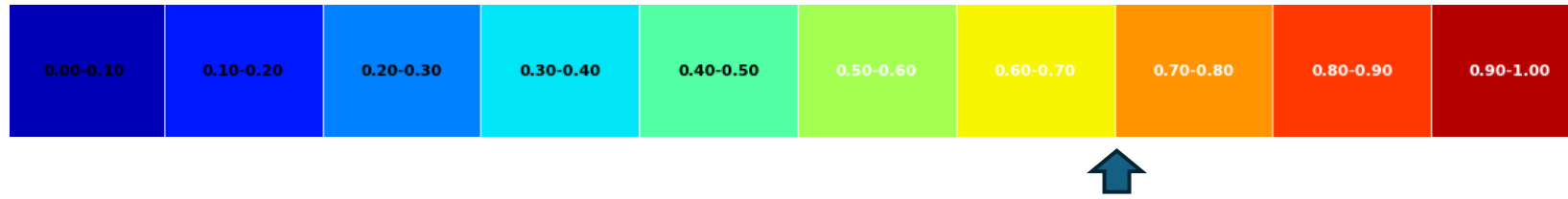
AI 스코어



Astromapper 매핑 결과

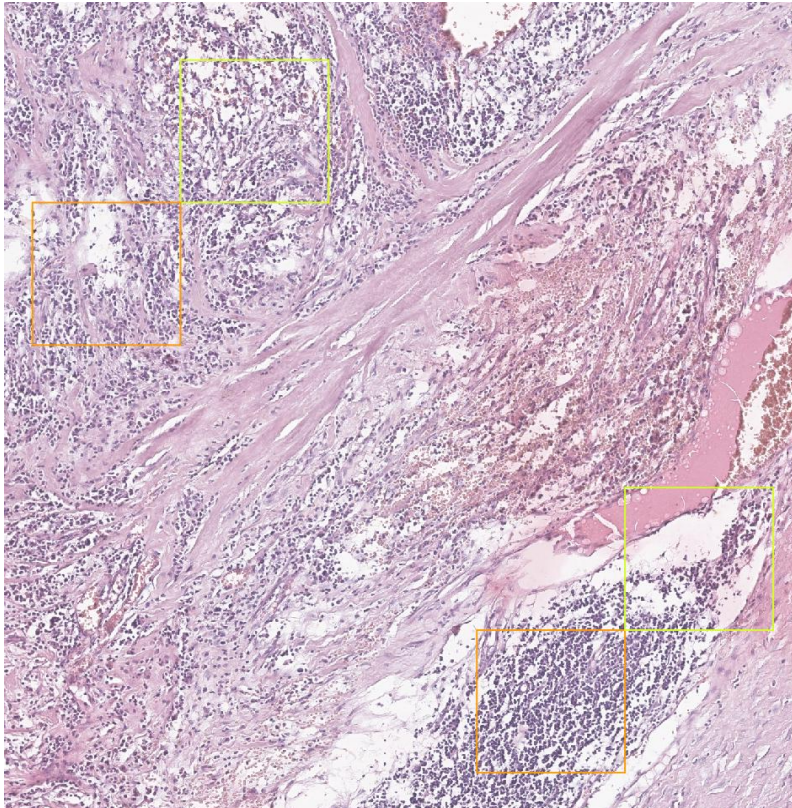


OmixAI 1-152-19 환자 샘플 high-score t-cell regions

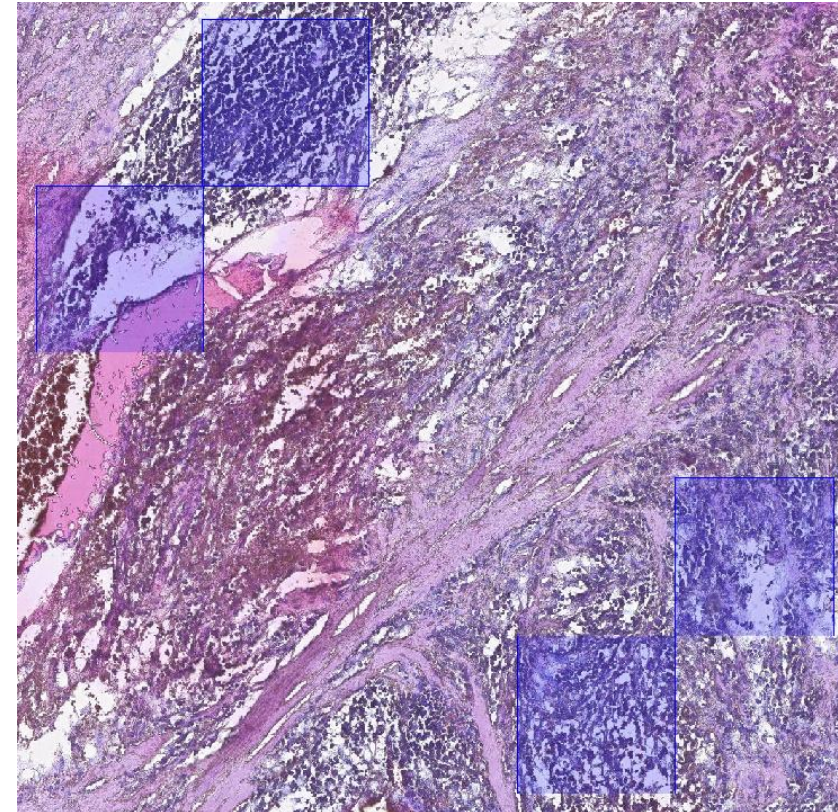


- Section **g1** => **High Score** & **T-cell**

AI 스코어



Astromapper 매핑 결과

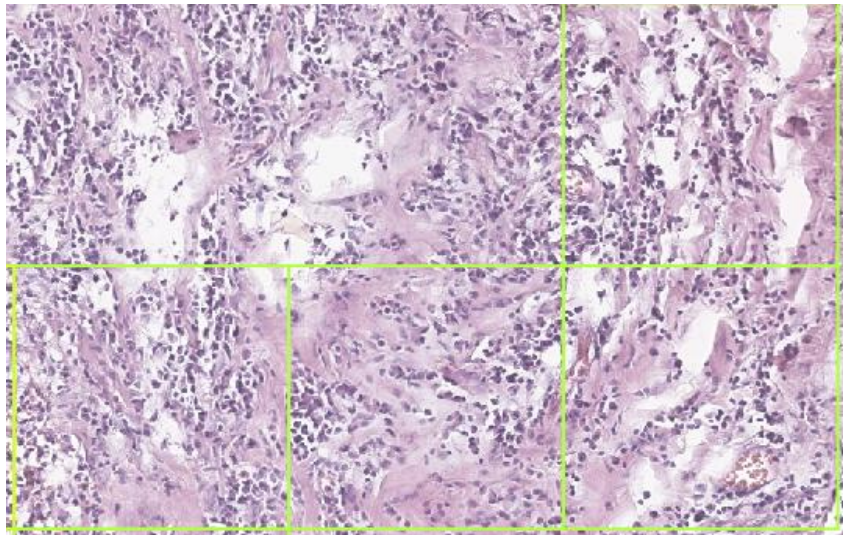


OmixAI 1-152-19 환자 샘플 high-score t-cell regions

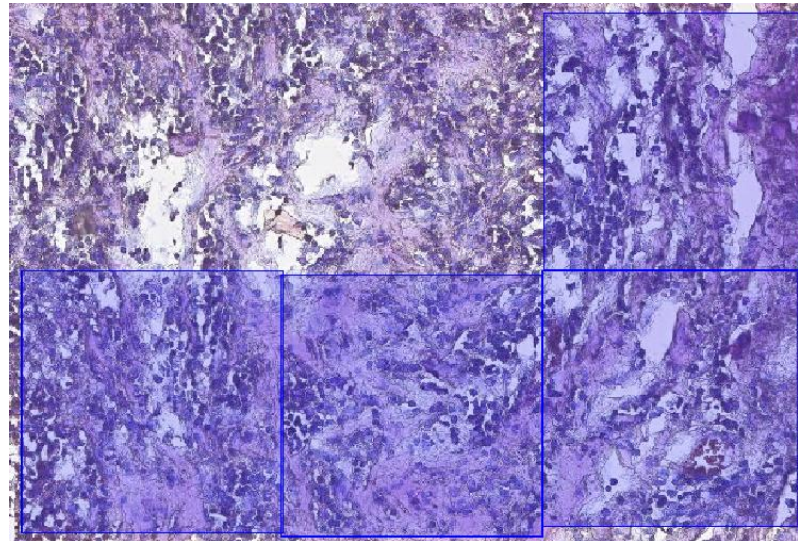


- Section **g2** => **High Score** & T-cell

AI 스코어



Astromapper 매핑 결과

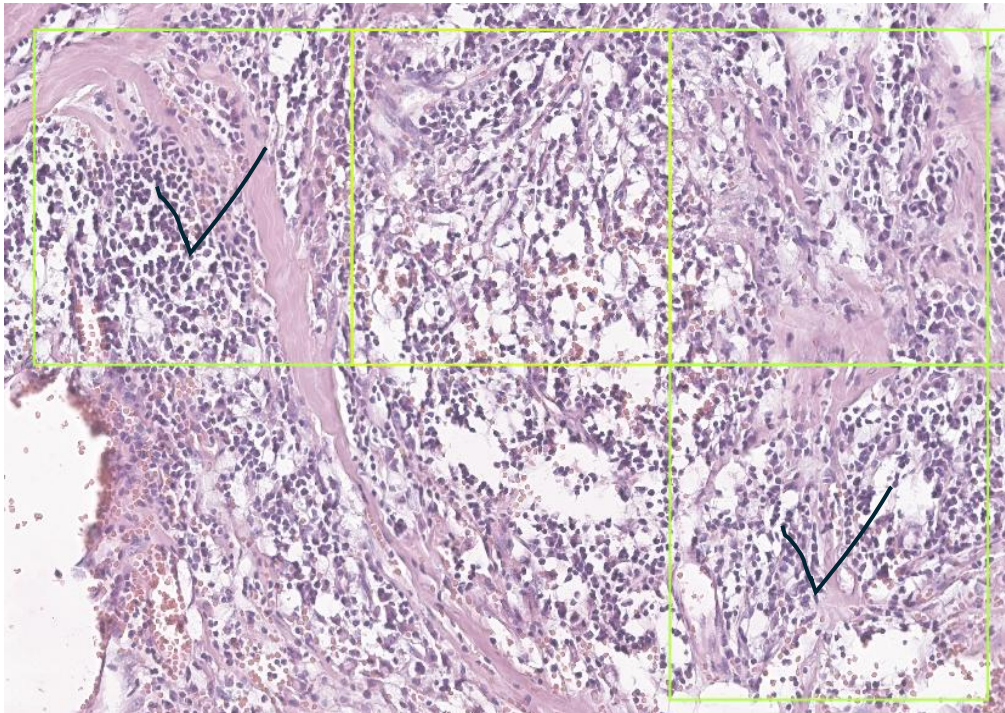


OmixAI 1-152-19 환자 샘플 high-score t-cell regions

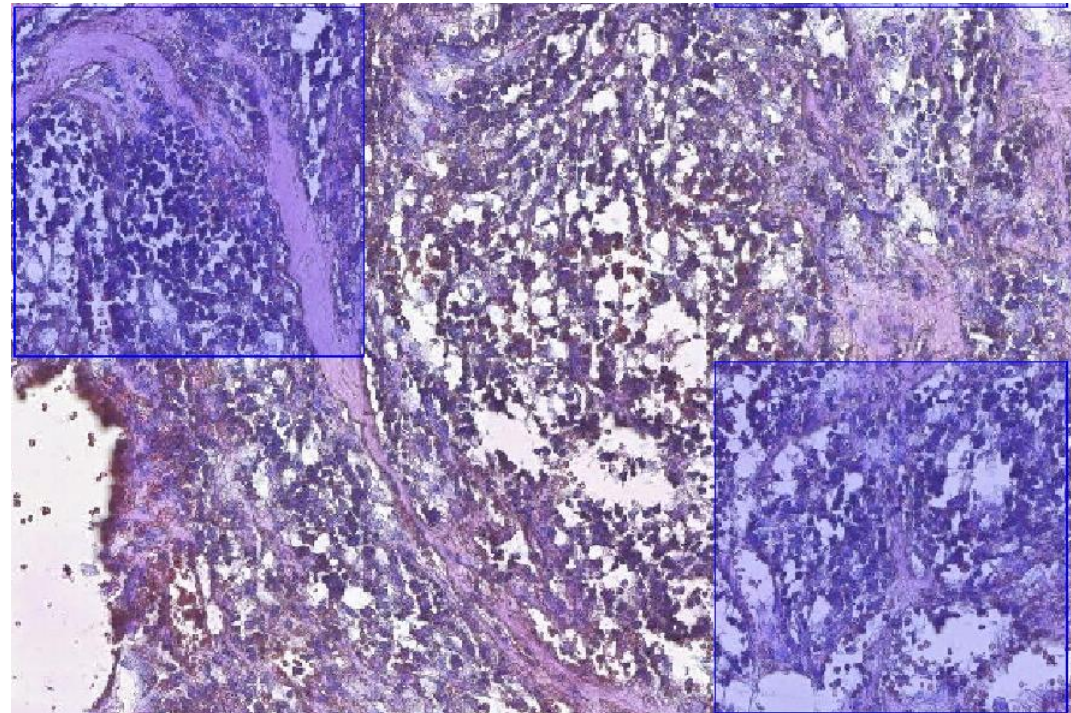


- Section **g3** => **High Score** & T-cell

AI 스코어



Astromapper 매핑 결과

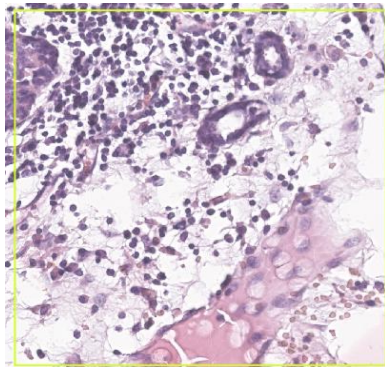
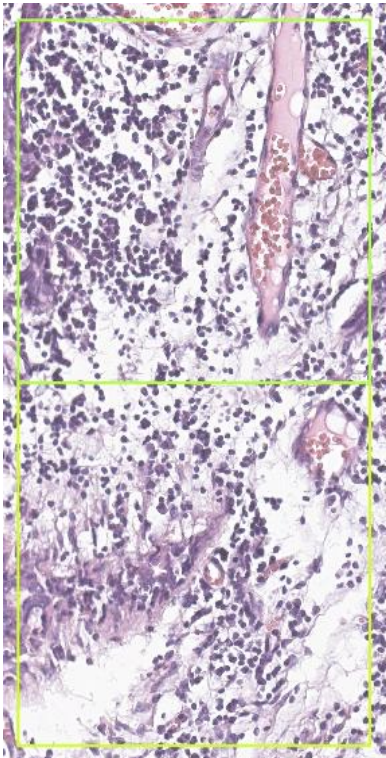


OmixAI 1-152-19 환자 샘플 high-score t-cell regions

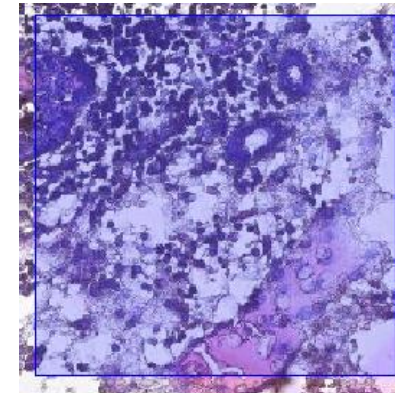
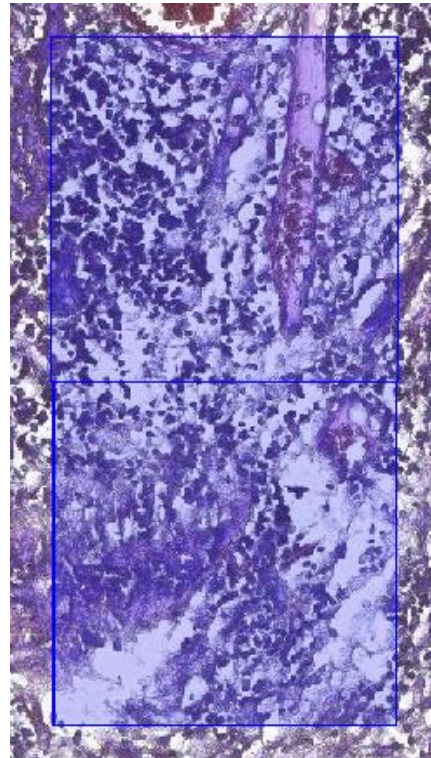


- Section **g4** => **High Score** & T-cell

AI 스코어



Astromapper 매핑 결과

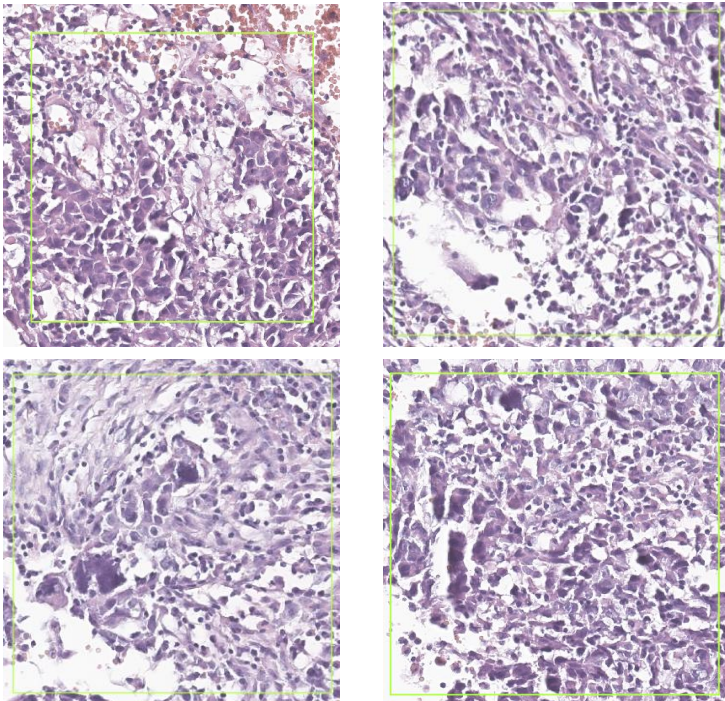


OmixAI 1-152-19 환자 샘플 high-score t-cell regions

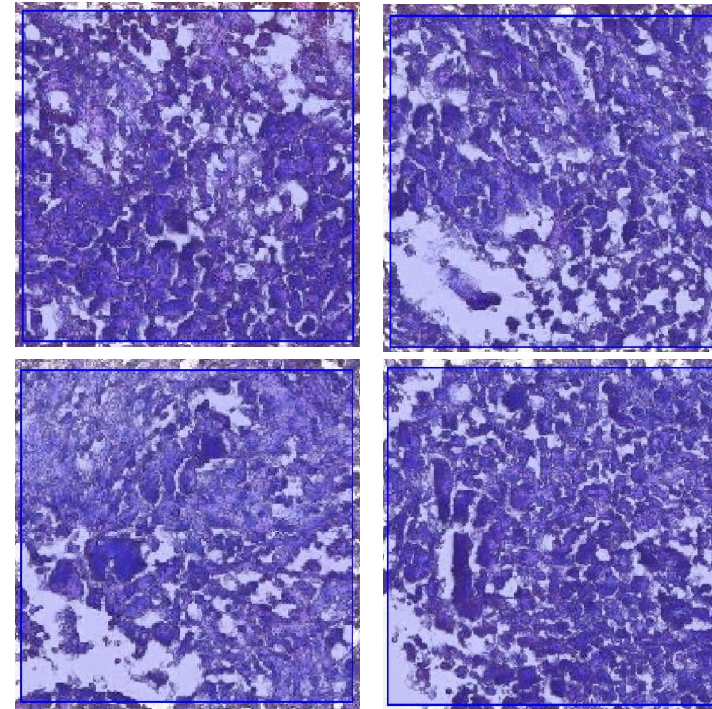


- Section **g5** => **High Score** & T-cell

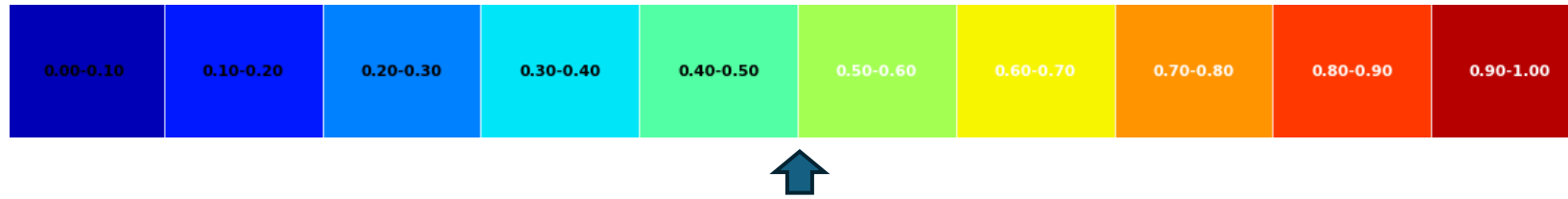
AI 스코어



Astromapper 매핑 결과

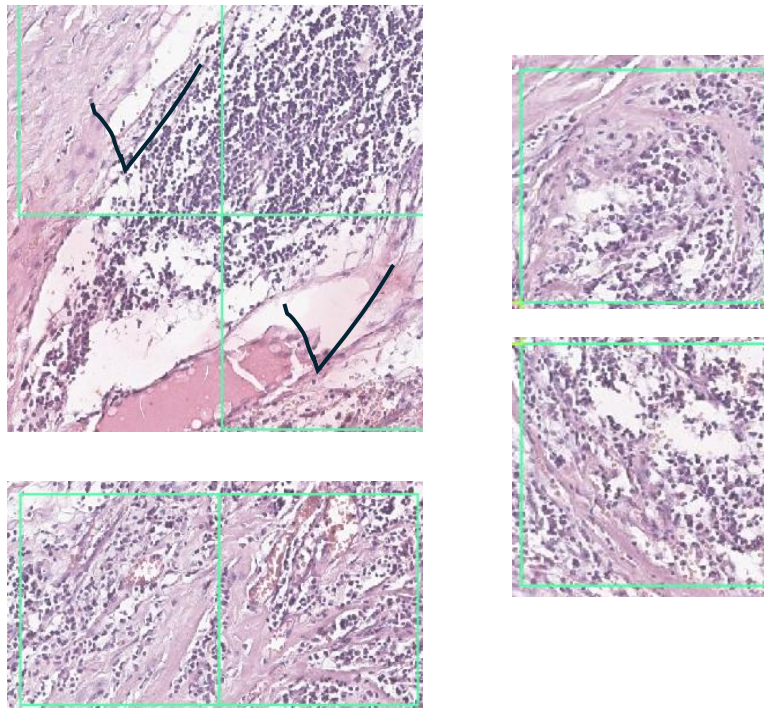


OmixAI 1-152-19 환자 샘플 high-score t-cell regions

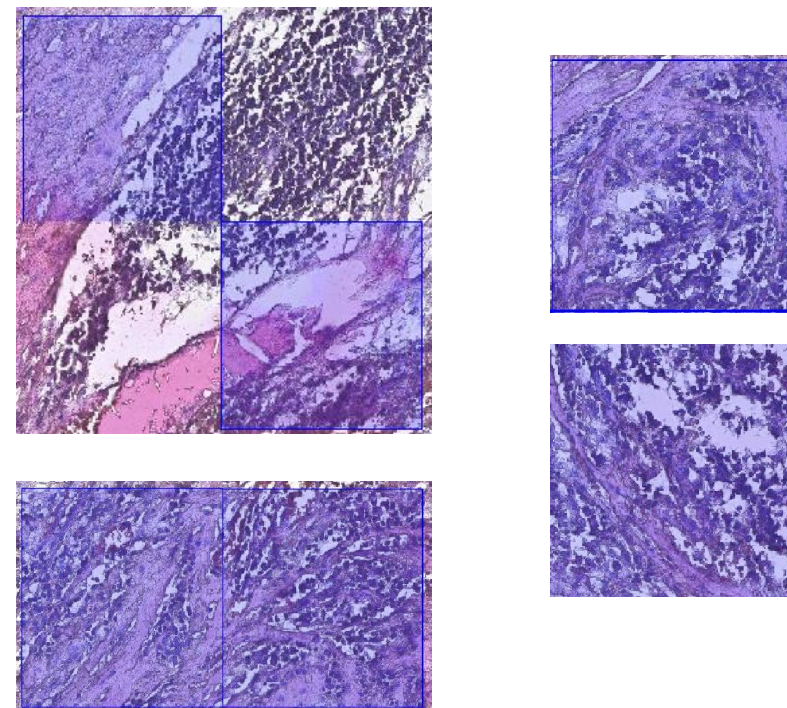


- Section **g6** => **High Score** & T-cell

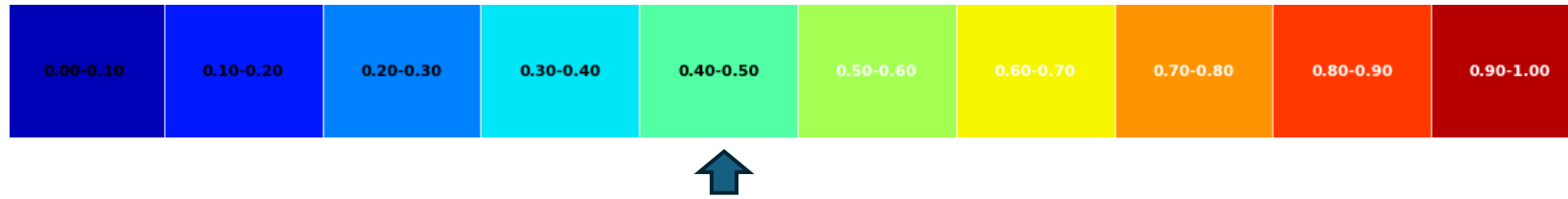
AI 스코어



Astromapper 매핑 결과

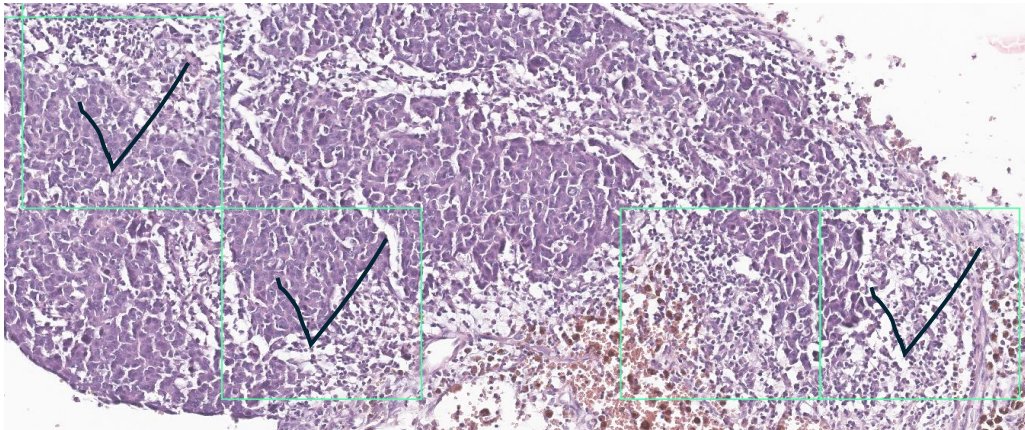


OmixAI 1-152-19 환자 샘플 high-score t-cell regions

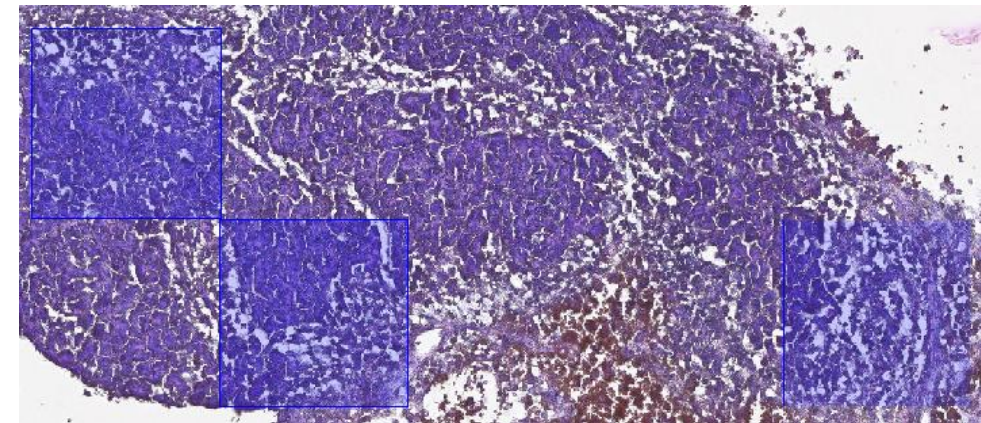


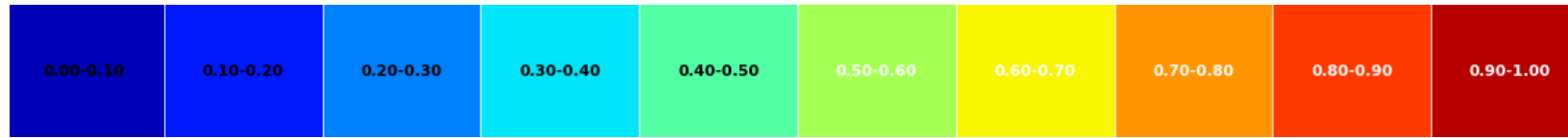
- Section **g7** => **High Score** & T-cell

AI 스코어



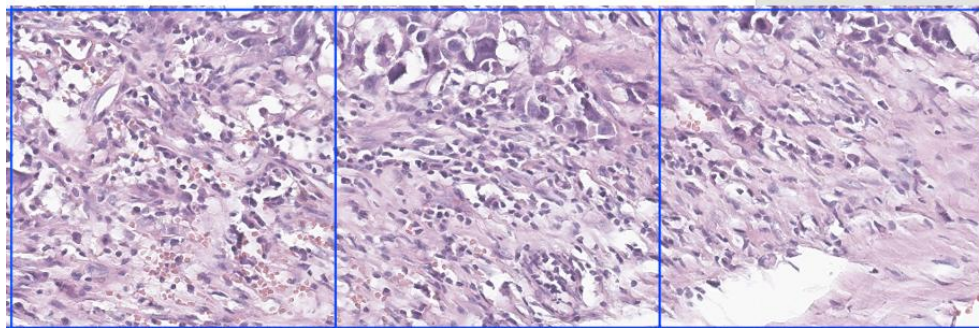
Astromapper 매핑 결과



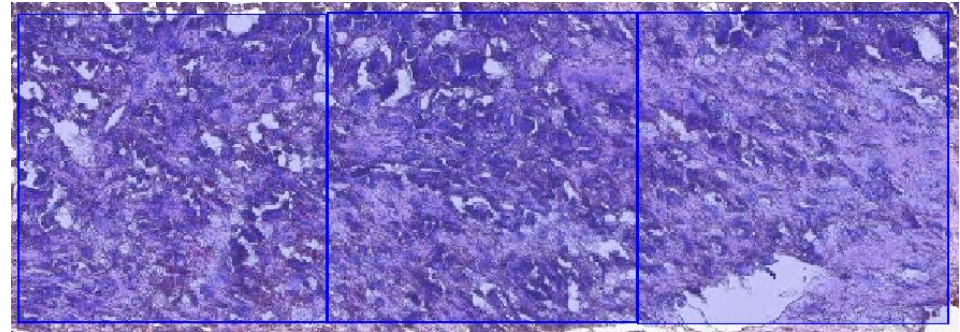


- Section **h1** => Low Score & T-cell

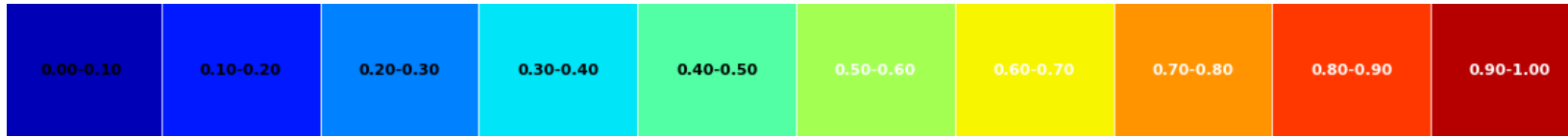
AI 스코어



Astromapper 매핑 결과



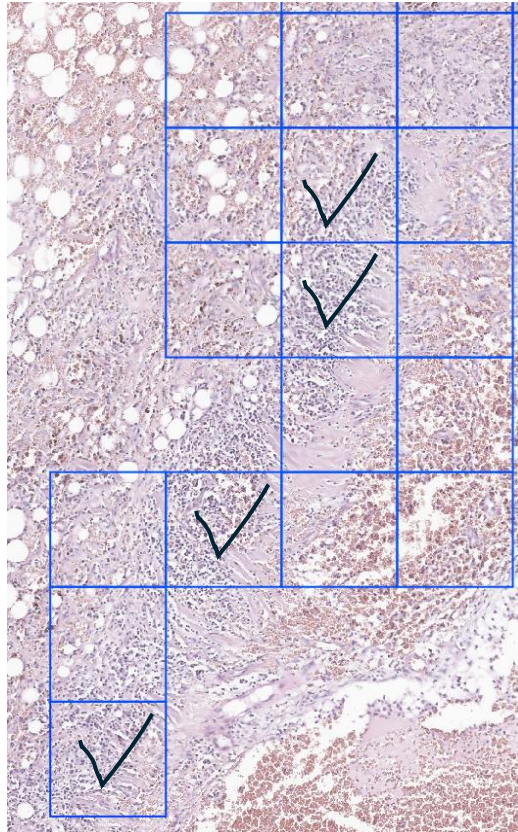
OmixAI 1-152-19 환자 샘플 low-score t-cell regions



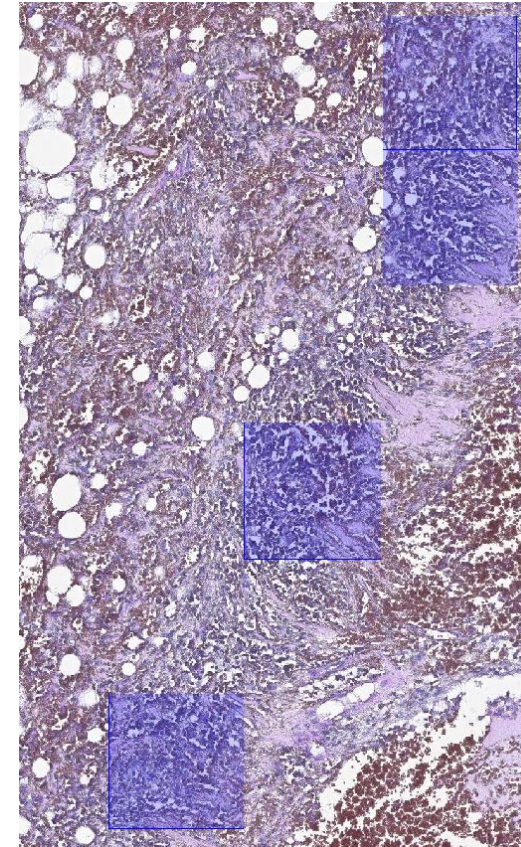
- Section **h2** => Low Score & T-cell

특징 : Blood cell이 포함됨

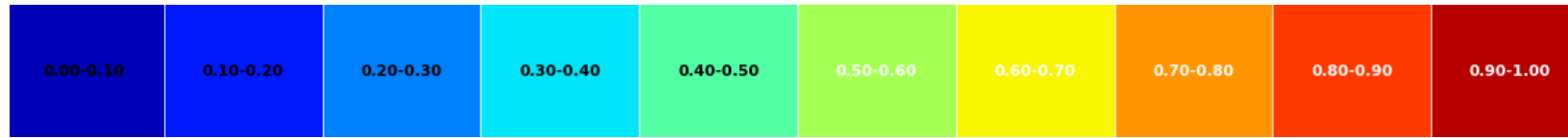
AI 스코어



Astromapper 매핑 결과

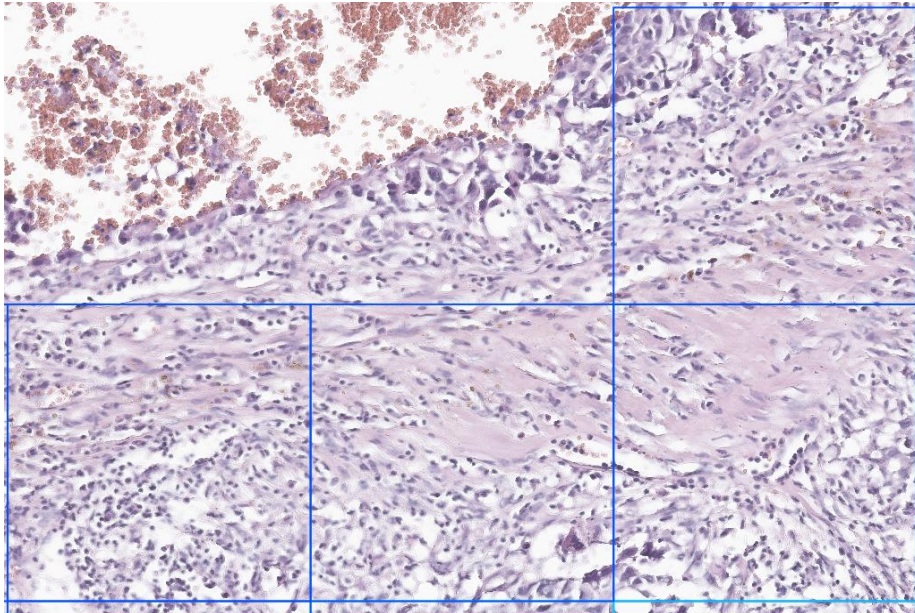


OmixAI 1-152-19 환자 샘플 low-score t-cell regions

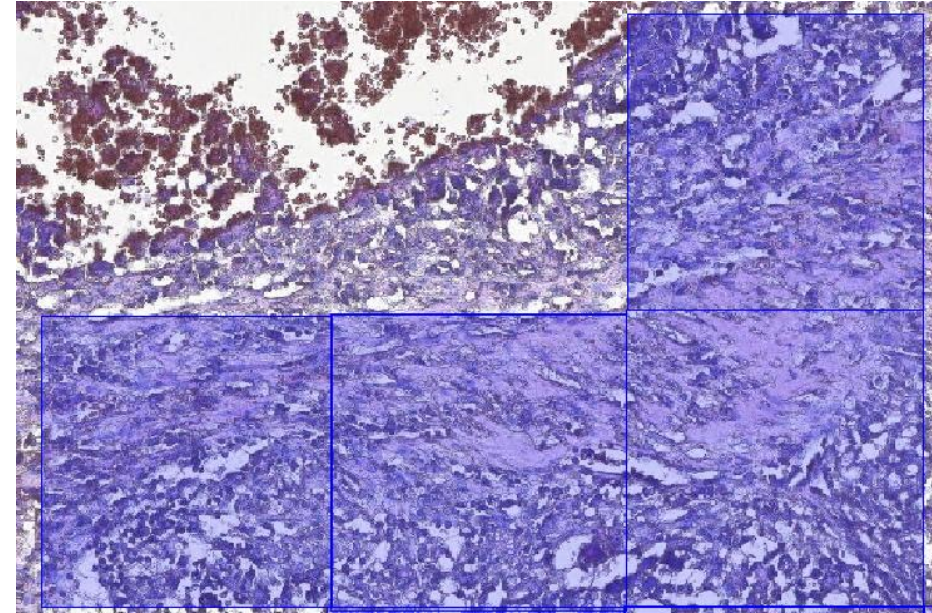


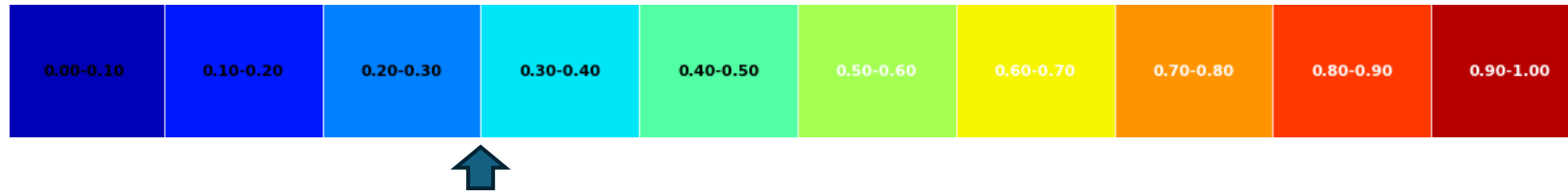
- Section **h3** => Low Score & T-cell

AI 스코어



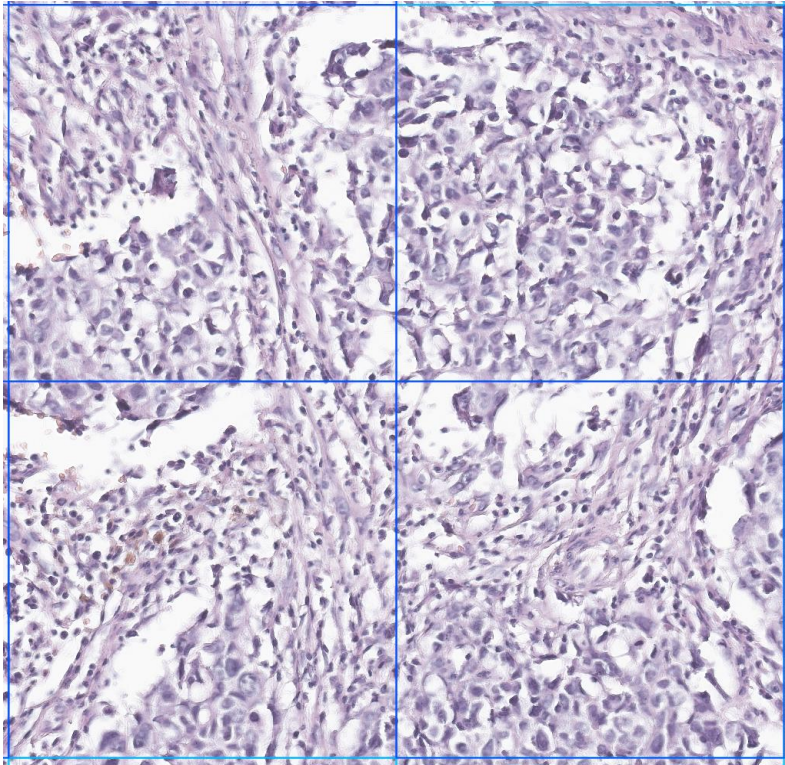
Astromapper 매핑 결과



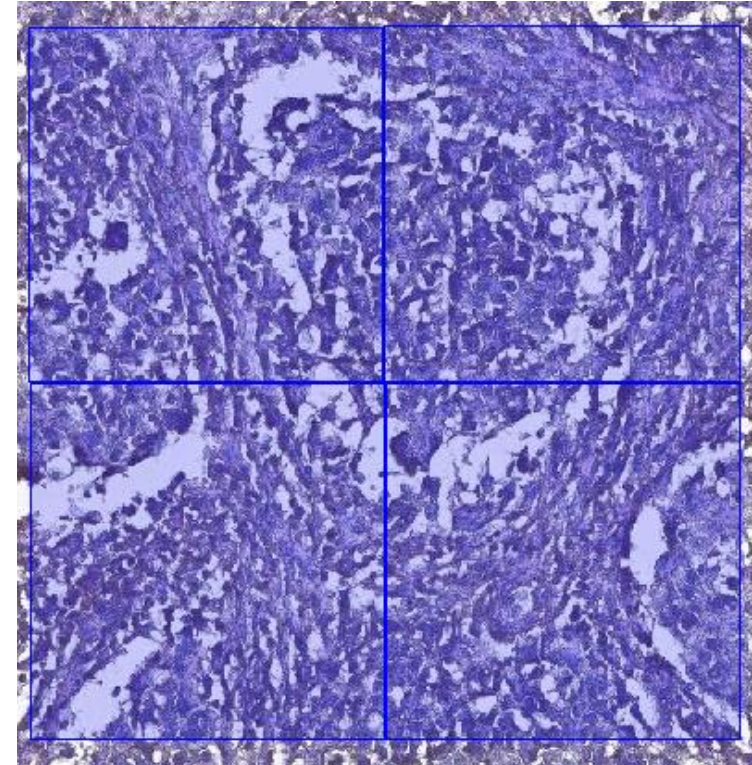


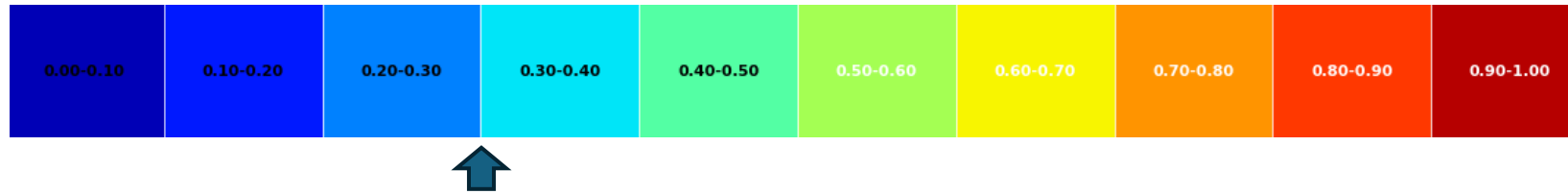
- Section **h4** => Low Score & T-cell

AI 스코어

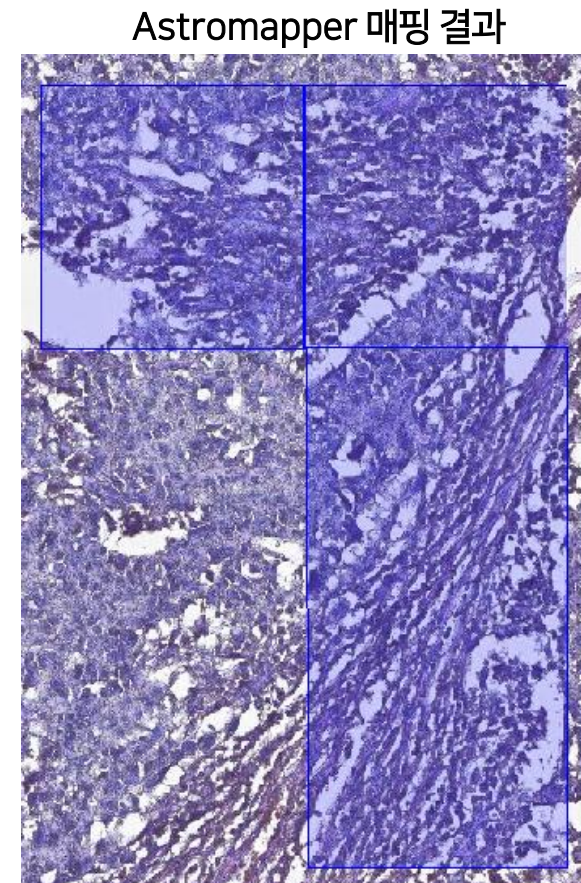
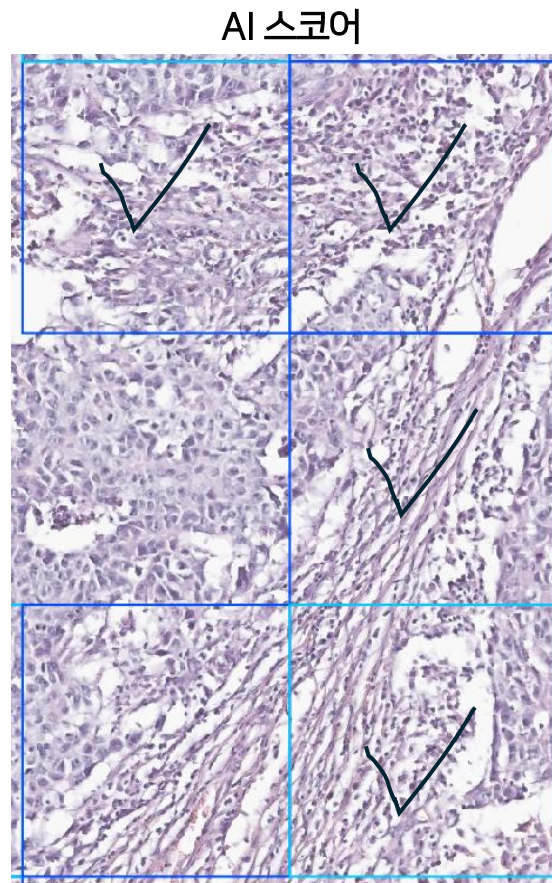


Astromapper 매핑 결과

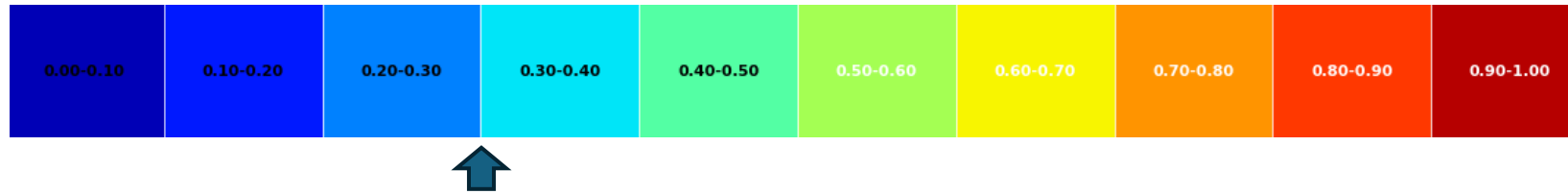




- Section **h5** => Low Score & T-cell

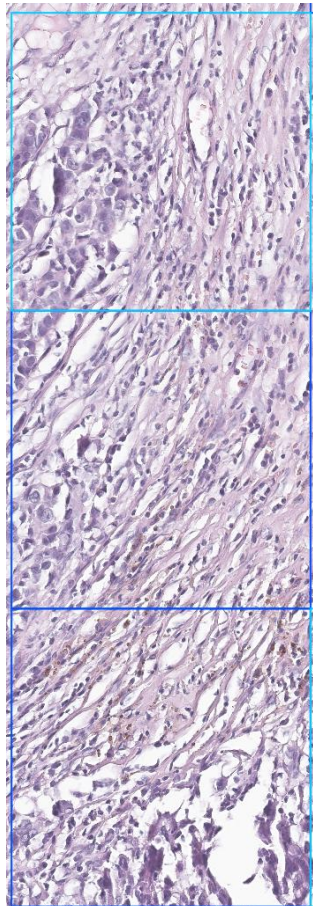


OmixAI 1-152-19 환자 샘플 low-score t-cell regions

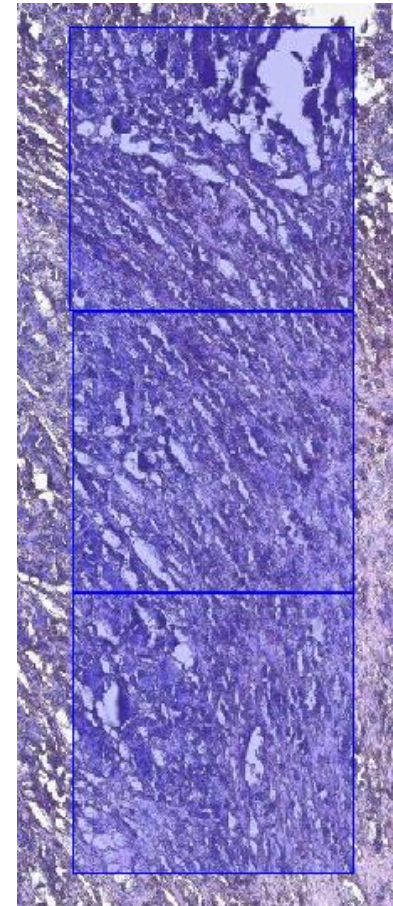


- Section **h6** => Low Score & T-cell

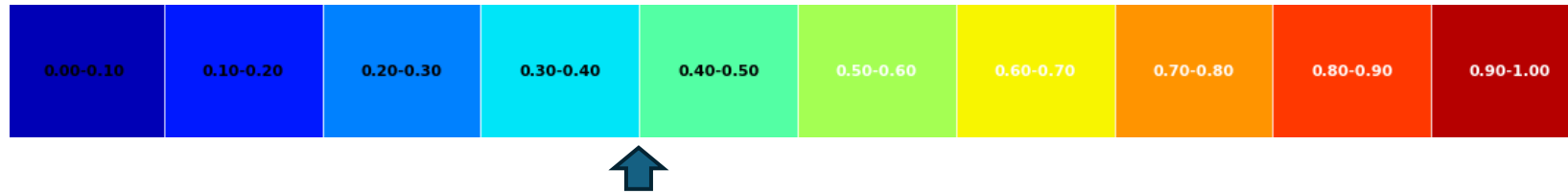
AI 스코어



Astromapper 매핑 결과

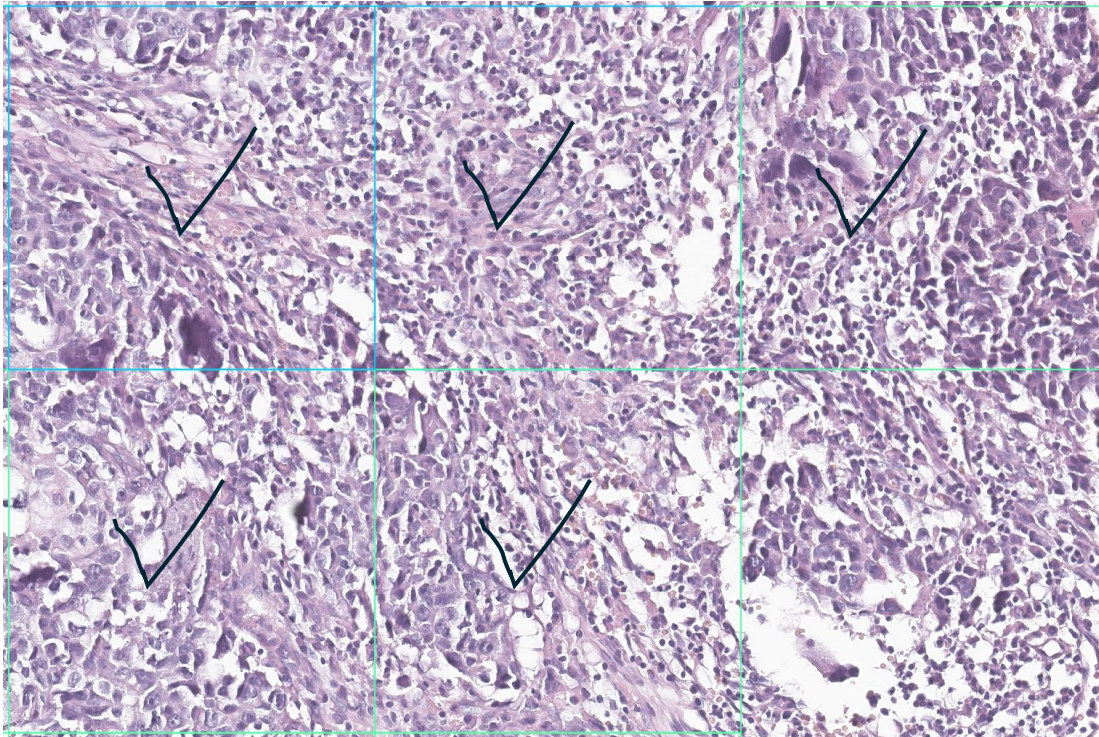


OmixAI 1-152-19 환자 샘플 low-score t-cell regions

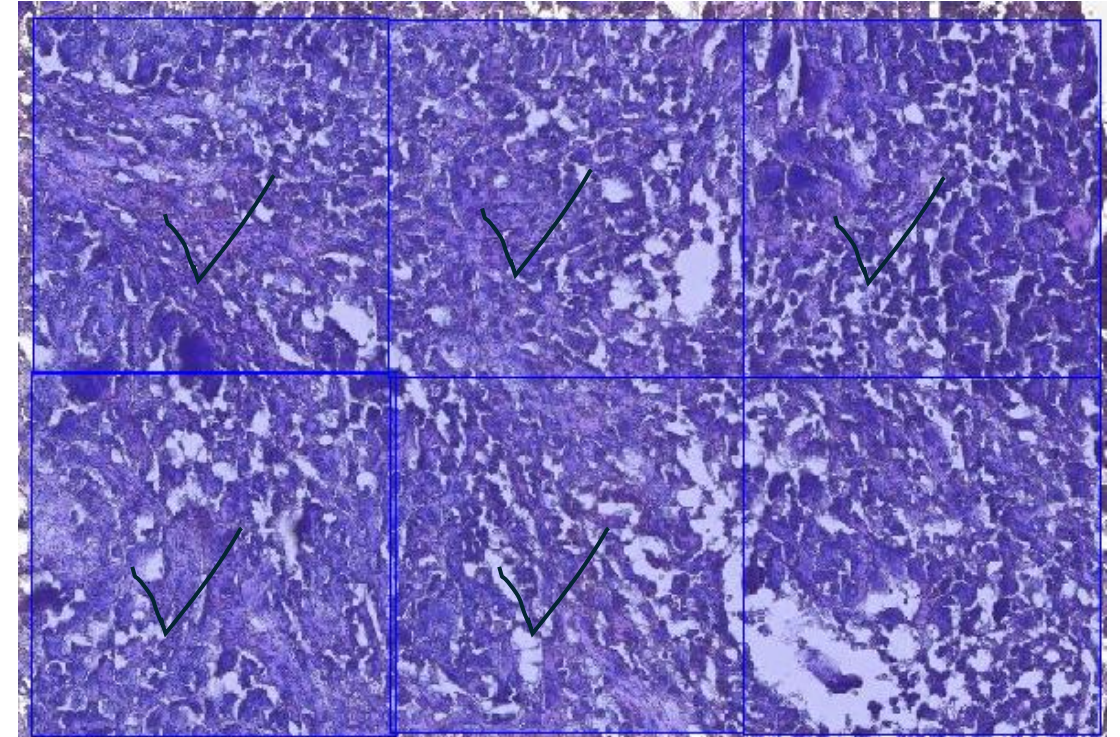


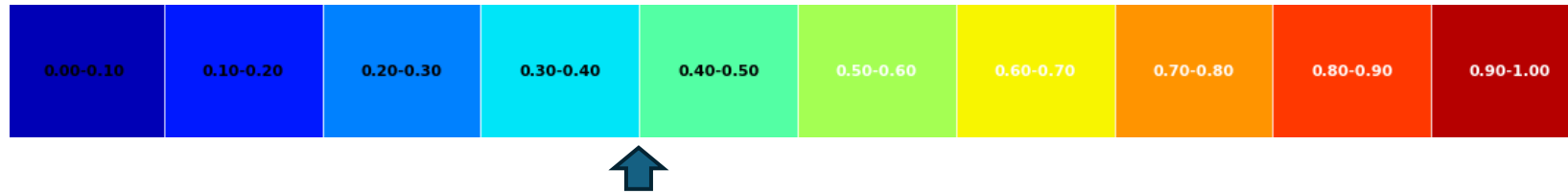
- Section **h7** => Low Score & T-cell

AI 스코어



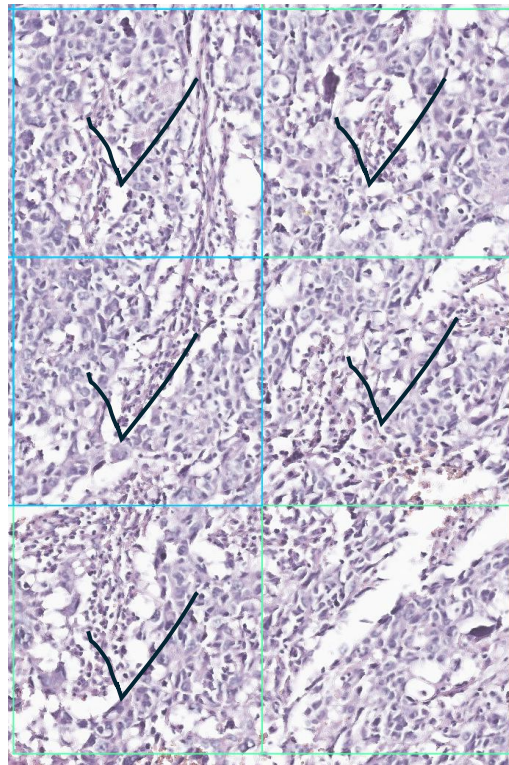
Astromapper 매핑 결과





- Section **h8** => Low Score & T-cell

AI 스코어



Astromapper 매핑 결과

